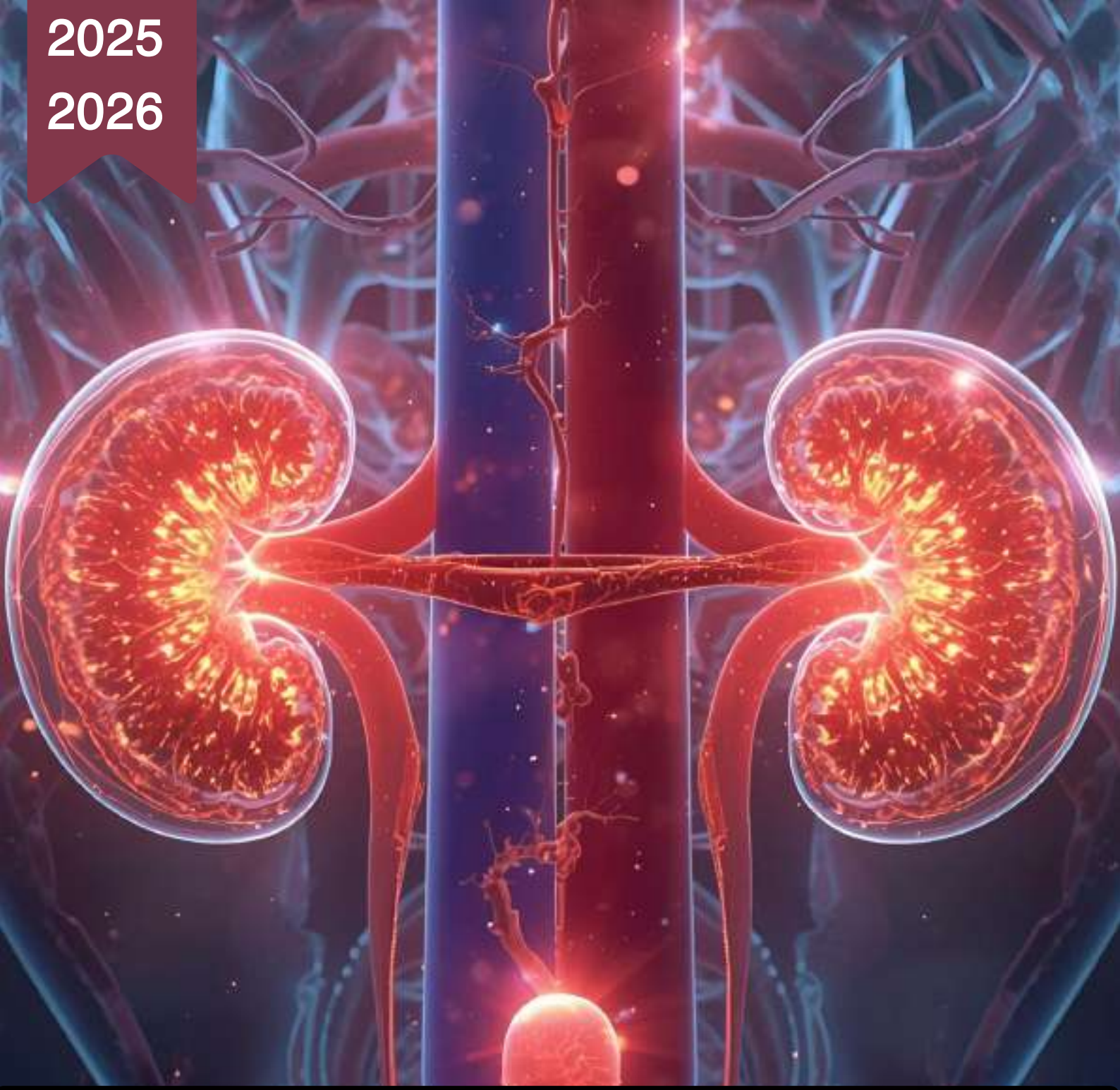
2025
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RENAL SYSTEM

RNS- 315

SOHAG UNIVERSITY
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Renal System

Anatomy & embryology development

RNS-315

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1 Urinary System

- Theurinary system organs include the two kidneys, renal pelvis, two ureters, bladder, and urethra.

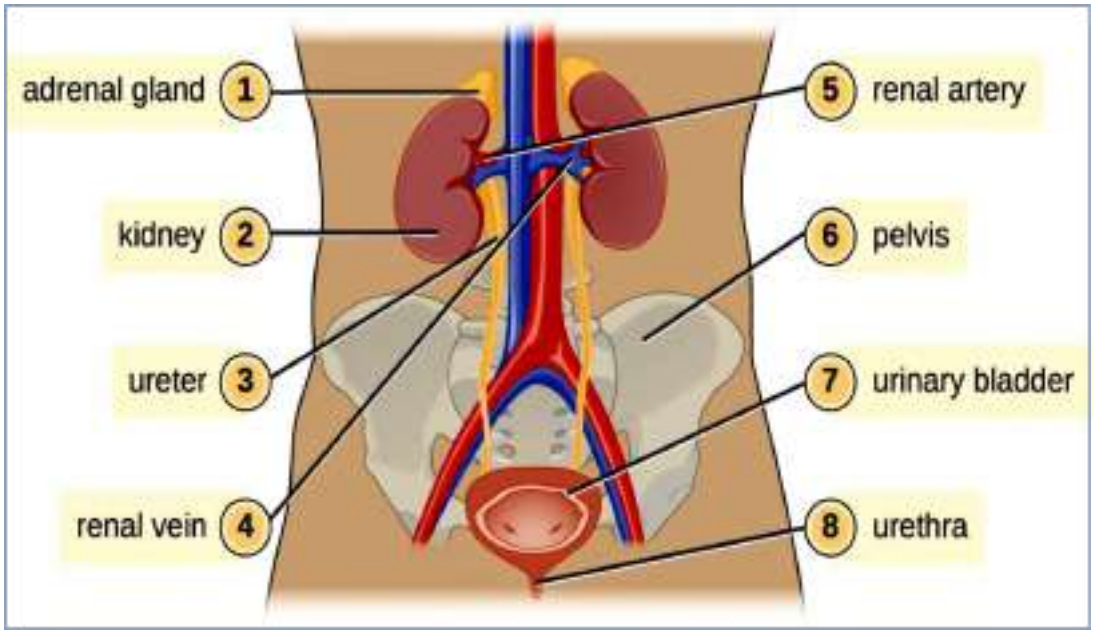


Figure (1): showing urinary system structure.

The kidneys

Site : Located below the ribs toward the middle of the back, the right kidney is positioned slightly lower than the left.

Size: Each of which is about 11 cm long, 6 cm wide, 3 cm thick. The average weight is 150 gm in male & 135 gm in female, each kidney has a fibrous capsule.

Shape : the kidney is bean-shaped organ and has 2 Poles, 2 borders and 2 surfaces.

- Two poles: upper and lower: the upper pole is slightly thicker.
- Two borders: lateral and medial.
- The lateral border: is convex.
- The medial border: is concavo-convex. It is convex at both ends, but is concave in the middle where it presents a vertical slit called the hilum of the kidney.
- Two surfaces: anterior and posterior.

The anterior surface	The posterior surface
Faces anterolateral Covered with peritoneum	Faces posteromedial No peritoneal covering

- The kidneys remove urea from the blood through tiny filtering units called nephrons. Each nephron consists of a ball formed of small blood capillaries,
- called a glomerulus, and a small tube called a renal tubule,

When a kidney is cut lengthwise, 2- regions become apparent:

1. Cortex: The outer region, which is light in color.
2. Medulla: It is a darker reddish-brown area, deep to the cortex
3. The renal pelvis: Is a funnel-shaped tube, continuous with the ureter leaving the hilum. Branching extensions of the pelvis form two or three major calyces. Each one subdivides to form several minor calyces, cup-shaped areas that enclose the papillae. The calyces collect urine, which drains continuously from the papillae, and empty it into the renal pelvis.

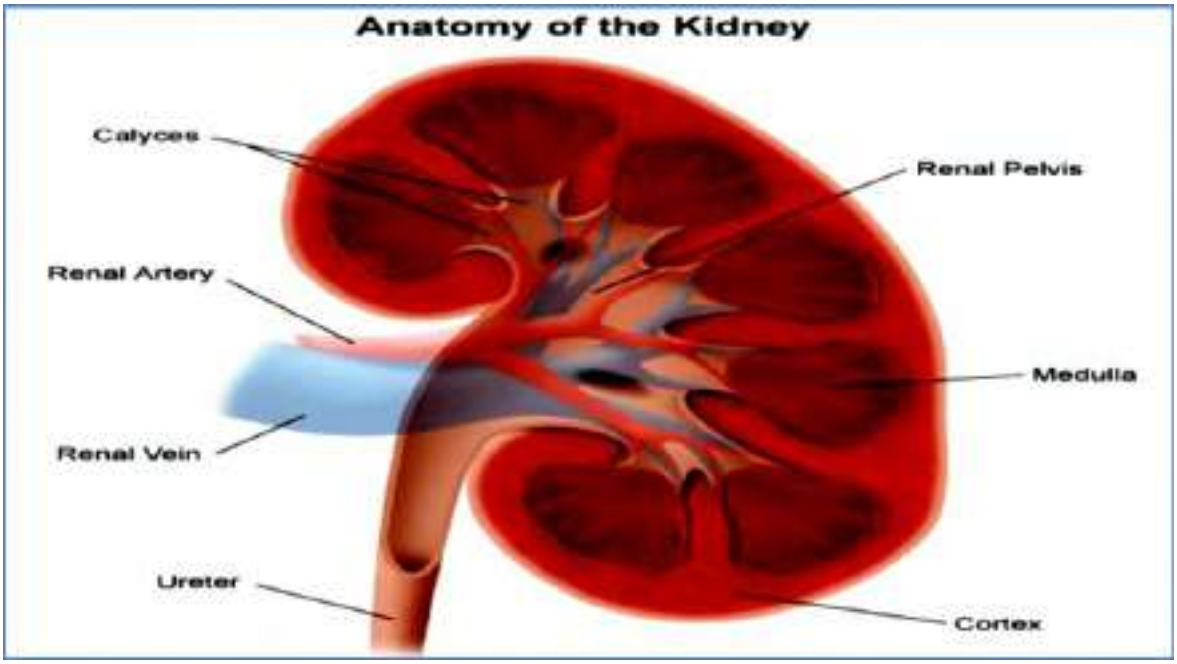


Figure (2): showing cut section of the kideny .

- The kidney functions are ;
  - Removing waste products and drugs from the body.
  - Balancing the body's fluids.
  - Releasing hormones to regulate blood pressure.

• Controlling production of red blood cells

□ The kidneys have the following coverings:

1. Fibrous capsule: This surrounds the kidney and is closely applied to its outer surface.

2. Perirenal fat: This covers the fibrous capsule.

3. Renal fascia: This is a connective tissue that lies outside the perirenal fat and encloses the kidneys and suprarenal glands.

4. Pararenal fat: This lies external to the renal fascia and is often in large quantities

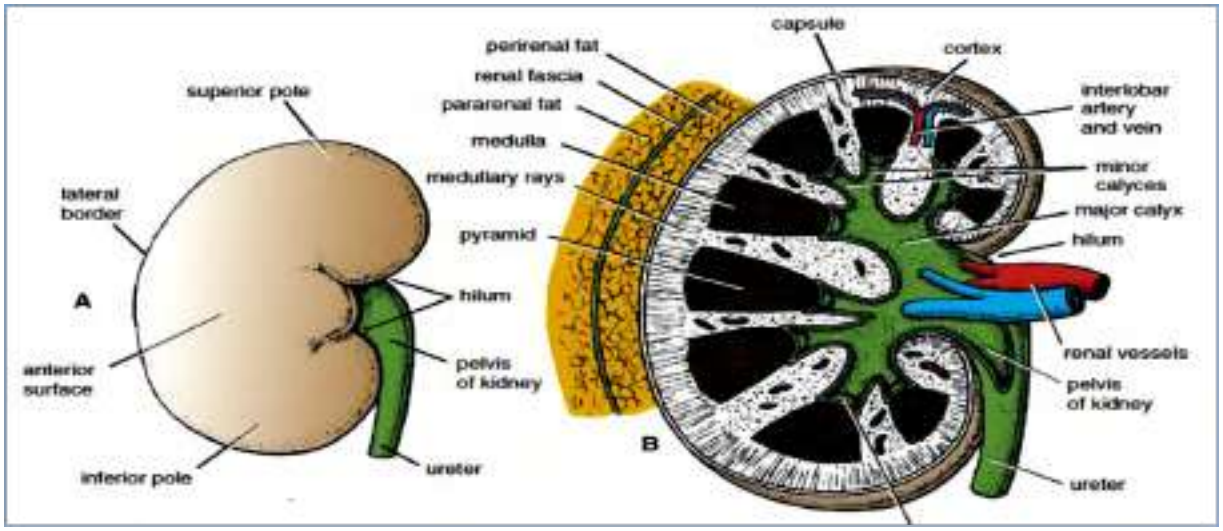


Figure (3): showing coverings of the kideny

□ Hilum of the kidney:

♦ It is a vertical opening at the middle of the medial border at the level of L1 vertebrae. It leads to sinus of the kidney.

♦It transmits the following (Member VAP) :

♦Renal Vein (anteriorly)

♦Renal Artery (middle)

♦Renal Pelvis (posterior and downward)

□ Relations:

♦ Posterior relations: Are the same in the right and left kidneys:

□ Four Muscles: Diaphragm: is related to upper end separating it 11th and 12th ribs (on the left side) or 12th rib (on the right side). Psoas major: medially,Quadratus lumborum: intermediate, Transversus abdominis: laterally.

□ Four Neurovascular structures: lies between kidney and the quadratus lumborum muscle, they include the followings: Subcostal vessels, Subcostal nerve (T12), Iliohypogastric nerve (L1), Ilioinguinal nerve (L1).

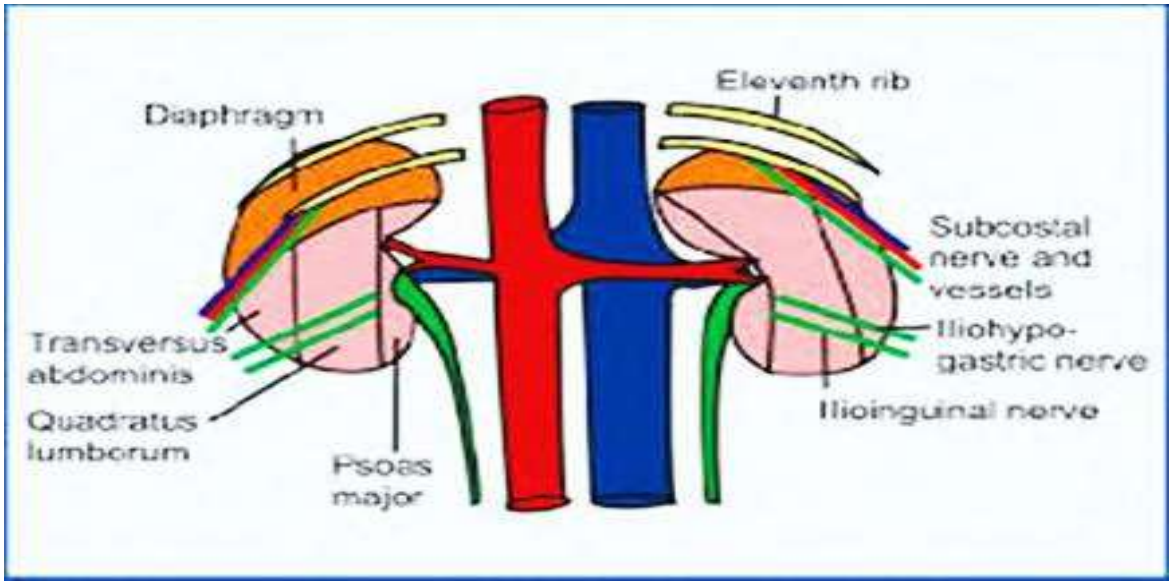


Figure (4): showing posterior relation of the kideny .

♦ Anterior relations: are different on both sides:

Right kidney	Left kidney
Right suprarenal gland: at its upper pole	Left suprarenal gland: from the upper pole to the hilum
2nd part of the duodenum: at the medial border	at the Stomach is related to the triangular area at the upper third between the areas for the pancreas, spleen and left suprarenal gland
Right lobe of the liver: is related to large area lateral to the second part of the duodenum	Spleen: close to the upper lateral part

Right colic flexure: below the hepatic area	Descending colon: is related to the anterior surface close to the lower part of the lateral border
lower medial area of anterior surface	Loops of jejunum & ascending branch medial area of anterior surface
Loops of jejunum & ascending branch of right colic artery: are related to of left colic artery: are related to lower	Body of pancreas and splenic vessels: are related to quadrilateral area at the middle third of the anterior surface

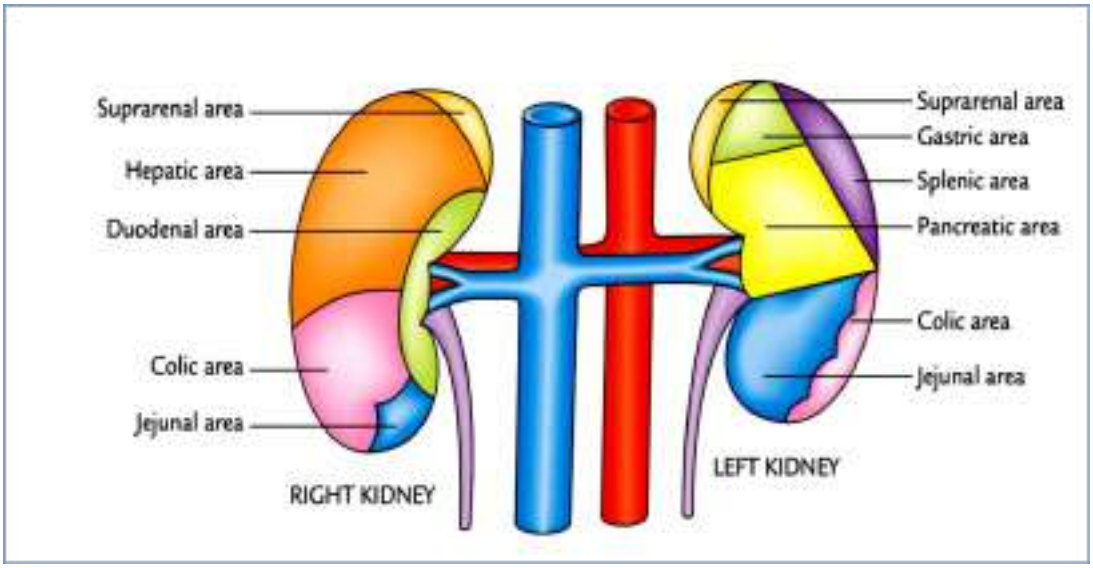


Figure (5): showing anterior relation of the kideny.

Blood supply of the kideny:

♦ **Renal arteries:** • Arise from the aorta at right angle, at the level of the upper border of L2. • The right renal artery is longer than the left and passes posterior to the IVC; the left artery passes posterior to the left renal vein.

♦ **Renal veins:**

•Terminate in I.V.C. The left is longer than the right one.

Lymphatic vessels of the kidney:

• Are drained into the lateral aortic lymph nodes around the origin of the renal artery.

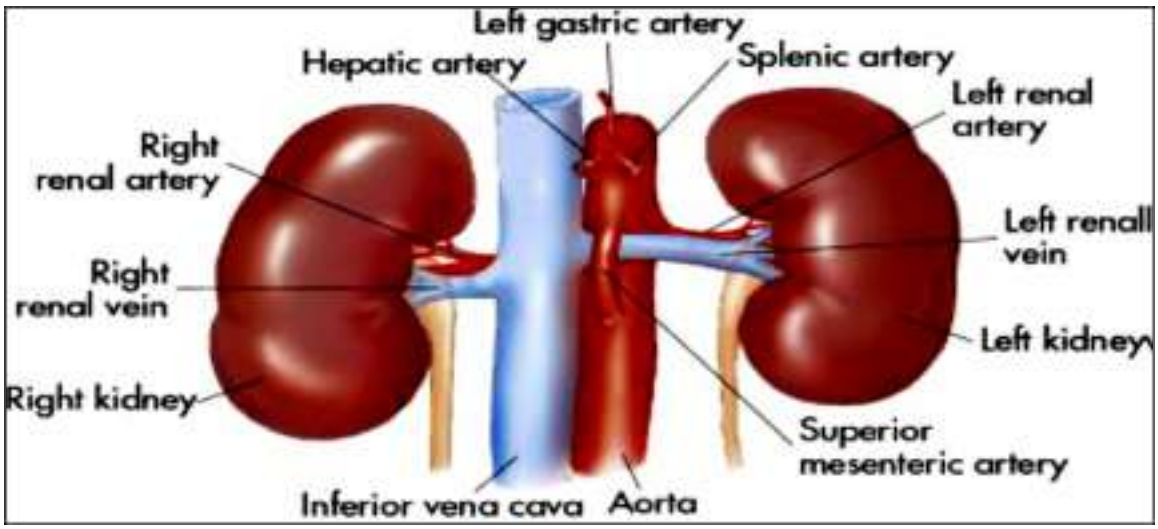


Figure (6): showing blood supply of the kideny.

Nerve supply of the kidney:

By renal plexus derived from coeliac plexus.

♦ **Sympathetic fibers:** are derived from the 10th, 11th, 12th thoracic and 1st lumber segments of the spinal cord.

♦ **Parasympathetic fibers:** are unknown probably from the vagi

2- URETERS

- Definition: muscular tube begins with the renal pelvis and extends from the kidney to the urinary bladder.
- Length: Ten inches (25 cm) long.
- Beginning: At the level of L2, the ureter begins inside the sinus of the kidney by union of the major calyces.
- End: In the wall of the bladder: it enters obliquely through the base of the bladder and opens by a slit-like orifice that acts as a valve.
- Function: Plays an active role in transporting urine.
- Course: It can be divided into three parts:

a- Renal pelvis (pelvis of the ureter): lies partly inside the renal sinus and partly outside along the medial border of the kidney, formed by the union of 2 - 3 major calyces.

b- Abdominal part: 12.5 cm long , starts at the lower end of the kidney descends downwards and medially, opposite the tips of the lumbar transverse processes (L2 - L5).Enters the pelvis by crossing pelvic inlet in front of the bifurcation of common iliac

Relations:

Posteriorly: (the posterior relations of two ureters are the same).

1-Medial part of the psoas major and its fascia.

2-Genitofemoral nerve (crosses behind the ureter).

3-Bifurcation of common iliac or beginning of external iliac artery.

Anteriorly: (different in right from the left side).

a. Right ureter:

1-Third part of duodenum.

2-Right testicular or ovarian vessels.

3-Right colic and ileocolic vessels.

4-Root of mesentery.

b. Left ureter:

1- Left testicular or ovarian vessels.

2- Left colic and sigmoid vessels.

3- Sigmoid mesocolon

Medially: (veins)

a. Right ureter: Inferior vena cava. b. Left ureter: Inferior mesenteric vein.

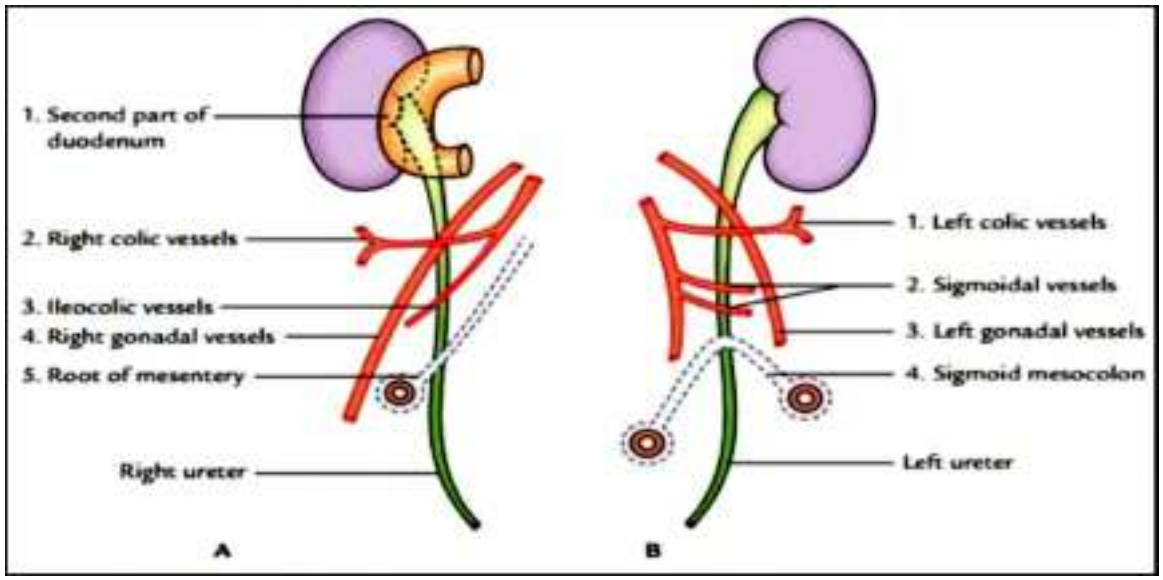


Figure (7): relations of the ureters

c- Pelvic part: 5 inches long Crosses the pelvic brim in front of the bifurcation of the common iliac artery; descends retroperitoneally on the lateral pelvic wall; and runs medial to the umbilical artery and the obturator vessels and posterior to the ovary, forming the posterior boundary of the ovarian fossa. Then it is accompanied in its course by the uterine artery In females which runs above and anterior to it in the base of the broad ligament of the uterus. Because of its location, the ureter is in danger of being injured in the process of hysterectomy. In males: passes posterior and inferior to the ductus deferens and lies in front of the seminal vesicle before entering the posterolateral aspect of the bladder.

Constrictions of the ureter:

1. At the lower end of kidney: at the junction between pelvis and the ureter proper.
2. At the site of crossing the bifurcation of common iliac artery.
3. Inside the wall of urinary bladder (intramural part).

Arterial supply of the ureter:

The ureter gets its arterial supply from many arteries, which join together to form a longitudinal anastomosis. These arteries are:

1. Renal artery: supply the pelvis of the ureter.
2. Testicular or ovarian artery.
3. Abdominal aorta.
4. Common iliac artery.
5. Internal iliac artery.
6. Inferior vesicle artery (supplies the lower most part of the ureter)
7. Uterine artery (in the female).

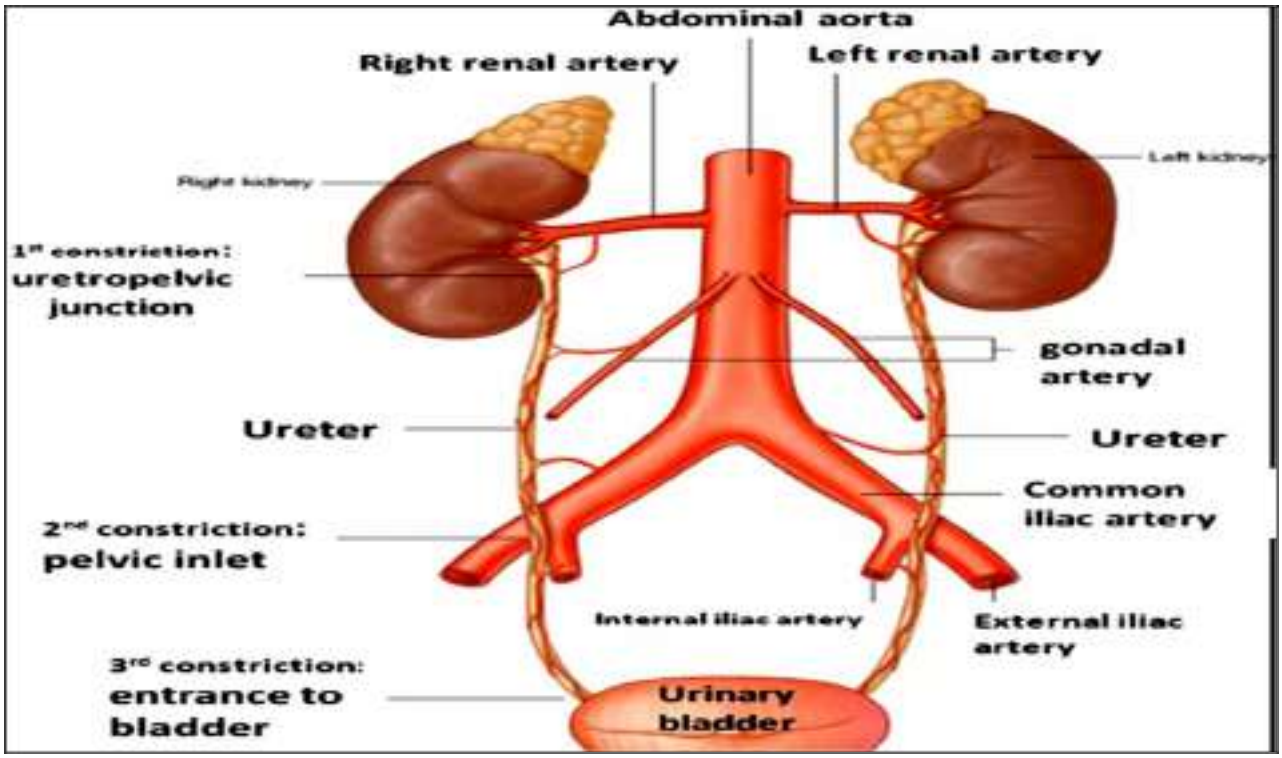


Figure (8): showing blood supply of the ureter.

Nerve supply:

♦ Sympathetic fibers: from the lower 3 thoracic and first lumbar spinal segments.

♦ Parasympathetic fibers: from 2nd, 3rd and 4th sacral segments of spinal cord.

Lymphatic drainage:

1. Lateral aortic lymph nodes 2. Common iliac lymph nodes.

3. External iliac lymph nodes. 4. Internal iliac lymph nodes.

3- URINARY BLADDER

Site: The bladder is the most anterior organ of the pelvis it is the site of urine storage so it is entirely situated in the pelvic cavity when empty but it expands superiorly into the abdominal cavity when full.

Shape : The empty bladder is shaped like a three-sided pyramid, It has an apex, a base, a superior surface, and two inferolateral surfaces.

a) The apex of the bladder :is directed toward the top of the pubic symphysis; attached to the umbilicus by median umbilical ligament

b) The base of the bladder: is shaped like an inverted triangle, The two ureters enter the bladder at each of the upper corners of the base, and the urethra drains inferiorly from the lower corner of the base. c) The inferolateral surfaces of the bladder:Not covered by peritoneum,
Related to the levator ani muscles and obturator internus muscles

d) The superior surface: is slightly domed when the bladder is empty; it balloons upward as the bladder fills and covered by peritoneum.

e) Neck of bladder: it is the most inferior part of the bladder and also the most "fixed" part because of tough fibromuscular bands, which connect the neck to the back of pubic bone.

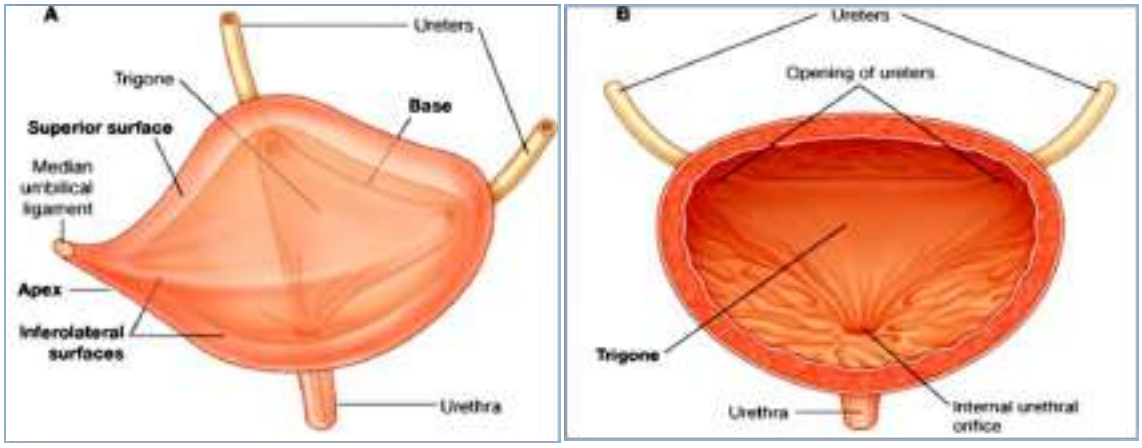


Figure (9): (A) shape of the urinary bladder, (B) the interior of urinary bladder.

Ligaments of the urinary bladder:

1. Median umbilical ligament: Continuouswithapexofthe bladder (itisthe embryonic urachus). 2. Puboprostatic (In the male) and pubovesical ligaments (In the female): ligaments extend from back of the bodies of pubic bones to the anterior neck of
the bladder.

3. Lateral ligaments of the bladder:
extends laterally from the side of the base of the bladder across the pelvic floor to the tendinous arch in side wall of the pelvis.
4. Posterior ligaments: Each extends backwards from the base of the bladder to the corresponding internal iliac vein.

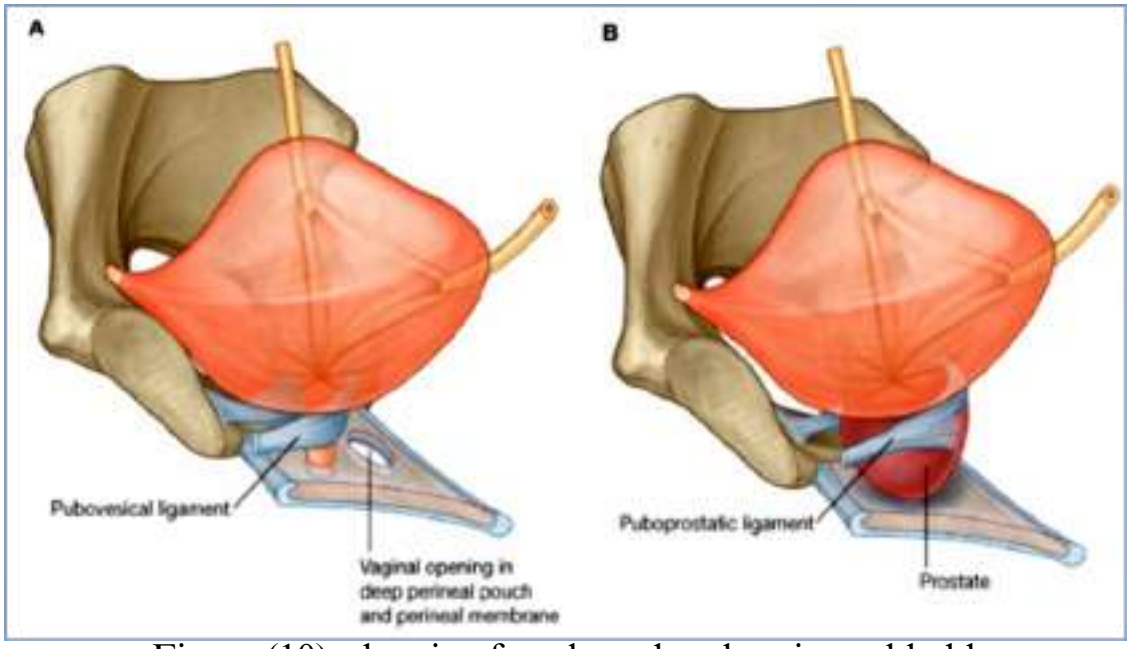


Figure (10): showing female and male urinary bladder

Blood supply:

• Arterial supply: Receives blood from the superior and inferior vesical arteries (and from the vaginal artery in females).

• Venous drainage: Its venous blood is drained by the prostatic (or vesical) plexus of veins, which empties into the internal iliac vein.

Nerve supply:

It is innervated by nerve fibers from the vesical and prostatic plexuses.
• The parasympathetic nerve (pelvic splanchnic nerve originating from S2–S4) stimulates to contract the musculature (detrusor) of the bladder wall, relaxes the
internal urethral sphincter, and promotes emptying.
• The sympathetic nerve (T11,T12,L1,L2) relaxes the detrusor of the bladder wall and constricts the internal urethral sphincter

4- URETHRA

It acts as a passage for urine from the urinary bladder to the exterior, but in male, it also serves as a passage for semen.

A. Male Urethra

Length: is about 18 to 20 cm long
Beginning, course, parts and end: it begins at the internal urethral orifice of the bladder; it is divided into 4 parts.

i. Preprostatic part.
The Preprostatic part of the urethra is about 1cm long, extends from the base
of the bladder to the prostate, and is associated with the internal urethral sphincter.

ii. Prostatic urethra

Is the widest part, about 3 cm long and is surrounded by the prostate gland. Its posterior wall has an elevation called the urethral crest.

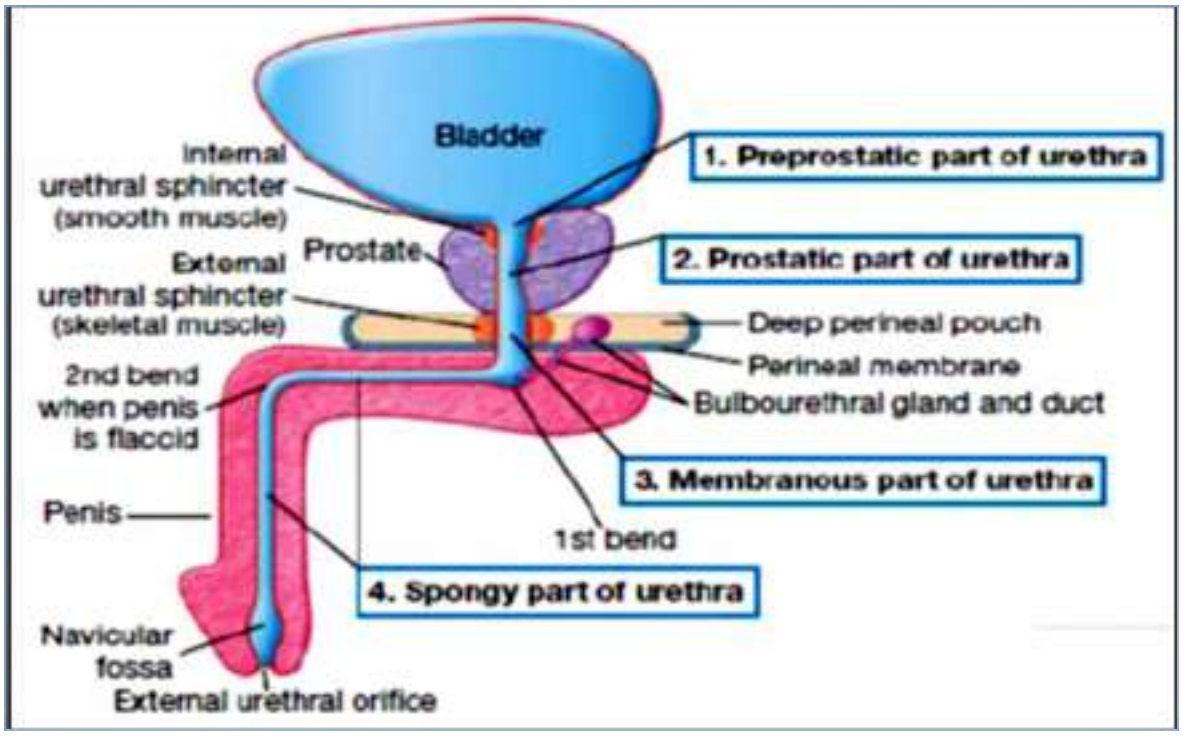


Fig. (11): male urethra

iii. Membranous urethra:
It is the shortest part 2 cm in length, passes through the urogenital diaphragm related to the deep transverse perineal muscle and sphincter urethrae muscle(also called external urethral sphincter), The external urethral sphincter completely surrounds the male urethra.

vi. Spongy urethra.
It is 15 cm long, surrounded by erectile tissue (the corpus spongiosum) of the penis. It is enlarged at the end of the penis to form the navicular fossa.

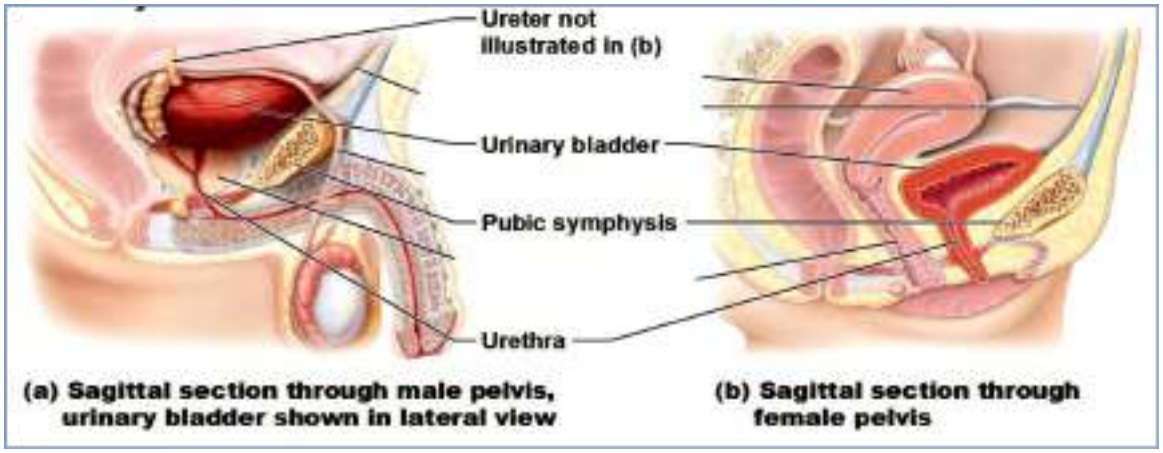


Figure (12): sagittal sections in male and female pelvis

B. Female Urethra

Length: The female urethra is 4 cm long.

Beginning: It begins at the internal urethral orifice of the bladder the passes through the urogenital diaphragm where it becomes related to the deep transverse perineal muscle and sphincter urethrae muscle

End: The female urethra terminates as the navicular fossa at the external urethral orifice which opens into the vestibule of the vagina between the labia minora just below the clitoris.

NOTE

Because of The fusion between the posterior surface of female urethra with the anterior wall of the vagina so the external urethral sphincter does not completely surround the female urethra---⊠ This may explain the high incidences of stress incontinence in women especially after childbirth.

Blood supply:

Prostatic and membranous parts supplied by inferior vesical, middle rectal and internal pudendal artery, Spongy urethra by urethral branches of the bulb from internal pudendal artery.

Lymphatic drainage:

The prostatic and membranous parts of the urethra drain into the internal iliac lymph node, The spongy urethra into superficial inguinal lymph node

Nerve supply:

- Prostatic urethra by inferior hypogastric plexus

• Membranous and spongy by perineal nerve

• Internal urethral sphincter by autonomic nerves

• External urethral sphincter by perineal nerve

Sphincters of the urethra:

	Internal urethral sphincter	External urethral sphincter
Site	Lies in the pelvis around the neck of the bladder and preprostatic part of the urethra	Lies in the perineum surrounds the membernous urethra in the deep perineal pouch
Structures	Formed of smooth muscle fibers	Formed of striated muscle fibers
Functions	• Acts involuntary • Well developed in male • Maintains continence of urine • In male: Contraction of this sphincter prevents retrograde movement of semen into the bladder during ejaculation	• Acts voluntary • Well developed in male and female • Maintains continence of urine

Development of the kidney

It develops from the intermediate mesoderm. Three successive kidneys are formed during intrauterine life. These are: Pronephros, mesonephric and metaphors or permanent kidney

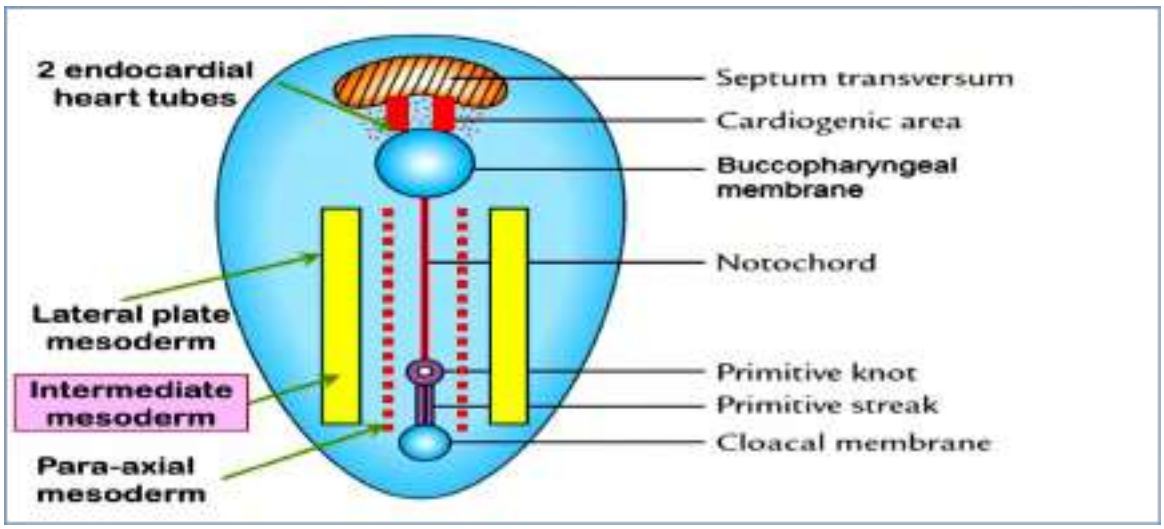


Figure. (13): showing the intermediate mesoderm

A- Pronephros

- Definition: It is the 1stkidneytoappearatthe beginning of the 4th week.
- Site: in the cervical region
- Structure: 7 to 10 solid masses known as nephrotomes that has
  - Medial end: have glomeruli
  - Lateral end: unite to form the Pronephric duct.
- Pronephric duct: extends downwards to open into the cloaca.
- Fate: By the end of the 4th week,
  - Pronephric tubules: degenerate completely, so it's functionless in humans.
  - Pronephric duct: persists & forms mesonephric duct so it id inductor for the second kideny.

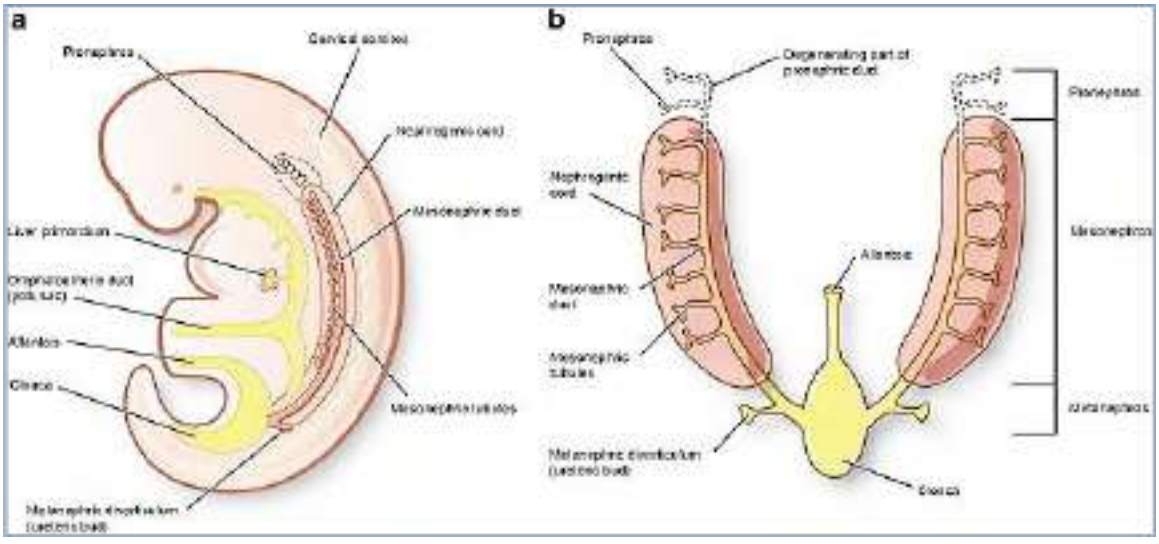


Figure (14): showing Pronephros development

B-Mesonephros

• It is the 2nd kidney, elongated excretory organ appear late in the fourth week, 70-80 mesonephric tubules are formed.

Site: the thoracic and upper lumbar regions of the intermediate mesoderm.

Structure: Each tubule elongates forming an S-shaped loop have 2 ends:

1- Medial end: invaginated by capillary tufts from the dorsal aorta forming glomerulus.

2- Lateral end , the tubules open into mesonephric or Wolffian duct

In the middle of the 2nd month, the mesonephros forms a large ovoid organ on each side of midline. Since the gonad develops on its side the ridge formed called urogenital ridge

The second kideny excrete urine 4 weeks until the perminant kideny develops.

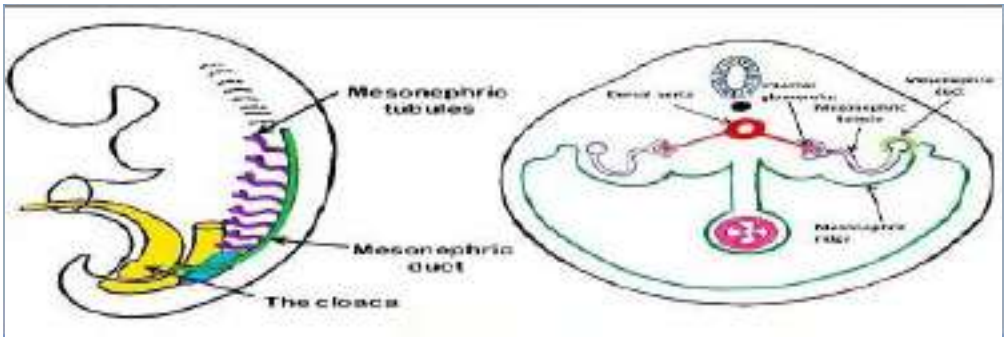


Figure (15): showing development of the mesonephric kidney.

Fate: differ in male and female:

	Male	Female
Mesonephric tubules	• Vasa efferentia • Head of epididymis. • Paradidymis	• Epoophron • Paroophron
Mesonephric duct	• Appendix of epididymis • Body & tail of epididymis. • Vas deferens • Seminal vesicle & ejaculatory duct.	• Gardener's duct.
	Ureteric bud & trigone of urinary bladder	

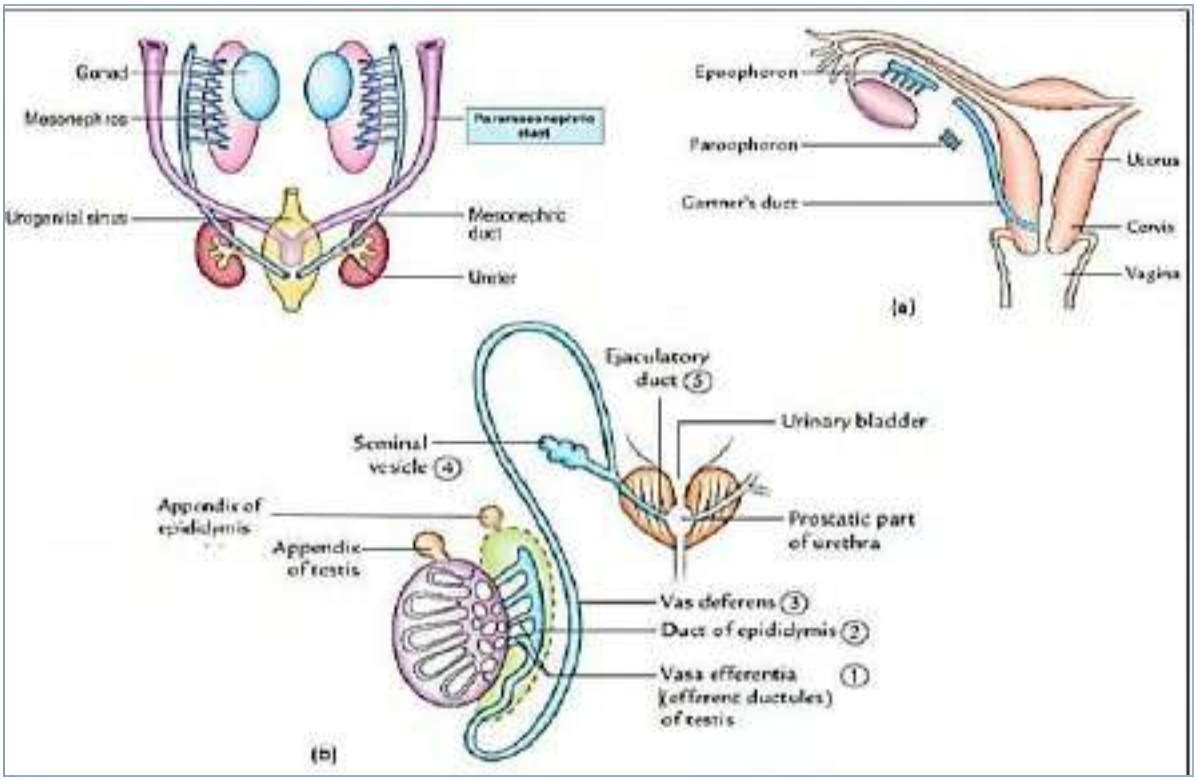


Figure (16): fate of mesonephric kidney in (a) female and (b) male.

C- Metanephros:

- **Definition:** It is the 3rd or permanent kidney; it appears in the 5th week.
- **Site:** lower lumbar and sacral regions of the intermediate cell mass.

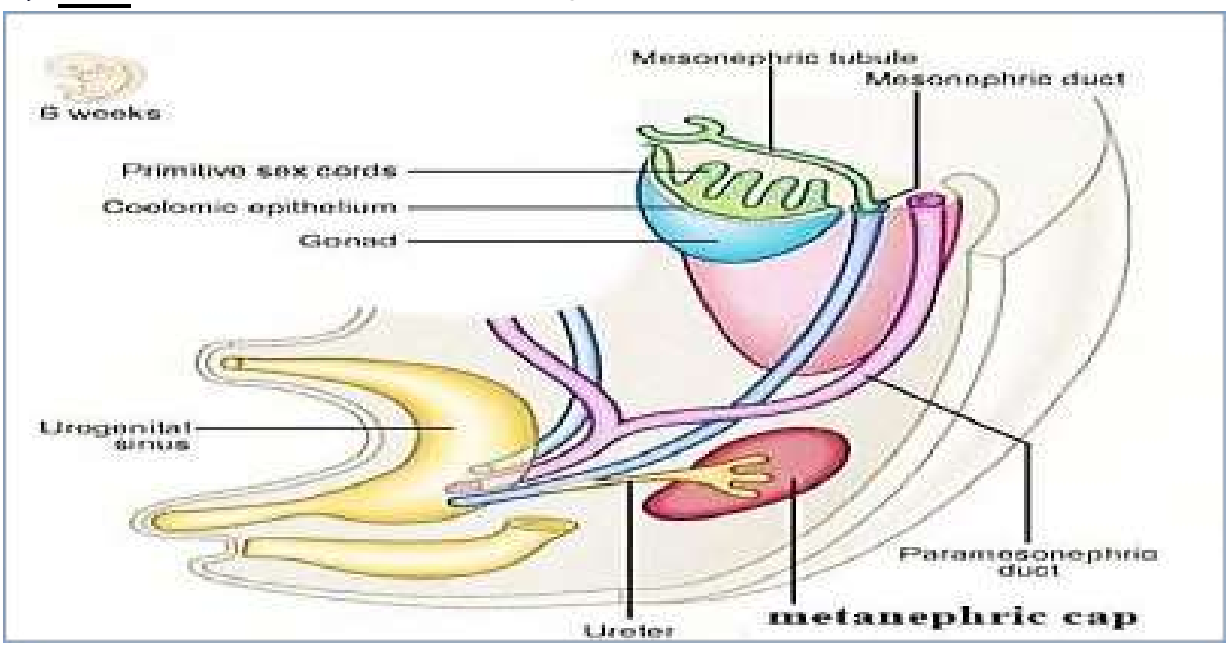


Figure (17): showing Metanephros development

Structure: It develops from 2 sources:

A- Metanephric cap (Metanephrogenic blastema), arise from the caudal part of intermediate mesoderm, it forms the nephrons.

B- Ureteric bud (Metanephric diverticulum), arise from the mesonephric duct near its entrance into cloaca, it forms the collecting partof the kideny it grows cranially to the cap expanding giving rise to therenal pelvis , major calyces , minor calyces and the collecting ducts.

Changesofthe metanephros during Growth:

1) Changes in shape: At first the kidney is lobulated, then becomes smooth at the end of 1st year of infancy

2) Ascent of the kidney:

Initially, the kidneys lie close to each other's in front of sacrum.

Later, the kidneys move upward to the abdomen at the 9th week

3) Change in the direction:

Initially the hilum of the kidney faces ventrally.

As the kidney ascends it rotates medially 90°.

4) Change in the blood supply: As it ascends it changes its blood supply:

At 1st from the common iliac artery and finally from abdominal aorta.

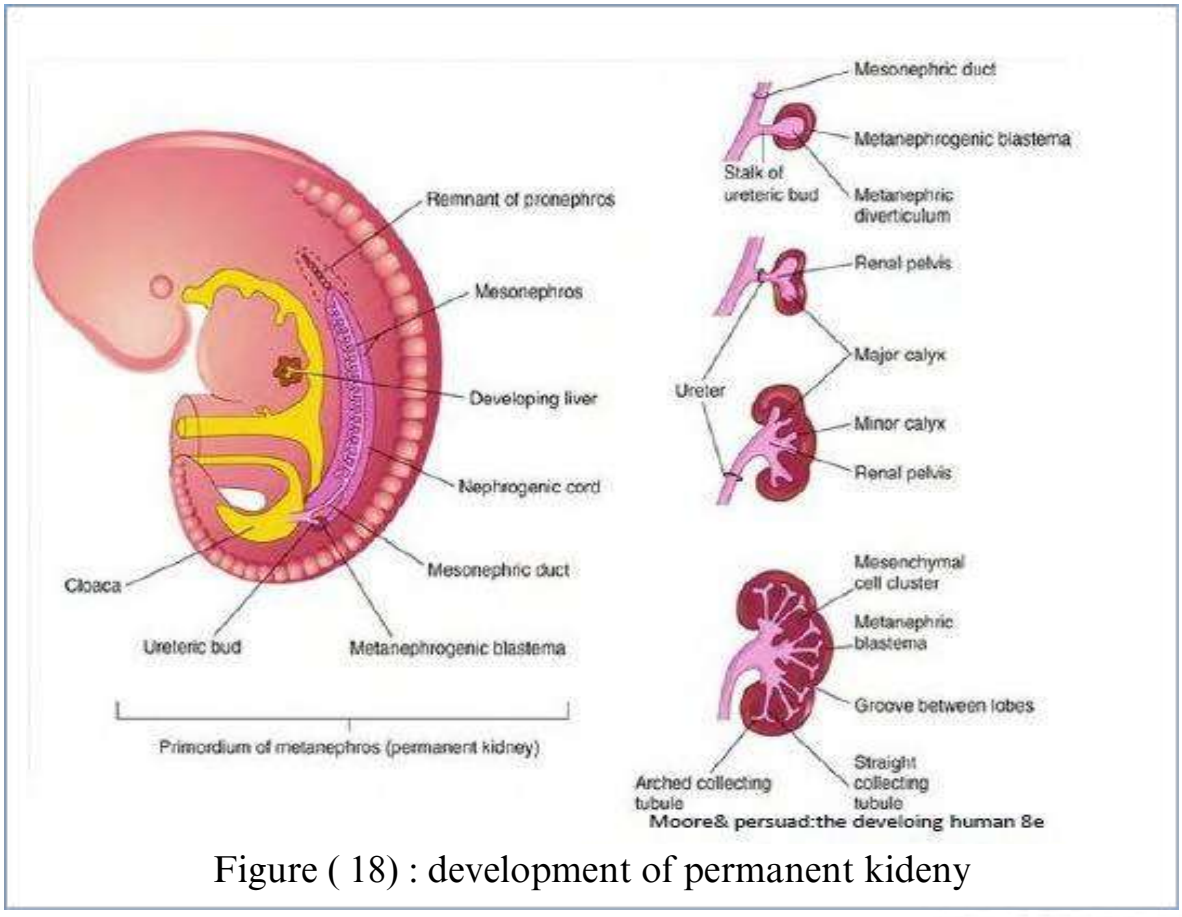


Figure (18) : development of permanent kideny

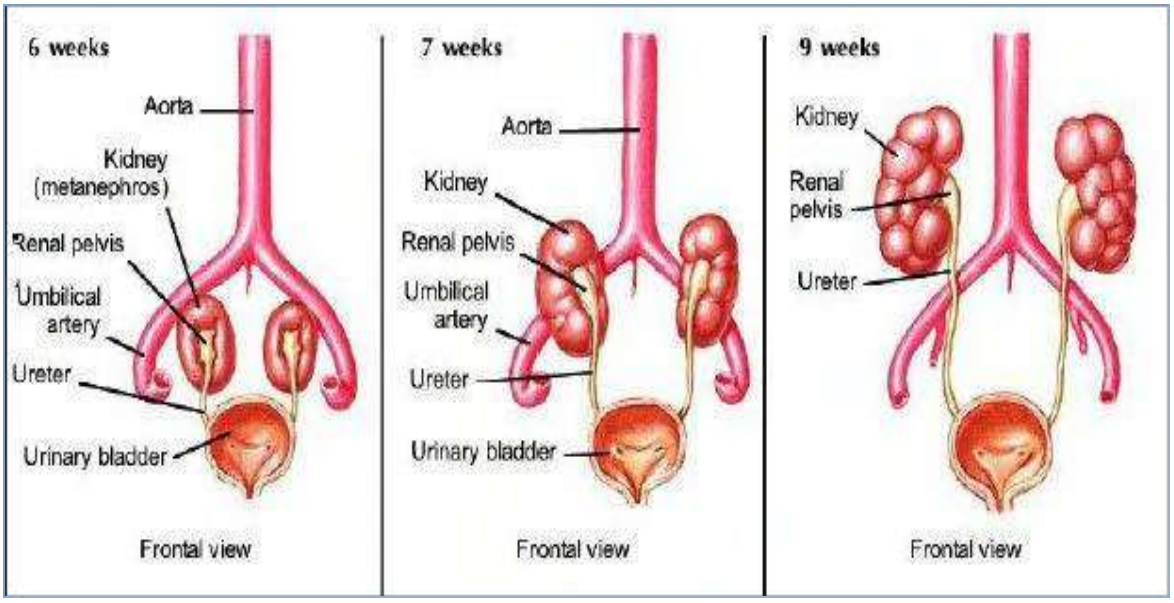


Figure (19): ascent of permanent kideny

Congenital anomalies of the kidney

1- Renal agenesis and hypoplasia: unilateral or bilateral absence of kidney. Cause: non development or degeneration of the ureteric bud or non-contact between ureteric bud and metanephric mesoderm

2-Ectopic kidney: one kidney may fail to ascend & remains in the pelvis. The ectopic kidney usually locates at the pelvis

3- Horse-shoe kidney: is due to fusion of the lower poles of the 2 kidneys across the midline as the normal ascent is prevented by the inferior mesenteric artery.

4- Congenital polycystic kidney: caused by failure of union between some collecting and the convoluted tubules, associated with fetal death, dialysis and renal transplantation.

5-Supernumerary kidney: It results from formation of two or more ureteric buds.

6- Mal-rotated kidney: The hilum of the kidney faces anteriorly, posteriorly, and laterally instead of medially.

7- Accessory renal arteries: Very common and 25% of kidneys have 2 or more arteries (surgical important).

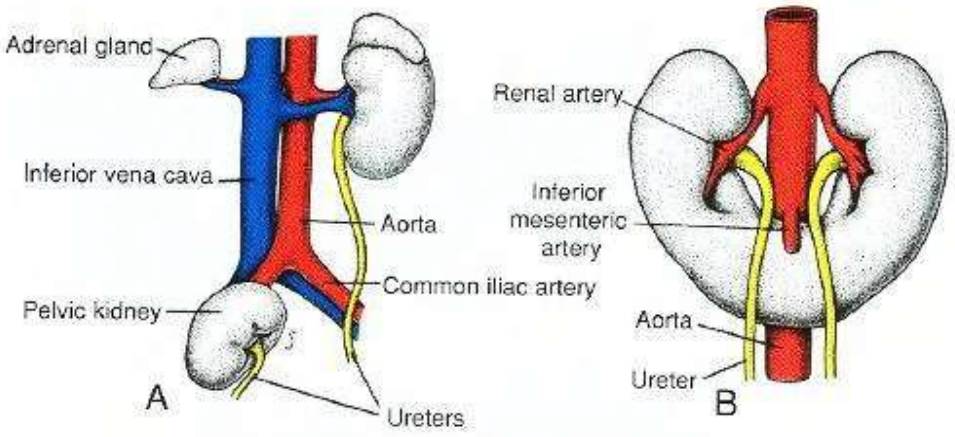


Figure (20): A- pelvic kideny B- horse shoe kideny

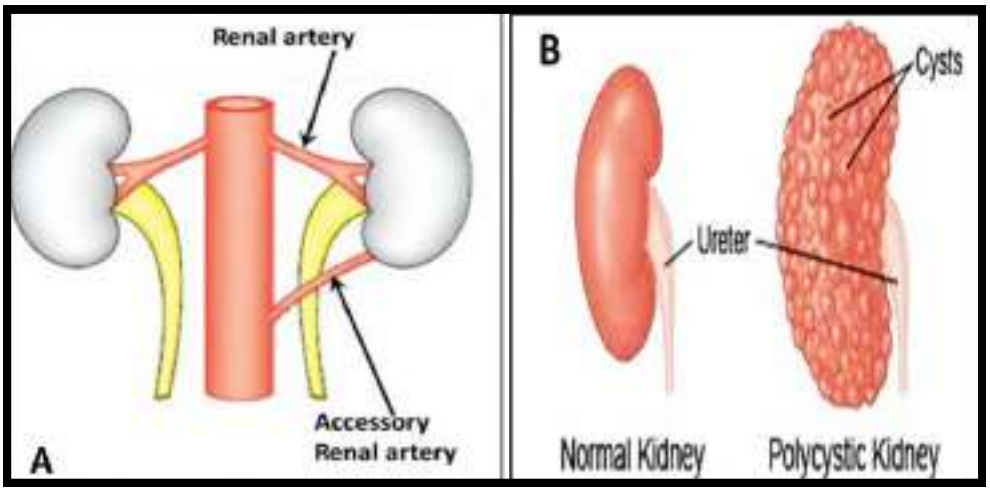


Figure (21): A- pelvic kideny , B- horse shoe kideny

Development of the ureter

It develops from the ureteric bud whicharises from the mesonephric duct Near its entrance in the cloaca, the budelongates dorsally and cranially to touch the metanephric cap, the upper endof the bud dilates and divides repeatedly into major calyces , minor calycesand collecting tubules

As the result, the 2 ureteric buds andthe 2 mesonephric ducts open separately into the urinary bladder.

As the result of ascent of the kidneys, theureters elongate.

- Congenitalanomaliesofthe ureter

1- Absentureter:noureteric budassociatedwithrenalagenesis

2-Double pelvis & double ureter: doubleureteric bud

3-Ectopic ureter: open into the vaginaorurethera

4-Congenital ureteric stenosis due tofailure of proper canalization of ureter: Associated with hydronephrosisand renal failure especially in bilateral cases.

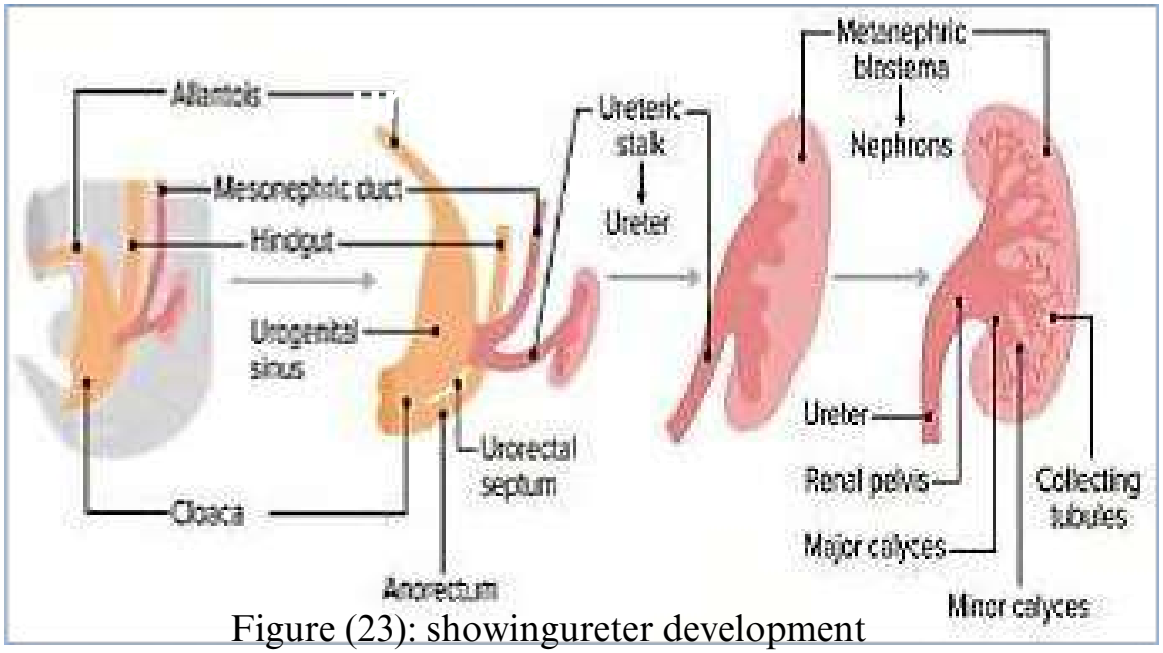


Figure (23): showingureter development

Development of the urinary bladder

The caudal end of the hind gut is expanded to form cloaca (endoderm in origin)

Cloaca is subdivided by the urorectal septum into ;

(a): dorsal part : gives recto anal canal

(b): ventral part: gives primitive urogenital sinus- which receive the opening of mesonephric ducts and the allantois at the apex and is subdivided into

- Cranial Vesico-uretheral part
- Caudal definitive urogenital part

So urinary bladder develops from :

Mucosa develops from the following sources:

1- Vesico-uretheral part of cloaca (endoderm) most of the urinary bladder.

2- Allantois (endoderm) gives urachus. the proximal part gives apex of urinary bladder , the distal part gives median umbilical ligament

3- Two caudal part of mesonephric ducts (mesoderm)trigon of urinary bladder.

Adjacent splanchnic mesoderm (mesoderm) the remaining layers of the urinary bladder.

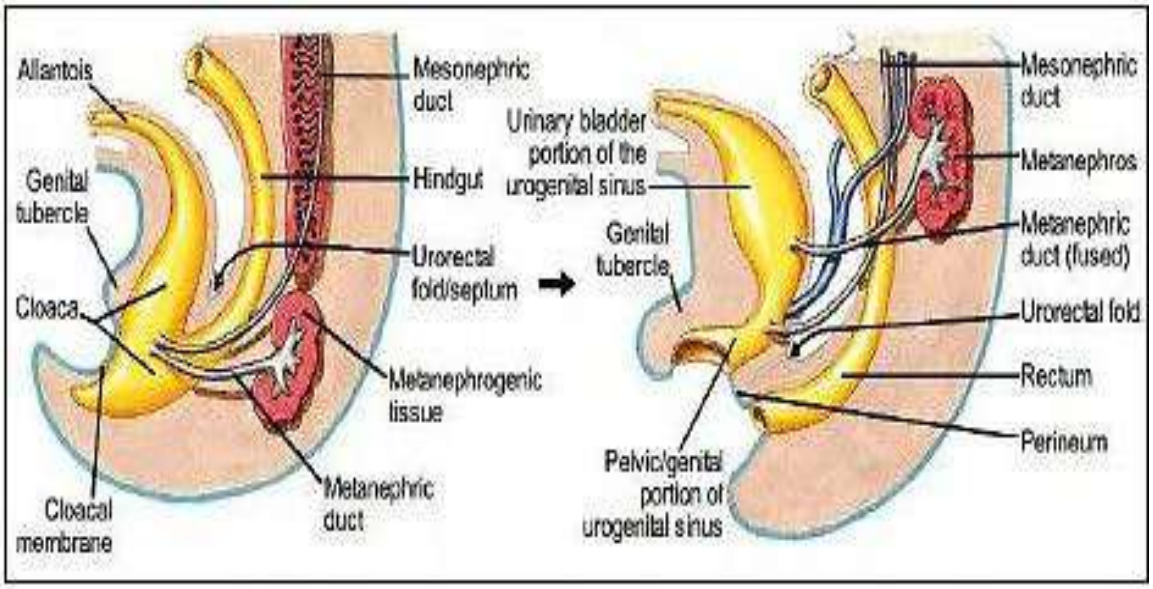


Figure (24): showing cloaca division and urinary bladder development

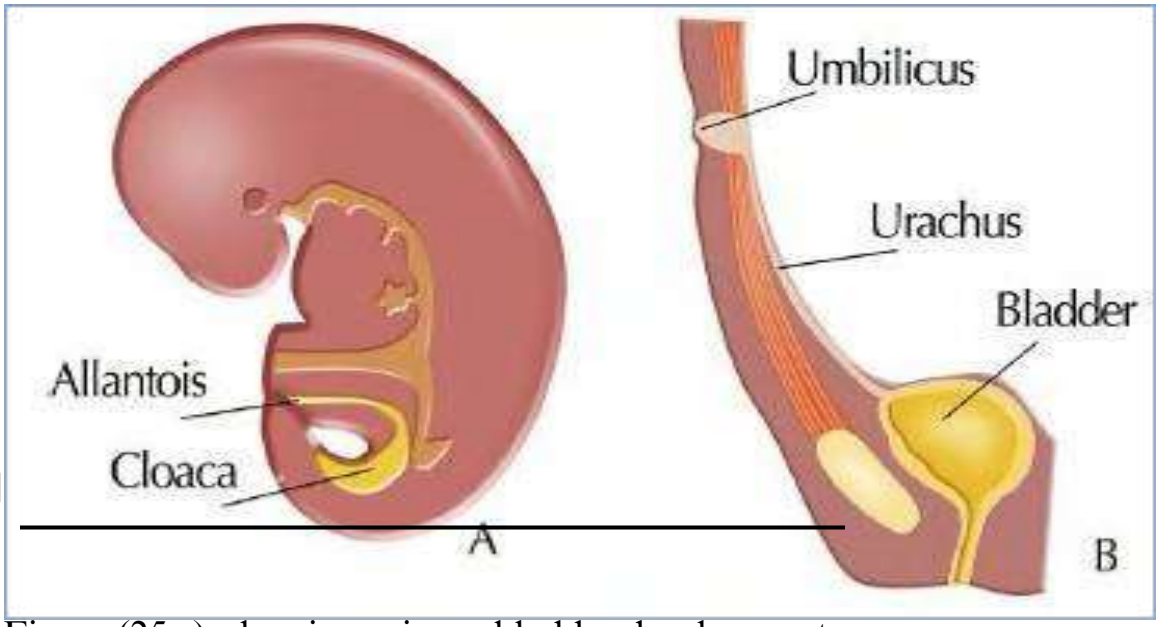


Figure (25): showing urinary bladder development

Congenital anomalies of the urinary bladder:

1-Ectopia vesicae (Exstrophy of the bladder): Ventral body wall defect in which the bladder mucosa is exposed Due to defective formation of the infraumbilical part of the anterior abdominal wall, usually associated with epispadius .

2- Uro-rectal fistula: Malformation of the urorectal septum can lead to several different types of fistula

3- -Anomalies of Urachus:

a) Urachal fistula: The entire urachus remains patent with subsequent discharge of urine through the umbilicus.

b) Urachal cyst: Only isolated part of the urachus fails to obliterate, the lining epithelium secret fluid.

c)Urachal sinus: Distal part of the urachus remains patent

d) Urachal diverticulum: persistent proximal part of urachus

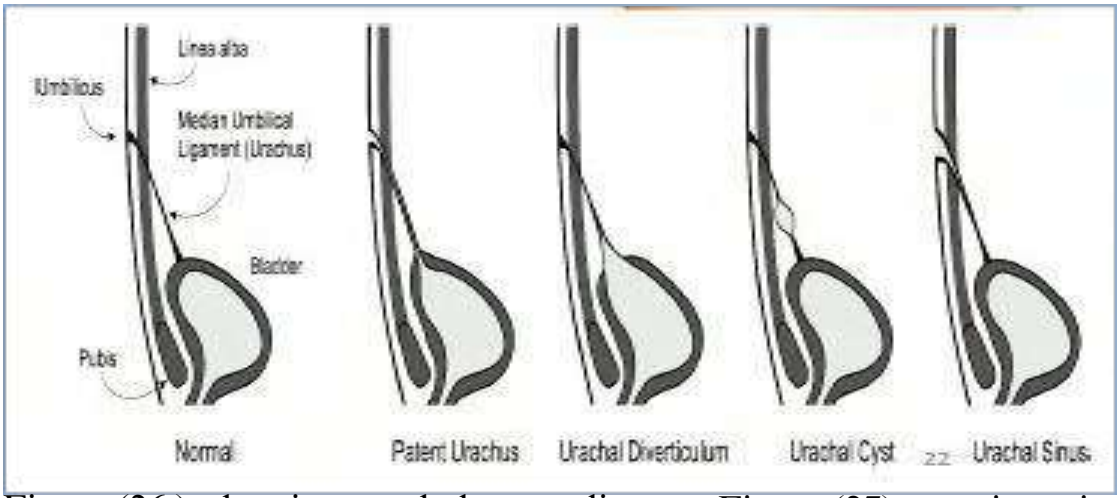


Figure (26): showing urachal anomalies

Figure (27):ectopia vesicae

Development of the urethra

Urethra	Embryonic source of development
Female urethra	• Caudal part of vesicourethral canal **except** its dorsal wall that is mesodermal in origin, derived from absorption of mesonephric ducts.
Male urethra ***1. Prostatic part*** (a) Above the level of opening of ejaculatory ducts (colliculus seminalis) (b) Below the level of openings of ejaculatory ducts	• Caudal part of vesicourethral canal **except** its dorsal wall that is mesodermal in origin, derived from absorption of mesonephric ducts. • Pelvic part of definitive urogenital sinus
2. Membranous part	• Pelvic part of definitive urogenital sinus
3. Penile part	• Phallic part of definitive urogenital sinus
4. Terminal part (which occupies the glans penis)	• Surface ectoderm

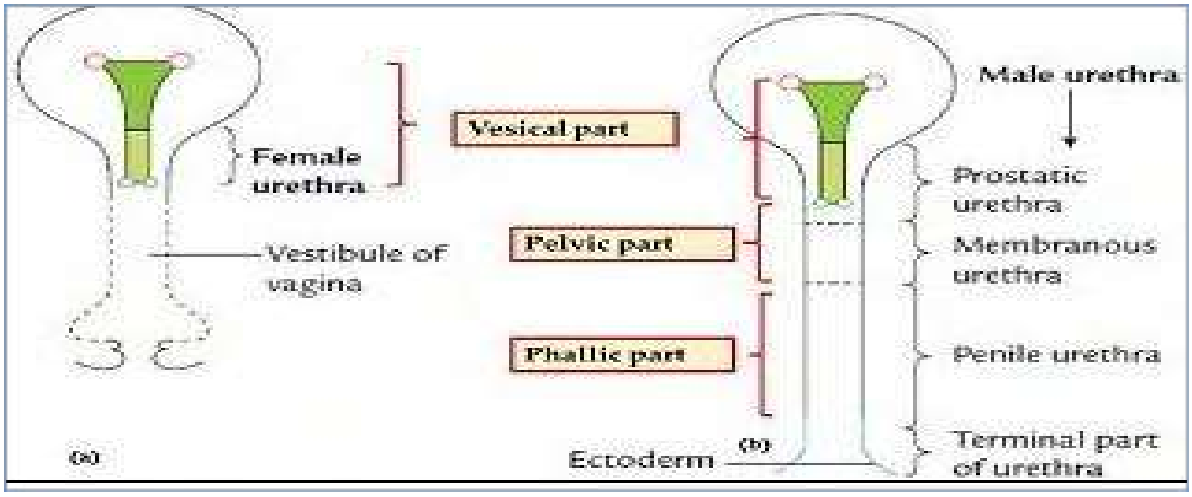


Figure (26): summarizing uretheral development

N.B. Only the lining epithelium of the urethra is derived from endoderm of urogenital sinus. Connective tissues and smooth muscles of urethra are derived from surrounding splanchnopleuric layer of intraembryonic mesoderm.

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2025



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MEDICINE

Renal System

Histology

RNS-315

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THE URINARY SYSTEM
The urinary system consists of 2 kidneys, 2 ureters, urinary bladder and urethra.

1 The kidney

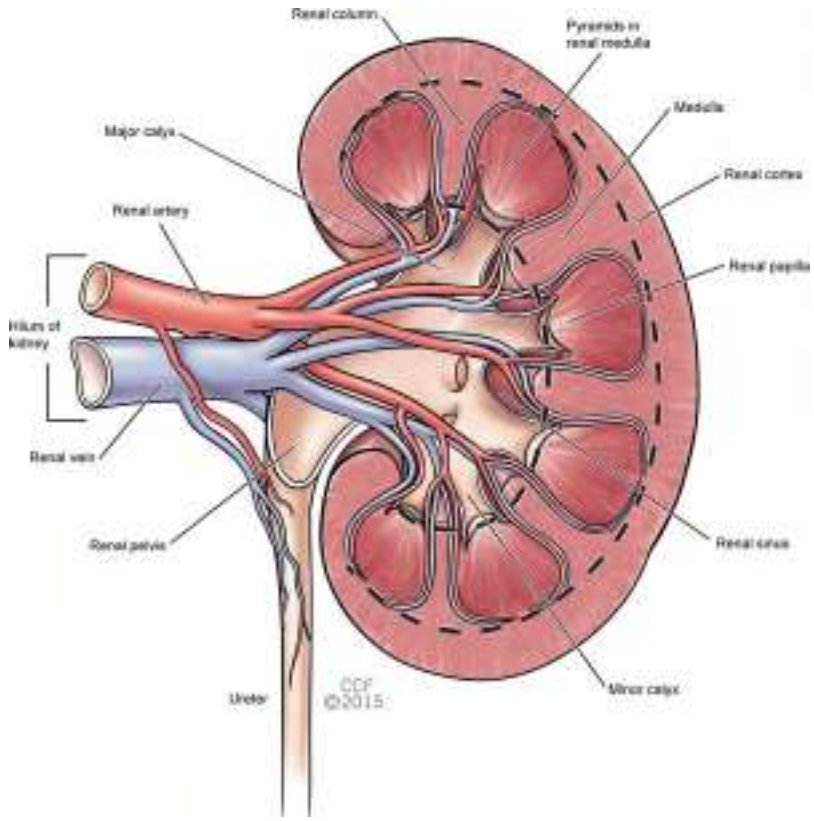


Fig. 1: General Organization of the kidney

Macroscopic structures

The kidney is bean shaped. It can be dividedinto an outer cortex and an inner medulla. (Fig. 1)

A) Capsule & hilus:

The kidney is coveredwith capsule of connective tissue. It has a convex lateral surface and a concave medialborder, the hilum where nerves enter, blood and lymph vessels enter and exit, and theureter exits.

B) Renal cortex:

The area between the kidney capsule and the base of the pyramids is called the cortex. It has a brown red color and a granular appearance.

C) Renal medulla:

Light gray in color consist of 10-18 medullary pyramids. From its base arise parallel rays penetrate the cortex called (medullary rays). They are collecting tubules which drain from the surrounding nephrons. Each renal pyramid with the overlying cortical tissue constitutes the **renal lobe**. The cortical tissue which is present in the area between medullary pyramids is called the **columns of Bertini.** From the base of each medullary pyramid, parallel arrays of tubules, the **medullary rays,** penetrate the cortex. Each medullary ray is surrounded by cortical tissue forming the **renal lobule.** The cortical tissue which is present in the area between medullary pyramids is called the columns of Bertini. From the base of each medullary pyramid, parallel arrays of tubules, the medullary rays, penetrate the cortex. Each medullary ray is surrounded by cortical tissue forming the renal lobule.

D) Renal sinus:

1. Renal pelvis.
2. Major calyces.
3. Minor calyces.

The renal pelvis, the expanded upper end of the ureter, is divided into 2 or 3 major calyces. Several small branches, the minor calyces, arise from each major calyx.

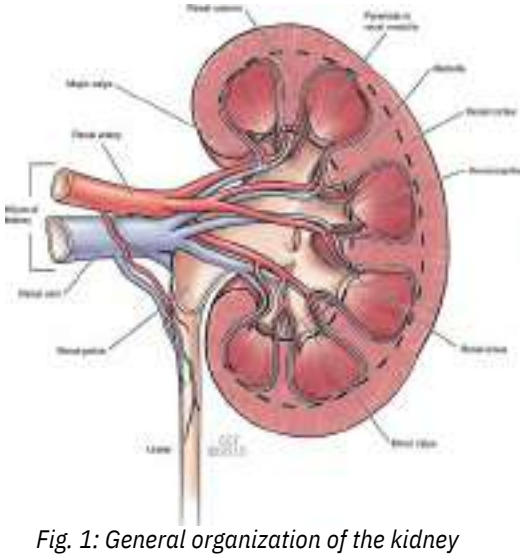


Fig. 1: General organization of the kidney

Histological structure:

A) Stroma:

- **Fibrous connective tissue capsule** surrounded by renal fat called perinephric fat.
- **Little connective tissue** present in the cortex between the nephrons and in the medulla (renal interstitium).
- **Supportive basement membrane** invests each kidney tubule from the renal corpuscle to the papillary duct.

B) Parenchyma:

The uriniferous tubules, consists of **nephron** and **collecting tubules**.

1- The nephron

- It is the active functional part of the kidney.
- It is the blood processing unit which serves to produce urine.

Length: 30-40 mm.
Number: 1-4 million in each kidney.
Components:

1. Renal corpuscle.
2. PCT (proximal convoluted tubules).
3. Loop of Henle.
4. DCT (distal convoluted tubules).

Types:

1. Cortical nephron(short) (85%): present in the cortex- they have a short loops of henle.
2. Juxta medullarynephron (long) (15%): present in the deep cortex near the medulla- they havelong loops of henle.

Renal Corpuscles

Renal (malpighian) corpuscles (Fig. 2)

Formed of:

A. Bowman's capsule.
B. Glomerulus.
C. Mesangium.

Each renal corpuscle has 2 poles:

- Vascular pole, where the afferent arteriole enters and the efferent arteriole leaves.
- Urinary pole, where the proximal convoluted tubule begins.

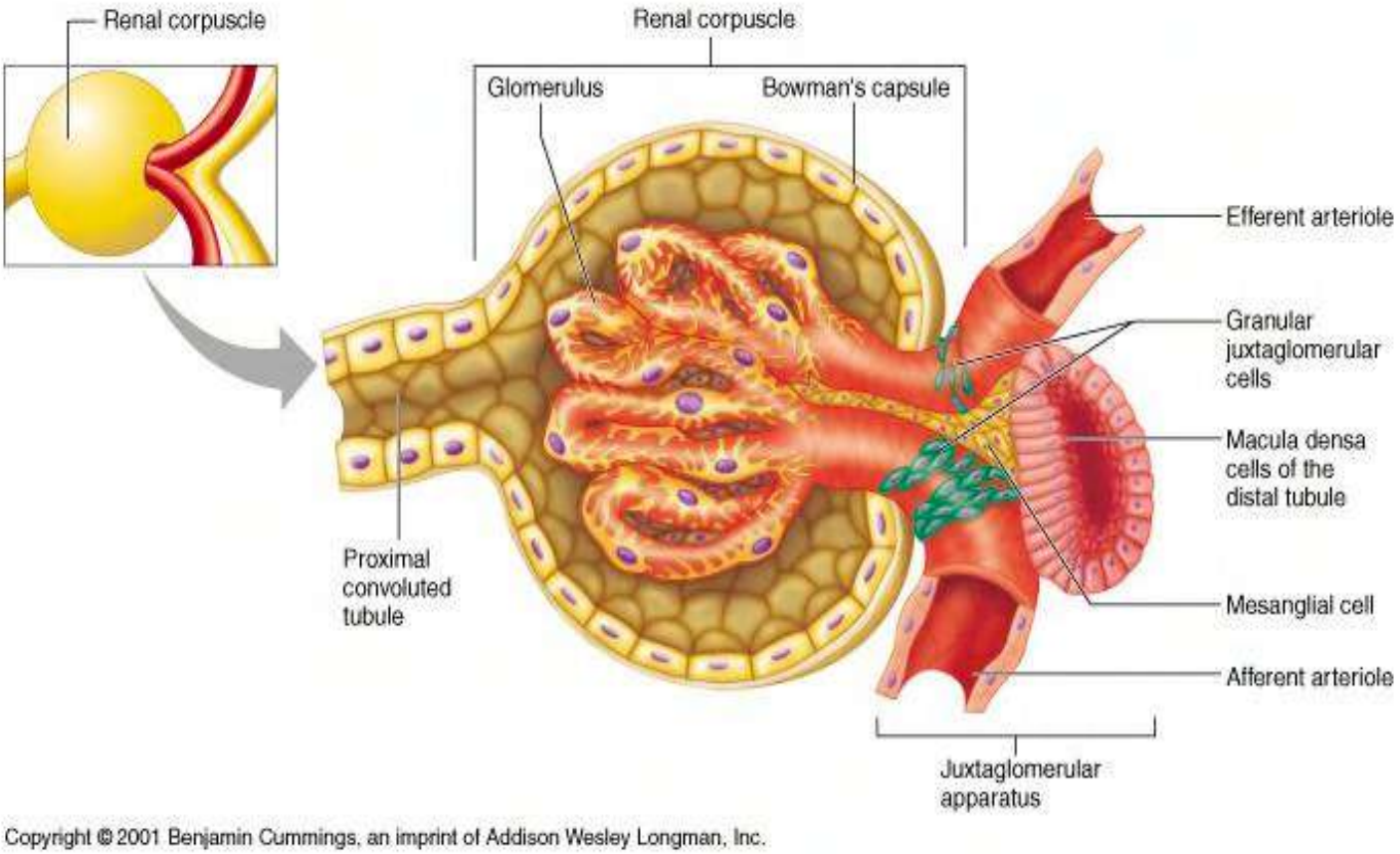


Fig .2: The renal corouscle

A) The Bowman's capsule (Fig. 2&3):
Double walled cup shaped capsule. It is formed of 2 epithelial layers. The internal layer or the visceral layer of the capsule envelops the capillaries of the glomerulus. The cells of this layer are called **podocytes**. The external or the parietal layer (capsular epithelium): formed of simple squamous epithelium. These cells increase in height to continue with the PCT. The space between the two layers is called (capsular space) or urinary space containing the filtrated liquid from the capillaries.

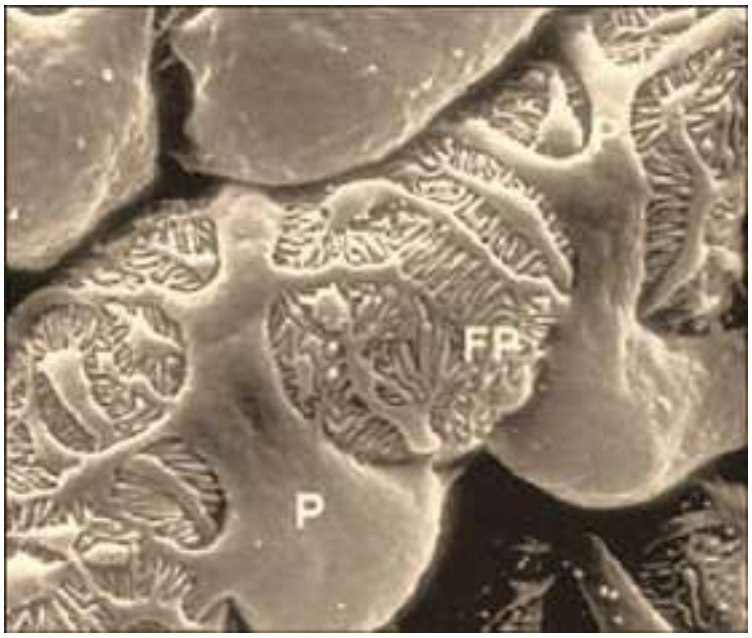


Fig. 3: Scaning electron microscopy of podocyte and its processes.

PODOCYTES (Fig. 3&4):

They are specialized epithelial cells. They are large and branched (like octopus). They have cytoplasmic processes called (**primary processes**). The podocytes are separated from the basement membrane of the glomerular capillary by (**subpodocytic space**) filled with glomerular filtration.

❖ The 1ry processes extend from the podocytes to the capillaries. From these processes another **2ry processes** extend and they are called pedicles.

❖ The 2ry processes embrace (surround) the BM of the glomerular blood capillaries. The pedicles of one cell interdigitate with the pedicles of the adjacent cells. The interdigitation leave a space between them called (filtration slit).

❖ Filtration slit is narrow gap between adjacent podocyte cellular processes, "pedicels". Its diameter is 25 nm which is closed by a diaphragm (as the fenestrated endothelium). Its thickness is 5-6 nm. It is very important in the filtration process of the blood plasma. Podocytes have bundles of actin microfilaments in their cytoplasm that give them a contractile capacity.

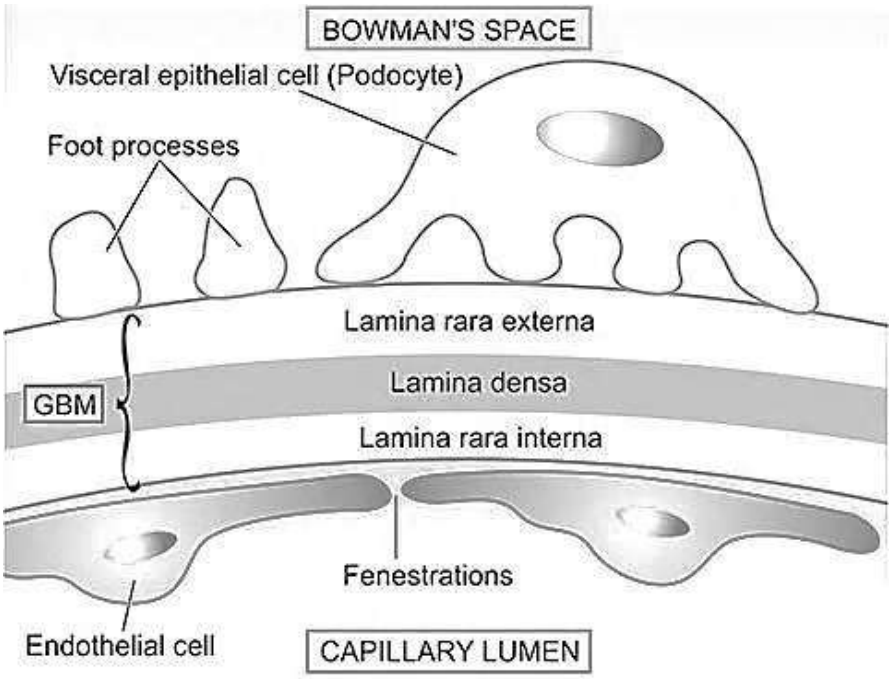


Fig. 4: A diagram showing the cell body of podocyte and its processes and blood urinary barrier.

B) The glomerulus (glomerular blood capillaries):

▪ After entering the renal corpuscle, the afferent arteriole usually divides into 2 to 5 primary branches, each subdividing into capillaries and forming the renal **glomerulus**. They will drain in the branches of efferent arterioles, to leave the renal corpuscles carrying filtered blood.

The **endothelial cells** of glomerular capillaries are of the fenestrated variety (i.e. interrupted by pores 60-100 nm), but they lack the thin diaphragm that covers the openings of other fenestrated capillaries.

1. **The glomerular basement membrane** is continuous and thick (320-340 nm). It results from fusion of 2 basement membranes one of podocyte and the other of glomerular endothelium.
It is formed of 3 layers:

1. Lamina densa: electron dense layer.The lamina densa is a meshwork of type IV collagen and laminin. They act as physical filter (selective macromolecular filter).
2. Lamina erara: electron lucent: (externa & interna).
Surround thelamina densa. They act as charge barrier (electric barrier). The BM is continuously renewed **by podocytes and endothelial cells while its turnover and maintenance obtain by podocytes.**

C) The mesangium:

▪The space between groups of glomerular capillaries where no podocytes or glomerular basement membrane are present is occupied by mesangium (Fig.5.).

▪It is a special type of C.T. consisting of mesangial cells and amorphous intercellular matrix (no fibers).

▪The mesangial cells are stellate shaped cells with a small and dark staining nucleus. They extend their processes between endothelial cells so as to reach the capillary lumen. They have many functions.

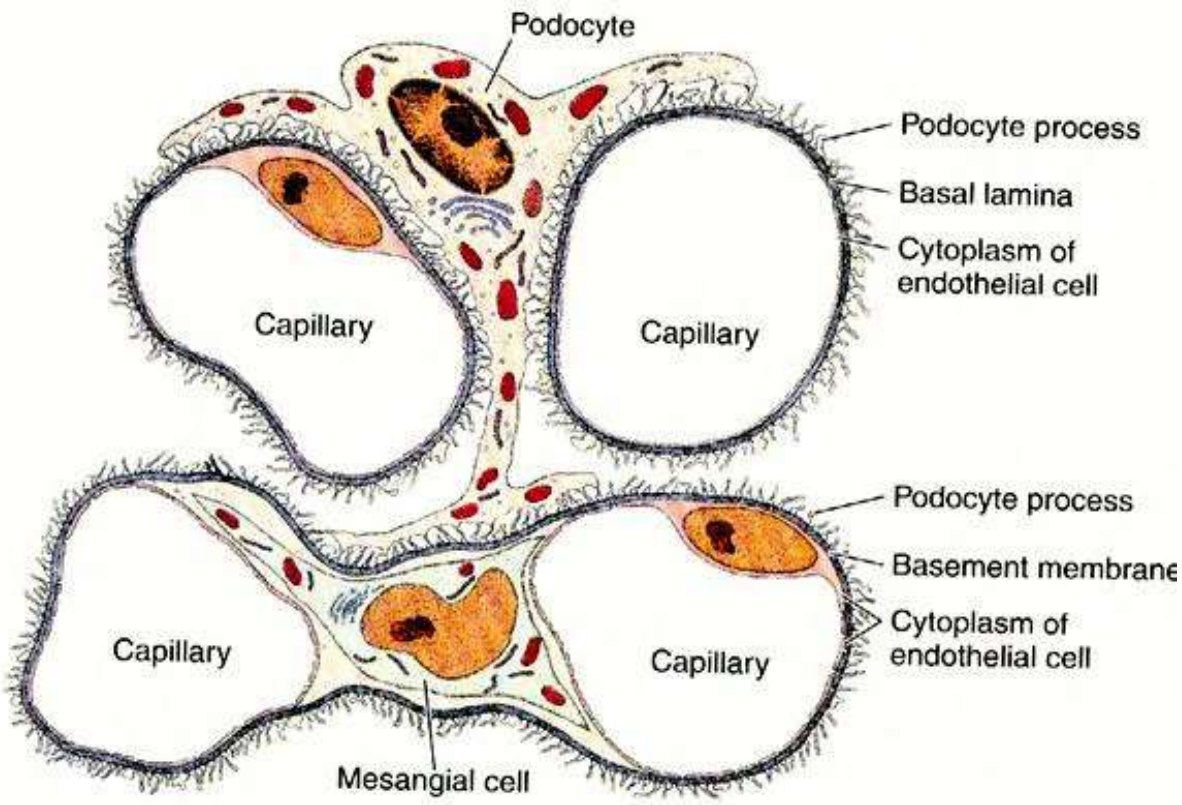


Fig.5: Mesangial cell located between capillaries enveloped by the basement membrane.

Functions of the mesengial cells:

1. Supportive function
2. Phagocytic: Theyphagocytose the exhausted glomerular basement membrane sharinginits continuous renewal.

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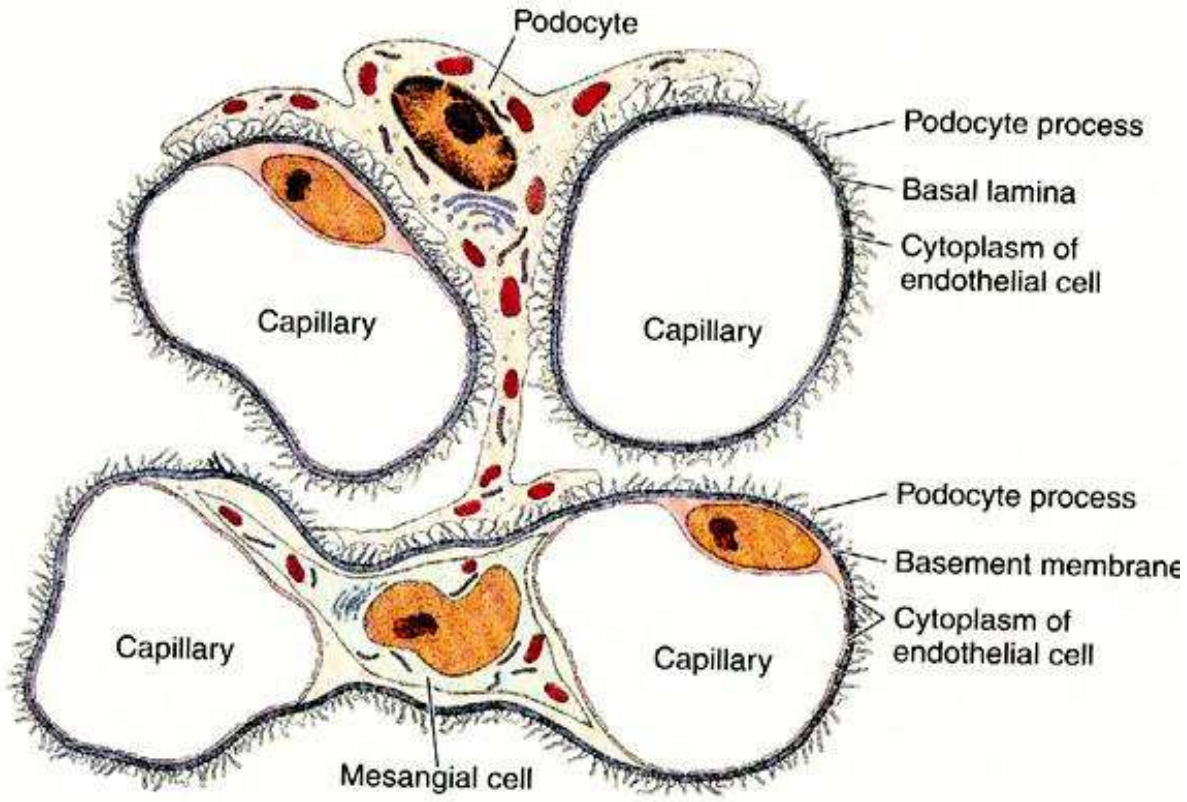


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The blood-renal (filtration) barrier (Figs. 4&6):
▪The filtration barrier between the blood in the glomerular capillaries and the renal filtrate in the Bowman's space.

▪It is formed of:

a. The glomerular endothelium (endothelium of the blood capillaries):
The fenestrae are not guarded with diaphragm. They hold the blood elements.
b. The thick glomeruar basement membrane (formed of 3 layers):
It holds the molecules of high molecular weight (like albumin). **and negative charged anions.**
c. The filteration slit diaphragm:
It stops the molecules of moderate molecular weight (protein molecules).

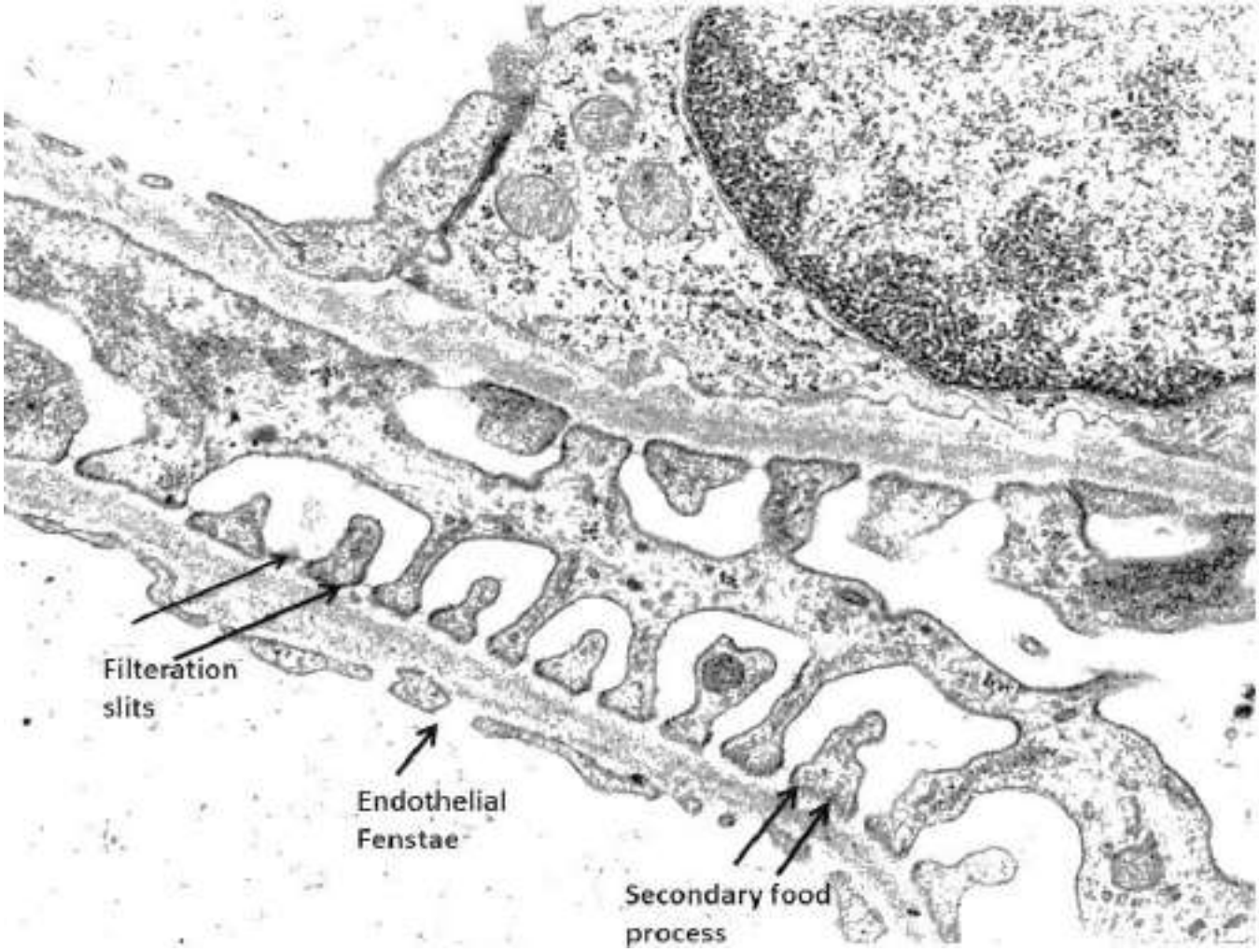


Fig .6: The filtration barrier

PCT

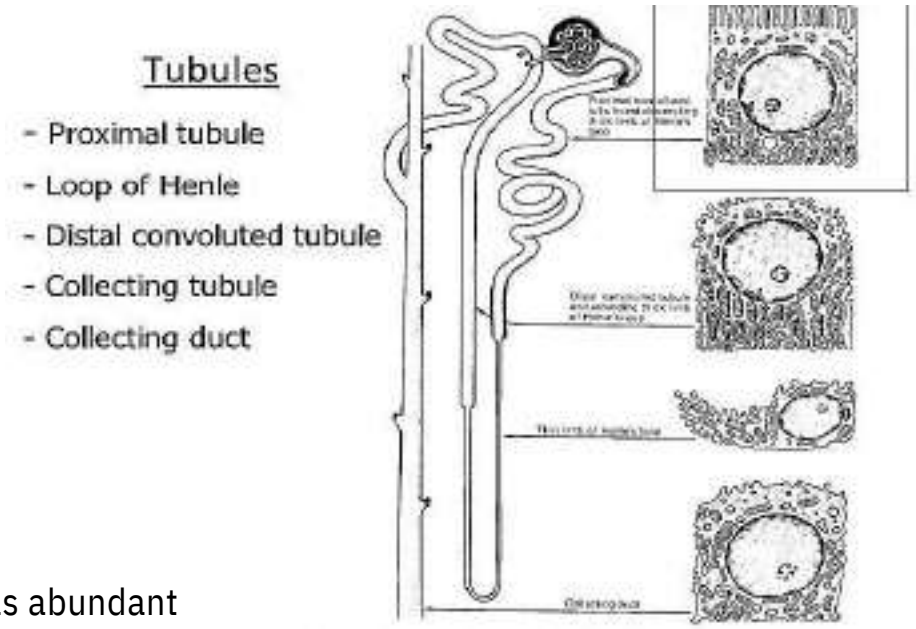
The Proximal convoluted tubules: (Figs.5. 6&7):
▪Present in the renal cortex.
▪It starts at the urinary pole of the renal corpuscle.
▪It is longer than the distal convoluted tubule andisthereforemorefrequentlyseen near renal corpuscles in the renal cortex.

By L/M:

- Lined with simple cuboidal epithelium.
- Because the cells are large, each transverse section of a proximal tubule contains only 3-5 spherical nuclei.
- The cytoplasm is strong acidophilic (due to the numerous mitochondria).
- Has apical brush border (due to the presence of microvilli).
- The basal border show basal striations due to:
  a. basal infolding
  b. parallel mitochondria
- Lateral border is indistinct (due to interdigitation of the adjacent cells)

By EM:

- The apical border of these cells has long closely packed microvilli and numerous canaliculi between the bases of the microvilli



- Lateral border shows interdigitation of the adjacent cells.

- The basal border of these cells has abundant membrane invaginations and lateral interdigications with neighboring cells.

Fig. 7: Different parts of the nephron (functional unite of the kidney) and the ultra-structure of the cells of its different parts.

▪ Mitochondria are concentrated at the base of the cell and arranged parallel to the long axis of the cell. This mitochondrial location and the increase in the area of the cell membrane at the base of the cell are characteristic of cells engaged in active ion transport.

▪ The basolateral membranes contain the Na+/K+-ATPase (sodium pump) responsible for actively transporting sodium ions out of the cells.

▪ Organelles: supranuclear Golgi, mitochondria and lysosomes.

The loop of henle

(Fig. 7):

▪ **Length**: 10-20 mm.

▪ **2 Types** are present:

- In the juxtamedullary nephrons: long loop of Henle extends in the medulla.
- In Cortical nephron: short loop of Henle.

▪ Henle's loop is a U-shaped structure consisting of:

▪ **Desending limb**: include

1. Initial thick segment similar to PCT (60 Um).
▪ Thin segment similar to blood capillaries; lined by simple squamous epithelium.

▪ **Ascending limb**:include

1. Thin segment similar to blood capillaries (lined by simple squamous epithelium).
2. Thick segment similar to DCT (cuboidal epithelium).

DCT

Distal Convoluted Tubule (Fig. 7):

▪Present in the renal cortex, 5 nm in length.

▪The distal convoluted tubules are lined with simple cubical epithelium and differ from the proximal convoluted tubules, because they have no brush border, no apical canaliculi, and smaller cells. Because distal tubule cells are flatter and smaller than those of the proximal tubule, more nuclei are seen in the distal tubule than in the proximal tubule.

▪ Cells of the distal convoluted tubule have elaborate basal membrane invaginations and associated mitochondria indicative of their ion transporting function.

▪ The cytoplasm is less acidophilic.

▪ The straight and convoluted parts of DCT have the same microscopic characters with exception of the point of contact between DCT and vascular pole of renal corpuscle (Macula densa of Juxtaglomerular apparatus).

2- The Collecting Tubules & Ducts

(Fig. 7): ▪Urine passes from the distal convoluted tubules to collecting tubules that join each other to form larger, straight collecting ducts (Papillary duct of Bellini), which widen gradually as they approach the tips of the medullary pyramids. ▪ The smaller collecting tubules are lined with cuboidal epithelium and as they penetrate deeper into the medulla, their cells increase in height until they become columnar. The cells are stained lightly by Hx&E.

- They have anelectron-lucent cytoplasm with few organelles.

- The intercellularlimitsofthe collecting tubule and ductcells areclearlyvisible in the light microscope

Juxta glomerular apparatus

(fig.8) The JG apparatus is a complex structure present where the distal convoluted tubule establishes contact with the vascular pole of the renal corpuscle of its parent nephron.
Composed of:

1. The macula densa.
2. Juxtaglomerular (JG) cells.
3. Lacis cells.

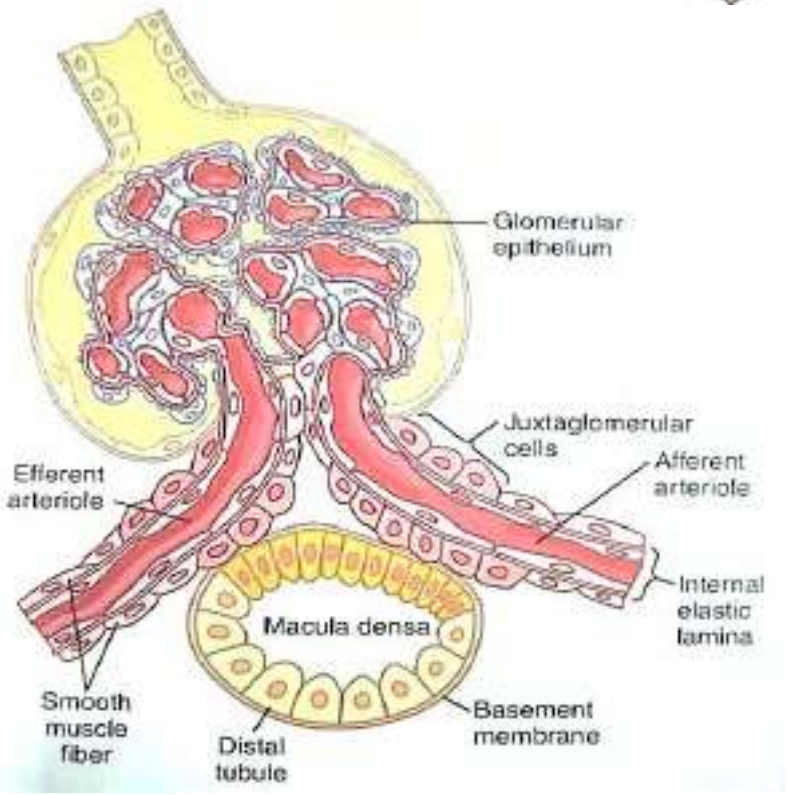


Fig. 8: the juxtaglomerular apparatus

A- The macula densa

▪Specialized spot in DCT when the distal convoluted tubule become in close contact with the vascular pole.
▪The wall of DCT shows certain changes;

– The cells are: tall columnar, narrow (crowded).
– The nucleus: deeply stained-close packed together.

By E/M:

▪ The cells have numerous mitochondria.
▪ The Golgi apparatus present between the nucleus and the base of the cells.
▪ The basal cell membrane show shallow folding.
▪ The free surface (luminal) is provided with microvilli.
▪ No Basement membrane. So the J G cells are in direct contact with the cells of macula densa.
▪This modified segment appears darker in microscopic preparations because of the close proximity of its nuclei.

Function: transmit to the Juxtaglomerular cells the osmolarity of the fluid in the distal convoluted tubules.

B- Juxta glomerular (JG) cells

Theyare modified smooth muscle cells that mainly present in the wall of afferent arterioles near the glomerulus (may be present in efferent arterioles).

The afferent arterioles have the following changes:

- The smooth muscle of the tunica media are modified and known as JG cells.
- JG cells are characterized by.

- The nucleus is rounded.
- The cytoplasm filled with secretory granules.

-When examined with the electron microscope, JG cells showcharacteristics of protein secreting cells, including an abundant rough endoplasmic reticulum, a highly developed Golgi complex, and secretory granules measuring approximately 10-40 nm in diameter.

- The internal elastic membrane of the afferent arteriole disappears in the area of the JG cells. So JG cells are in direct contact with the endothelial cells and the blood present in the lumen in one side and in direct contact with the macula densa in the other side.
- The cells are connected with Lacis cells by gap junction.

▪ Function:

-Secrete renin enzyme to increase the blood pressure.

C- The Lacis cells

The Lacis cells (Extraglomerular mesangial cell) (=JG mesangial cells= polar cushion cells = polkison cells):

These are aggregation of small light-staining cells whose functions are not well understood.
They are found between afferent and efferent arterioles and macula densa..

They seem to be similar to the mesangial cells.

Blood Circulation

1. **Glomerular circulation**: (fig.9)

Each kidney receives blood from its renal artery, which usually divides into 2 branches; anterior and posterior before entering the organ.

These branches give rise to

1. interlobar arteries that branch again to form the arteries located between the renal pyramids.

2. At the level of the corticomedullary junction, the interlobar arteries form the arcuate arteries

3. Interlobular arteries

4. The afferent arterioles which supply blood to the capillaries of the glomeruli .

5. the efferent arterioles

6. peritubular capillary network that will nourish the proximal and distal tubules .

7. vasa recta arise from efferent arterioles in juxtamedullary nephrons

8. Veins follow the same course as arteries

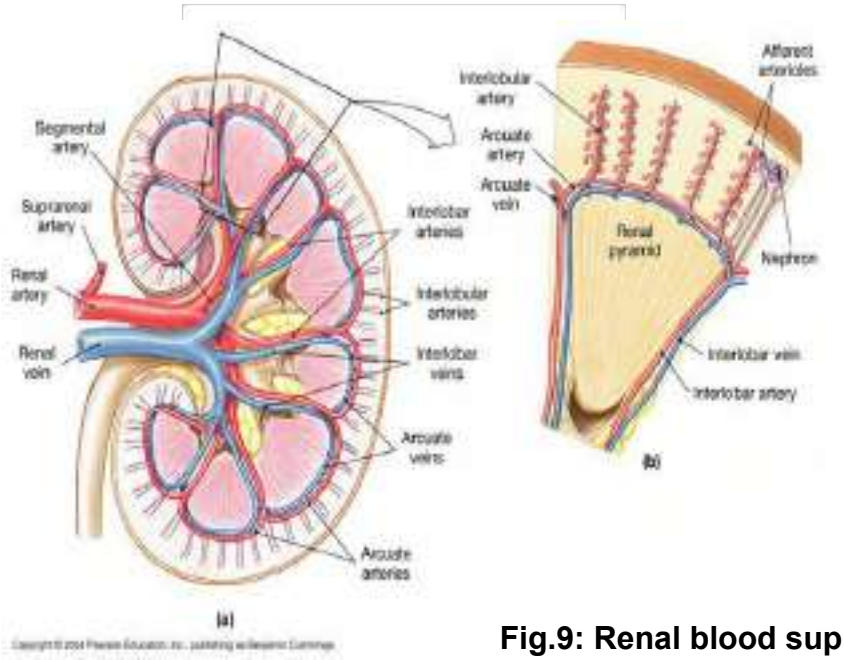


Fig.9: Renal blood supply

2. Non glomerular circulation:
Small amount of blood supplying the renal tissue do not pass through the glomeruli including:
Capsular plexus: from the terminal branches of the interlobular arteries

Renal Interstitium:

The space between uriniferous tubules and blood and lymph vessels is called the **renal interstitium.** It occupies avery small volume in the cortex but increases in the medulla.
The renal interstitium contains a small amount of connective tissue with fibroblasts, some collagen fibers and, mainly in the medulla, a highly hydrated ground substance rich in proteoglycan.

In the medulla, secreting cells called, **medullary interstitial cells**, are found. They contain cytoplasmic lipid droplets and are implicated in the synthesis of prostaglandins and prostacyclins (old theory). The medullary interstitial cells secrete **medullipin 1** hormone which is activated in the liver into **medullipin 2** hormone which is vasodilator (↓ blood pressure).

Collecting system

BLADDER & URINARY PASSAGES

(Fig., 10) ▪The calyces, renal pelvis, ureter, and bladder have the same basic histologic structure.
▪The mucosa of these organs consists of **transitional epithelium** and a lamina propria of loose-to-dense connective tissue. Surrounding the lamina propria of these organs is a dense woven sheath of smooth muscle.
▪The transitional epithelium of the bladder in the undistended state is 5 or 6 cells in thickness; the superficial cells are dome shaped cells that bulge into the lumen. These cells are frequently polyploid or binucleate. When the epithelium is stretched, as when the bladder is full of urine, the epithelium is only 3 or 4 cells in thickness, and the superficial cells become squamous.

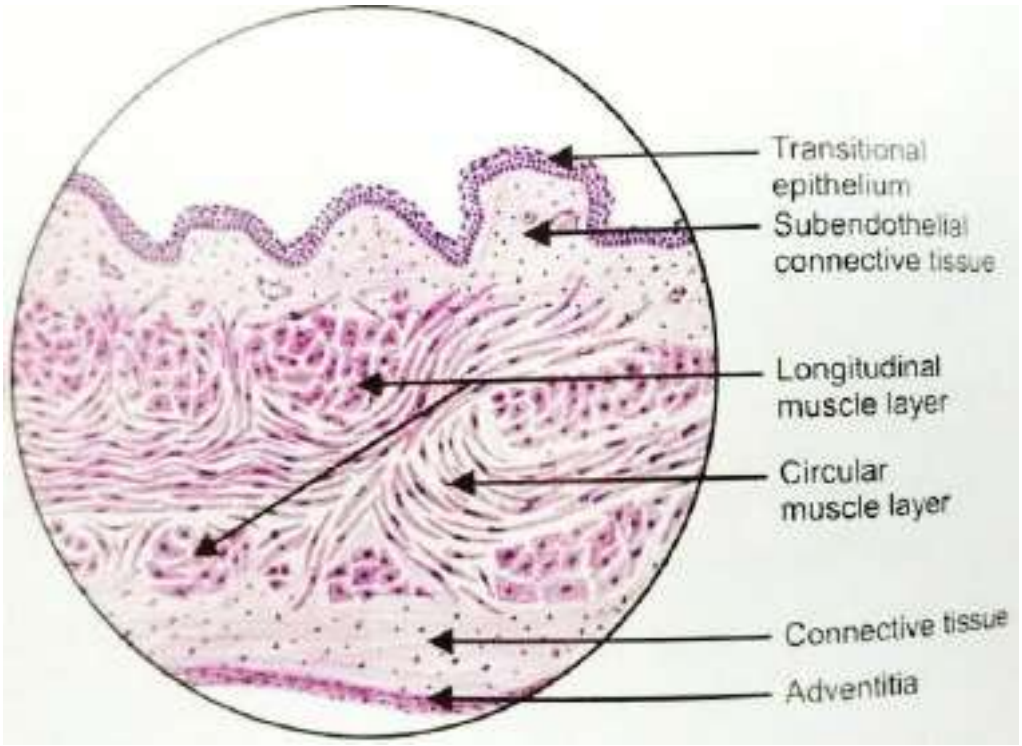


Fig.10: Histological structure of urinary bladder

▪The superficial cellsofthetransitional epithelium have a special membrane of thick plates separated bynarrowbands of thinner membrane that are responsible for the osmotic barrier betweenurine and tissue fluids. When the bladder contracts; the membrane folds alongthethinner regions and the thick plates invaginate to form fusiform cytoplasmicvesicles. These vesicles represent a reservoir of these thick plates that can be storedinthe cytoplasm of the cells of the empty bladder and used to cover the increasedcellsurface in the full bladder. This luminal membrane is assembled in the Golgicomplex and has an unusual chemical composition; cerebrosideisthe major component of the polar lipid fraction.

The muscular layers in the calyces, renal pelvis, and ureters have a helical arrangement.
As the ureteral muscle cells reach the bladder, they become longitudinal. The muscle fibers of the bladder run in every direction (without distinct layers) until they approach the bladder neck, where 3 distinct layers can be identified: **The internal longitudinal layer**, distal to the bladder neck, becomes circular around the prostatic urethra and the prostatic parenchyma in men. It extends to the external meatus in women. The fibers form the true involuntary urethral sphincter. **The middle layer ends at the bladder neck,** and the outer longitudinal layer continues to the end of the prostate in men and to the external urethral meatus in women.

- The ureters pass through the wall of the bladder obliquely, forming a valve that prevents the backflow of urine. The intravesical ureter has only longitudinal muscle fibers.
- The urinary passages are covered externally by an adventitial membrane, except for the upper part of the bladder, which is covered by serous peritoneum

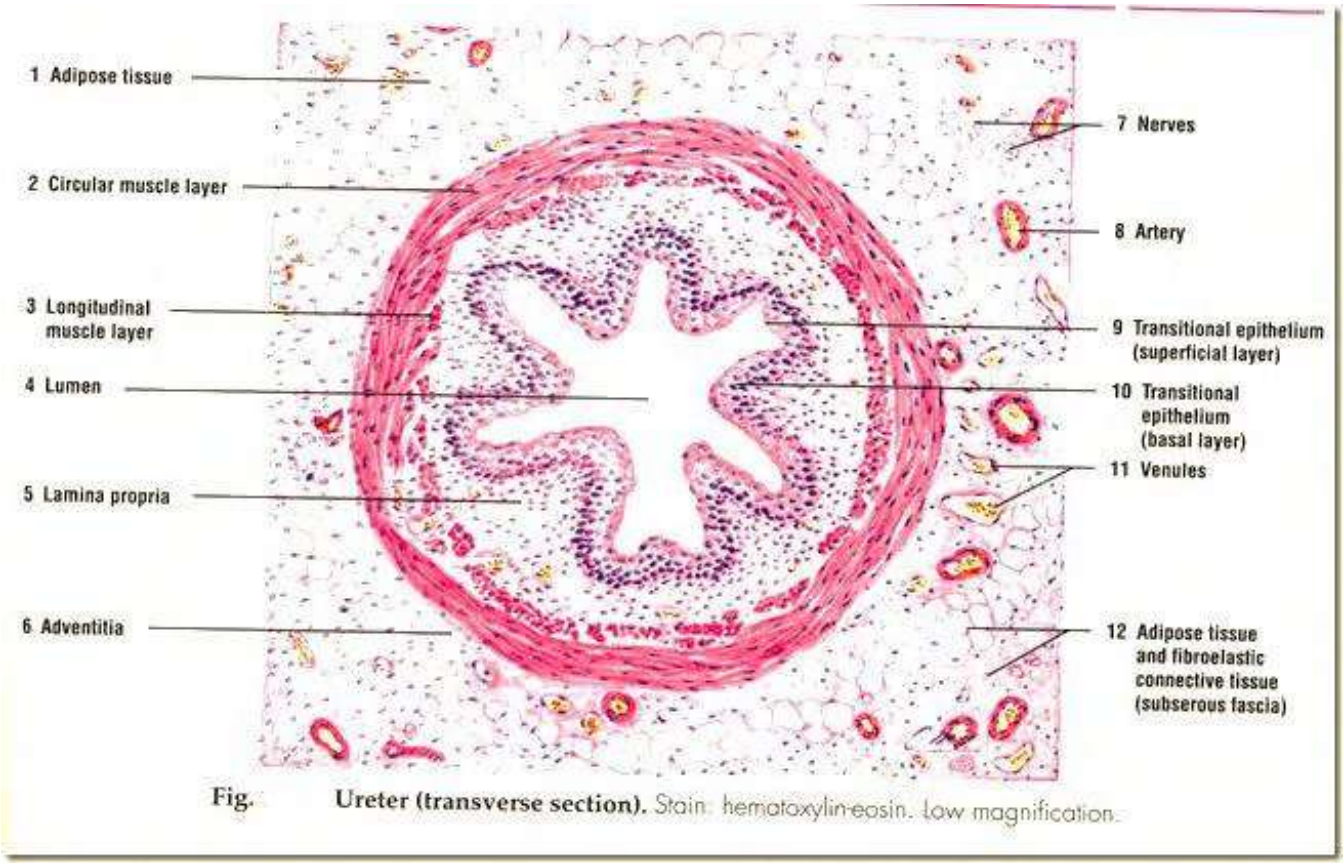


Fig. 12: histological structure of the ureter

URETHRA

The urethrais a tube that carries the urine from the bladder to the exterior. In men, sperm also pass through it during ejaculation. In women, the urethra is exclusively a urinary organ.

Male urethra:

The male urethra consists of 4 parts: prostatic, membranous, bulbous, and pendulous.

1. The prostatic urethra passes through the prostate. Proximal to the ejaculatory duct it is lined with transitional epithelium. Distal to the ejaculatory duct it is lined with pseudo stratified or stratified columnar epithelium.
2. The membranous urethra extends through the perineal membrane and is lined with stratified or pseudostratified columnar epithelium.
3. The penile urethra are located in the corpus spongiosum of the penis. The epithelium of this portion of the urethra is mostly stratified columnar. The urethral lumen dilates distally, forming the fossa navicularis which is lined with stratified squamous (non-keratinized) epithelium.

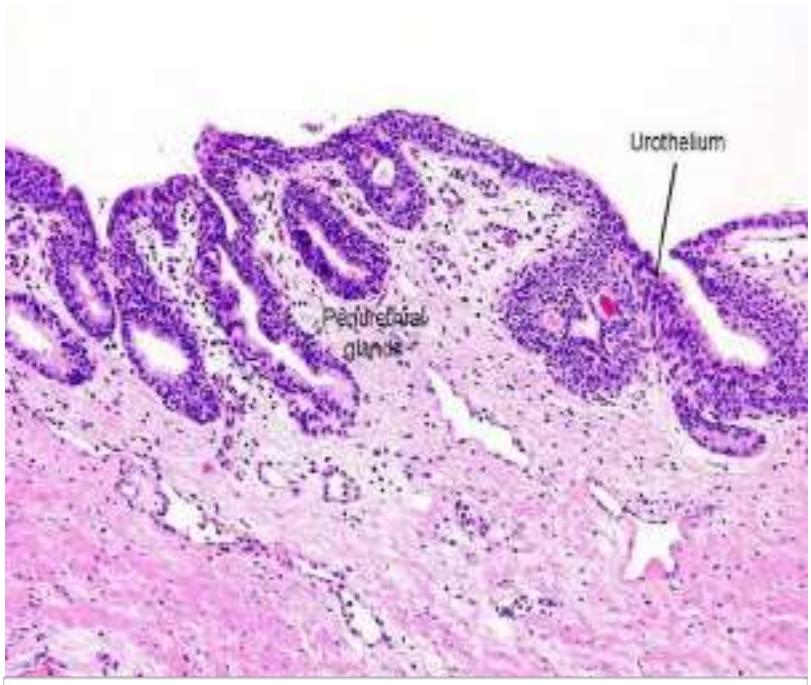


Fig.13: A section of male urethra

Litre's glands are mucous glands found along the entire length of the urethra but mostly in the penile part. The secretory portions of some of these glands are directly linked to the epithelial lining of the urethra; others have excretory ducts.

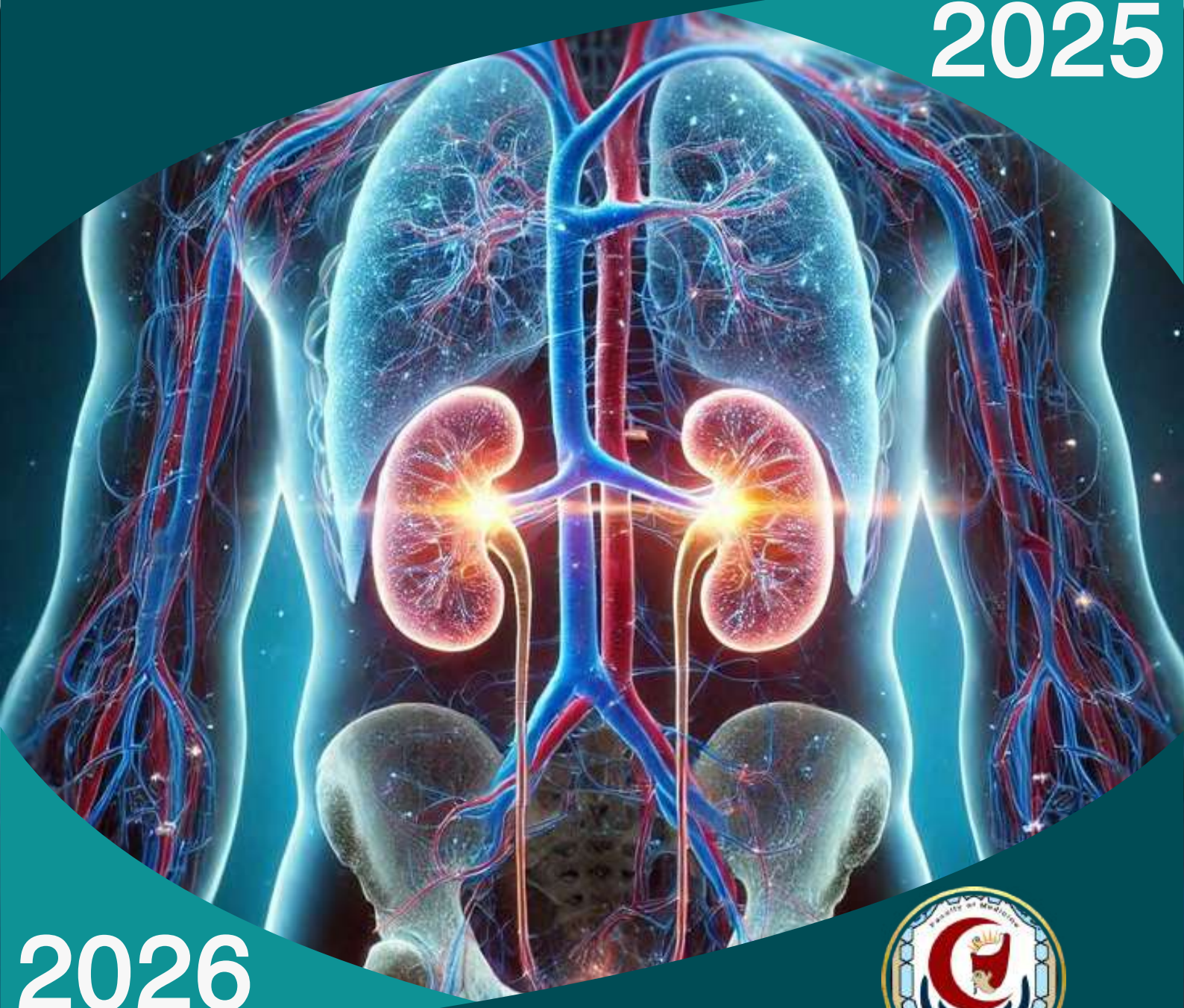
Female urethra:

- The femaleurethra is a tube 4-5 cm long, formed of:
- Mucosa formed of
  - o Epithelium.
- Transitional epithelium in its initial part.
- Pseudostratified and stratified columnar epithelium in its main part.
- stratified squamous epithelium in its terminal part.
- The mid part of the female urethra is surrounded by an external striated voluntary sphincter.
  - o Corium of highly vascular connective tissue.
  - o Musculosa; very thin and poorly developed smooth muscles
  - o Adventitia.

References:

- **Basic Histology**
- **Oxford Handbook of Medical Sciences 2nd ed (2011)**
- **Lippincott Illustrated Review: Integrated systems**
- **First Ahe basic sciences**

2025



2026

SOHAG UNIVERSITY
FACULITY OF
MEDICINE

Renal System

Physiology

RNS-315

Page	Lecture
1	Functional organization of the Kidney and renal blood flow
10	Mechanism of urine formation: I- Glomerular filtration
14	Mechanism of urine formation: II-Reabsorption and secretion
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33	Renal regulation of acid-base balance and the micturition reflex
41	Micturition and Micturition reflex

1 Phyisology of the kidney

Functions of the kidney

1. Excretory function (Urine formation):

The kidney excretes:

a) Waste productsas: Urea,Uric acid, creatinine,bilirubin,… & others

b) Excess substances as: excess water and electrolytes.

c) Drugs and metabolites

2. Regulatory (Homeostatic function)

a- Regulation of water and electrolyte balance.

b- Regulation of acid base balance.

c- Regulation of blood pressure (long- term regulation):
by regulating the volume of extracellularfluid &Through renin-angiotensin mechanism.

d- Regulation of RBCs and platelets count (Hemopoietic function):
through the release of erythropoietin & thrombopoietin.

3. Endocrine function

a) **Erythropoietin.**

b) **Thrombopoietin.**

c) **Renin.**

d) **Active form of vitamin D (1,25 dihydroxycholecalciferol).**

e) **Prostaglandins.**

4. Gluconeogenesis

During prolonged fasting.

Functional anatomy of the kidney

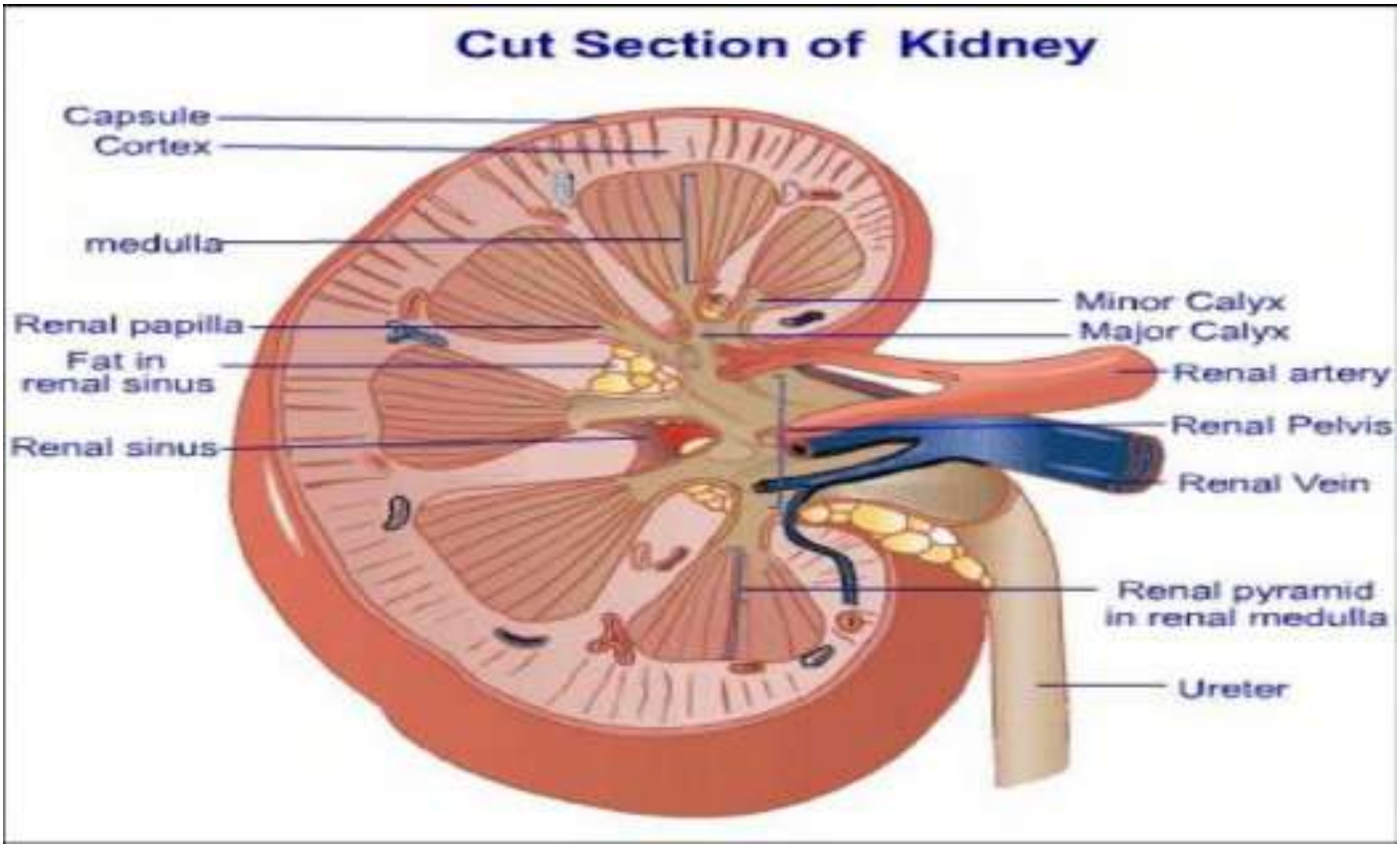
1. Cortex:

characterized by being
a) Red because it is richly supplied by blood.

b) Granular because it contains the glomeruli of the nephrons.

2. Medulla:

characterized by being
A) Pale due to its less blood supply.
B) Striated because it contains the renal tubules.
C) It is formed of 10-15 triangular shaped structures called pyramids; the apex of each pyramid forms the renal papilla which drain into minor calyces then into major calyces then into renal pelvis.



The nephron

Definition

The nephron is the functional unit of the kidney, where the processes of urine formation take place. There are about 1-1.3 million nephrons in each kidney. After the age of 40, nephrons decreased by 10% every 10 years due to aging. When there are only 500,000 nephrons remaining, renal failure develops.

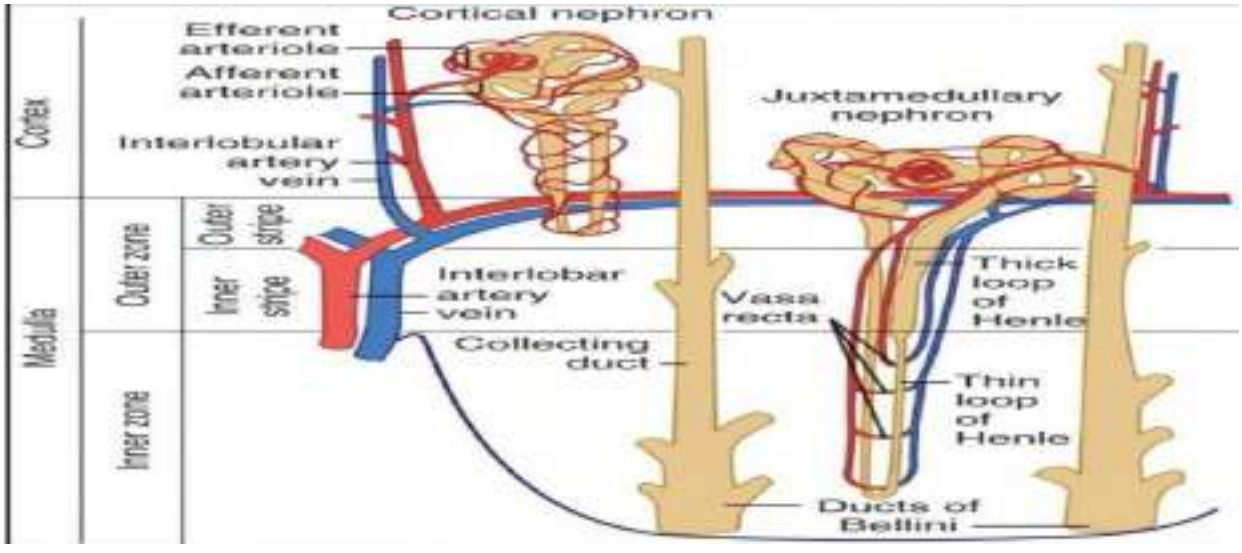
Compostion

1- Vascular component: Afferent arteriole, glomerular capillaries, efferent arteriole, and peritubular capillaries. The diameter of the efferent arteriole is lesser than that of the afferent arteriole. This makes the glomerular capillaries a high-pressure capillary set.

2- Tubular component: Glomerular capillaries are involved in Bowman's capsule

(Glomerular capillaries and Bowman's capsule together are called renal corpuscle). which continues in the proximal tubule, loop of Henle, distal tubule, collecting tubule, and collecting duct multiple nephrons may share the same collecting duct.

- All glomeruli, PCTs, DCTs and cortical collecting ducts lie in the cortex.
- The loop of Henle, medullary collecting ducts and vasa recta lie in the medulla.



Types

Cortical (superficial) nephrons

A) Found in the superficial layer of the renal cortex.
B) Forms **about 80%** of the whole nephrons.
C) These nephrons have shorter loop of Henle that **does not dip into the medulla**.
D) Surrounded by peritubular capillaries.
E) Play **minor role** in urine concentration

Juxtamedullary nephrons

A) Found in the inner part of the renal cortex, close to medulla. B) Form **about 20%** of the whole nephron. C) Their loops of Henle **dip deeply in the medulla.** D) surrounded by special peritubular capillaries that are straight and known as **Vasa Recta.** E) Play **major role** in urine concentration.

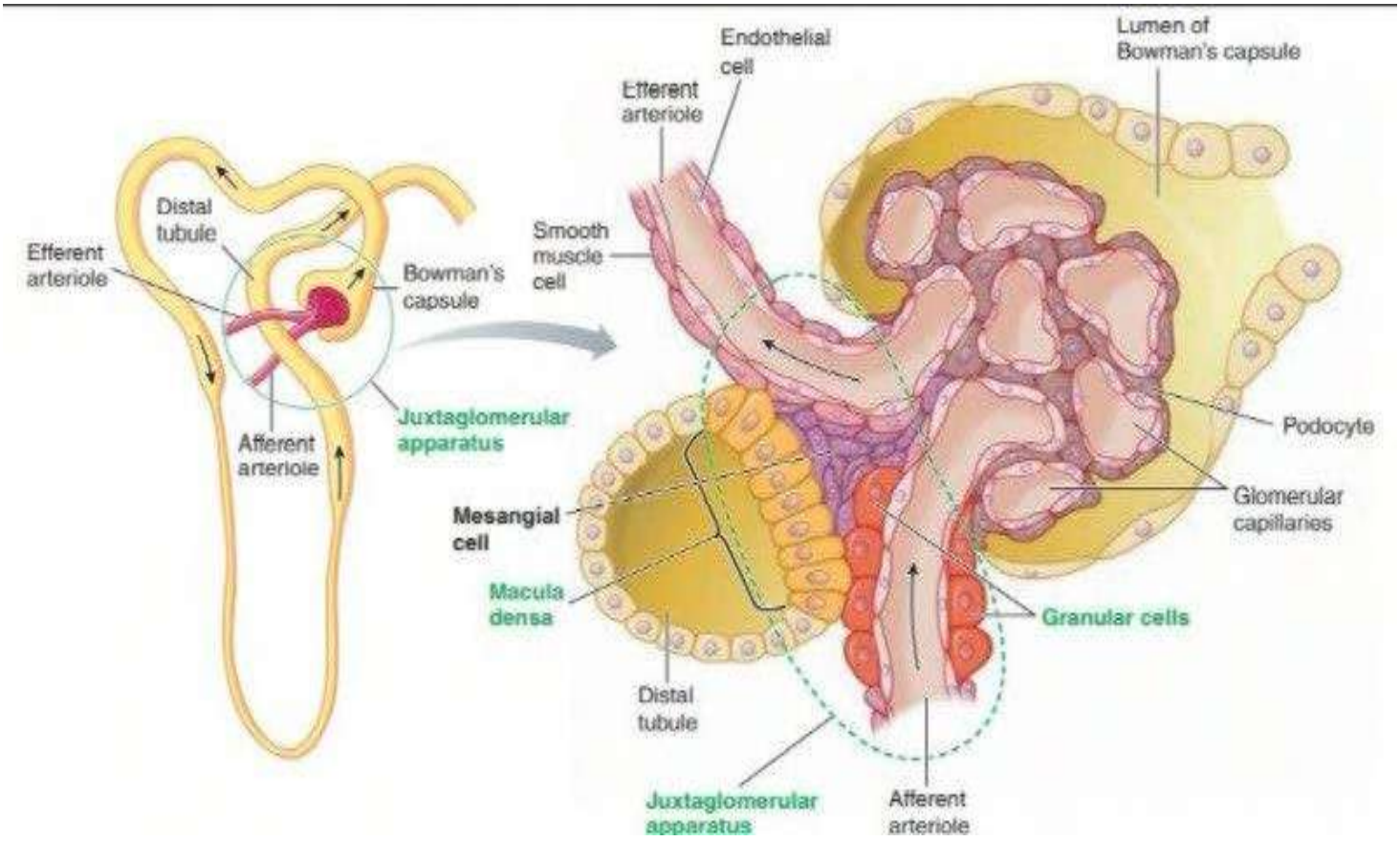
Juxtaglomerular apparatus

- The juxtaglomerular apparatus is a specialized structure situated at a point where the initial part of the distal tubule passes through the fork formed by the afferent and efferent arterioles of the same nephron.
- **The Function of juxtaglomerular apparatus** is Regulation of the glomerular blood flow and glomerular filtration rate through secretion of Renin & Prostaglandins.
- Itconsists of 3 specialized cells that involve:

1- Macula densa of distal nephron (Chemoreceptors, stimulated by decrease sodium chloride concentration in tubular fluid).

2- Juxtaglomerular cells (Granular cells), located in the media of afferent arteriole: act as Renin-secreting cells.

3- Lacis cells (extraglomerular mesangial cells): are situated in the triangular region bound by afferent arteriole, efferent arteriole and macula dense.



Renin

- Secreted by Juxtaglomerular cells.
- Along with angiotensins, renin forms the renin-angiotensin system, which is a hormone system that plays an important role in the maintenance of blood pressure.
- Secretion of renin is stimulated by ↓ NaCl at the macula densa.
- This occurs when there is: decreased load of sodium and chloride in macula densa, fall in arterial blood pressure, Reduction in the ECF volume,Increased sympathetic activity.

Renin-angiotensin system

- ✓ When renin is released into the blood,it converts angiotensinogen (inactive protein secreted by the liver) into angiotensin I (inactive).
- ✓ Angiotensin I is converted into angiotensin II (the most active), by the activity of angiotensin-converting enzyme (ACE) secreted from lungs.
- ✓ Angiotensin II Then it is rapidly degraded into angiotensin III (less active) by angiotensinases, which are in RBCs and vascular beds in many tissues.
- ✓ Angiotensin III is converted into angiotensin IV (less active).

Actions of angiotensin II

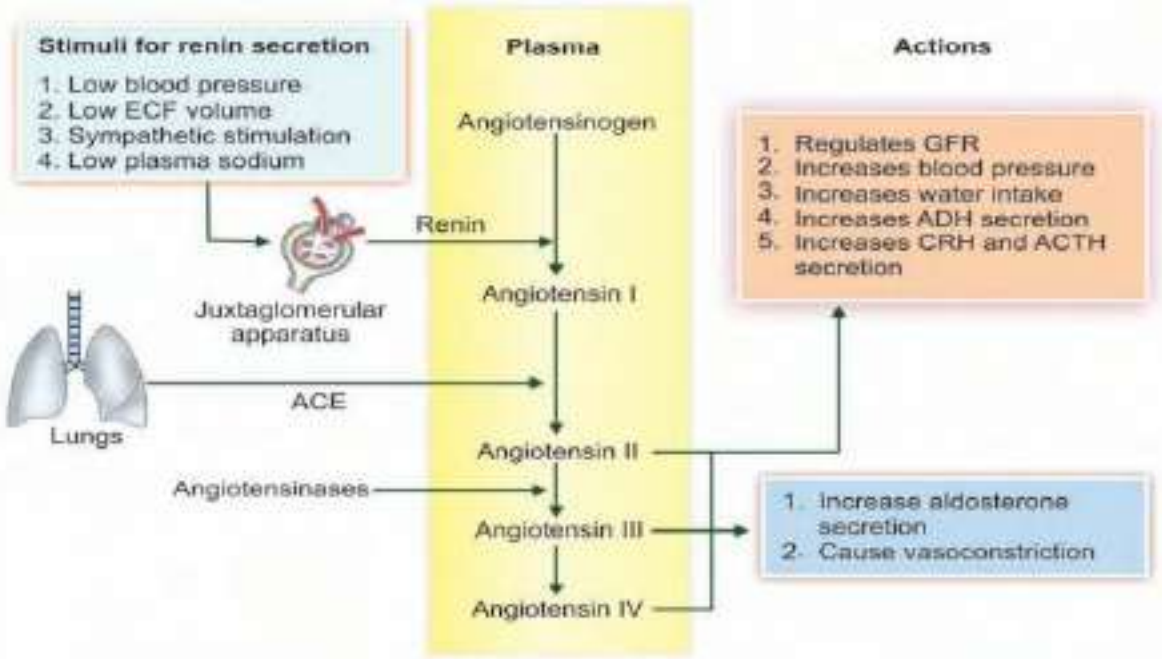
- ➢ On blood vessels: increases arterial blood pressure by directly acting on the blood vessels and causing vasoconstriction. It is a potent constrictor of arterioles.
- ➢ On adrenal cortex: It stimulates the adrenal cortex to secrete aldosterone which increases retention of sodium and water, which is also responsible for elevation of blood pressure.
- ➢ On the kidney:

1) It constricts the efferent arteriole more than the afferent arteriole increase the glomerular pressure.

2) It contracts the glomerular mesangial cells leading to decrease in surface area of glomerular capillaries and filtration

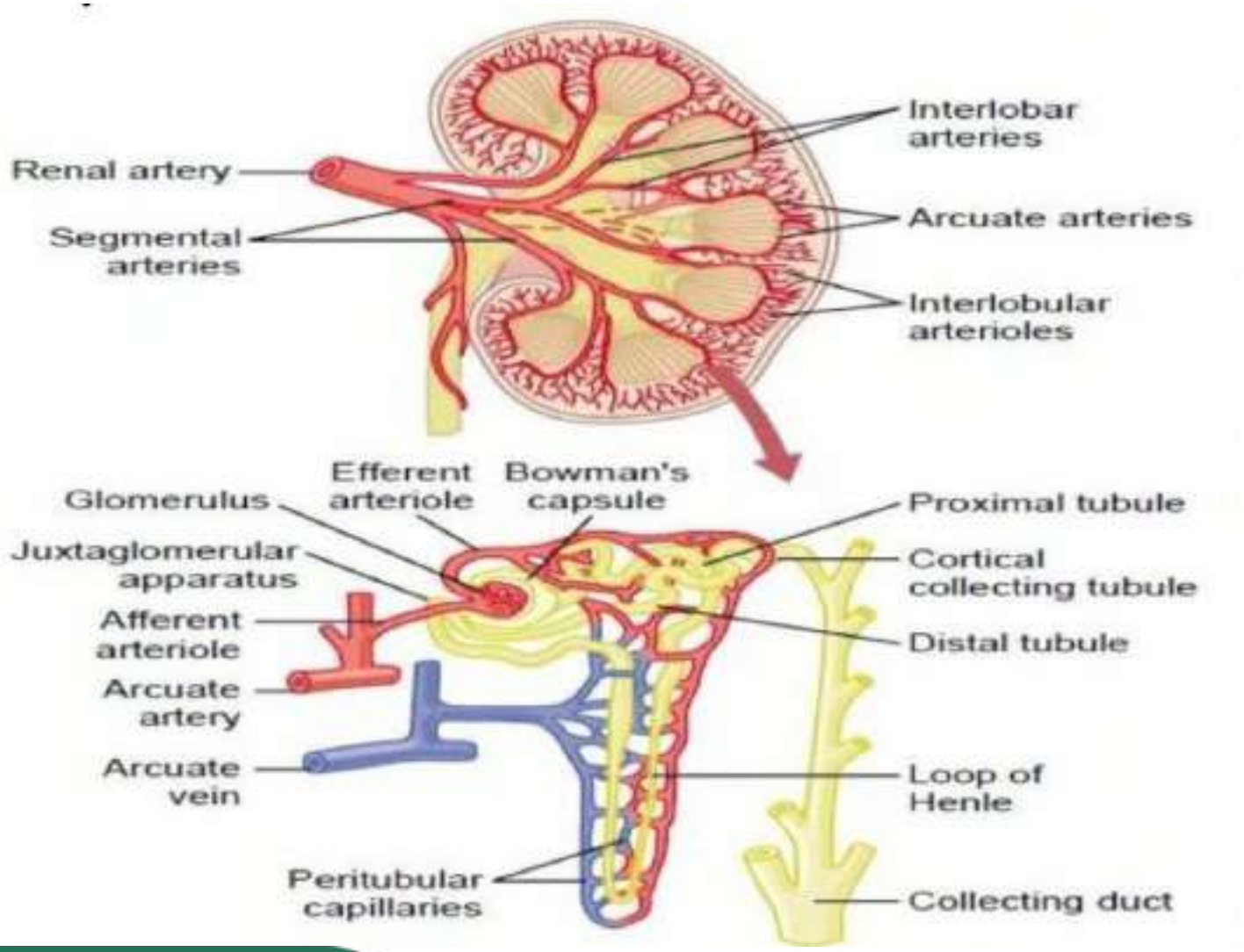
3) It increases sodium reabsorption from renal tubules. This action is more predominant on proximal tubules.

➢ On brain: It increases water intake by stimulating the thirst center.



Renal circulation

Renal artery → Segmental arteries → Interlobar arteries → Arcuate artery → Interlobular arteries → Afferent arterioles → Glomerular capillary tuft → Efferent arteriole, → Peritubular capillaries, vasa recta → Peritubular venules → Interlobular veins → Arcuate veins → Interlobar veins → Segmental veins → and finally the renal vein.



Renal regulation

- Renal blood flow is regulated mainly by autoregulation.
- The nerves innervating renal blood vessels do not have any significant role in this.
- Autoregulation is the intrinsic ability of an organ to regulate its own blood flow.
- Autoregulation is present in some vital organs in the body such as brain, heart and kidneys.
- It is highly significant and more efficient in kidneys.

Renal autoregulation

- Renal autoregulation is the ability of the kidney to maintain constant renal blood flow (RBF) and glomerular filtration rate (GFR) and to allow continuous control of renal functions.
- This occurs between 60 mmHg and 180 mmHg.
- Two mechanisms are involved in renal autoregulation:
  1. Myogenic mechacnism
  2. Tubuloglomerular feedback.

1- Myogenic mechanism

A- Whenever the blood flow to kidneys increases, it stretchesthe elastic wall of the afferent arteriole → increases the flow of calciumions from extracellular fluid into the cells → contraction of smooth musclesin afferent arteriole, which causes constriction of afferent arteriole. So,the blood flow is decreased.

B- At low blood pressure, relaxation of the smooth muscles of the afferent arterioles → decrease of vascular resistance → maintain constant blood flow.

2- Tubuloglomerular feedback.

A- When GFR increases → the concentration of NaCl increases in the filtrate → macula densa releases adenosine → decreases the renin secretion → VD of the efferent arteriole & constriction of afferent arteriole → ↓GFR & RBF back to normal.
B- When GFR decreases → the concentration of NaCl decreases in the filtrate → Macula densa secretes prostaglandin (PGE2), bradykinin and renin → PGE2 and bradykinin cause dilatation of afferent arteriole, renin induces the formation of angiotensin II, which causes constriction of efferent arteriole → increase GFR.

Urine formation

Steps of urine formation

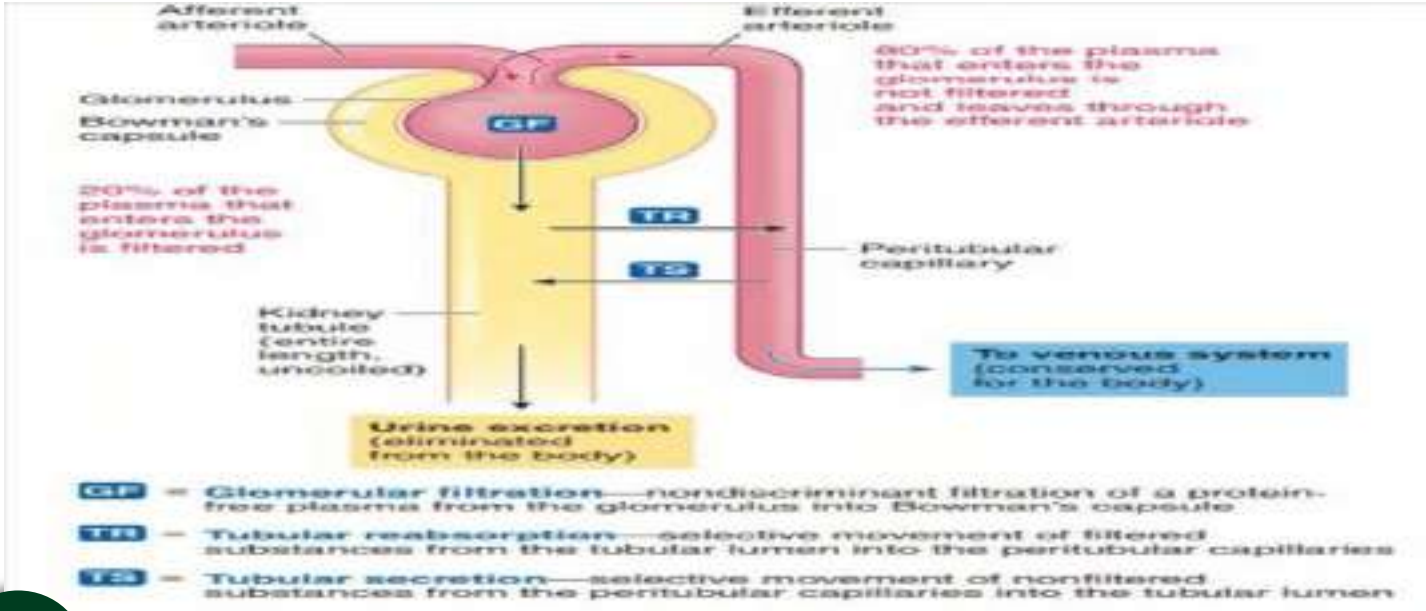
1-Glomerularfiltration(GF): which is non-selective movement of constituents of p lasma except the plasma proteins from the glomerular capillaries into the Bowman's capsule.

2- Tubular reabsorption (TR): A selective process, by which some substances of the filtrate are retained back to the blood.

3- Tubular Secretion (TS): A selective process by which some substances, that have not been filtered are secreted from the peritubular capillaries into the tubules.

➢ Whatever left over these three processes will be excreted in form of urine

Excretion = Filtration - Reabsorption + Secretion



Glomerular filteration

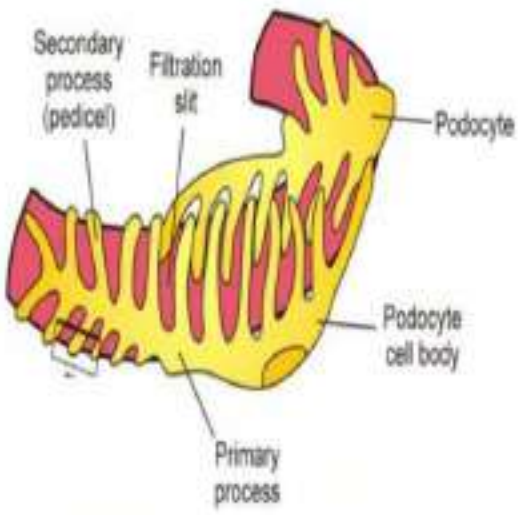
- Kidneys receive about 22-25 % of cardiac output, this is called Renal Blood Flow (RBF) which is approximately 1.1-1.3 L of blood in a minute.
- Plasma in this flow is about 625 ml. It is called Renal Plasma Flow (RPF).
- About 20 % of Plasma entering the glomerular capillaries is filtered into the Bowman's capsule i.e. GFR is about 125 ml/min (180 L/day).
- The urine flow is about 1 ml/min (about 1-1.5 liter /day) This means that kidney reabsorbs about 178.5 liters every day.

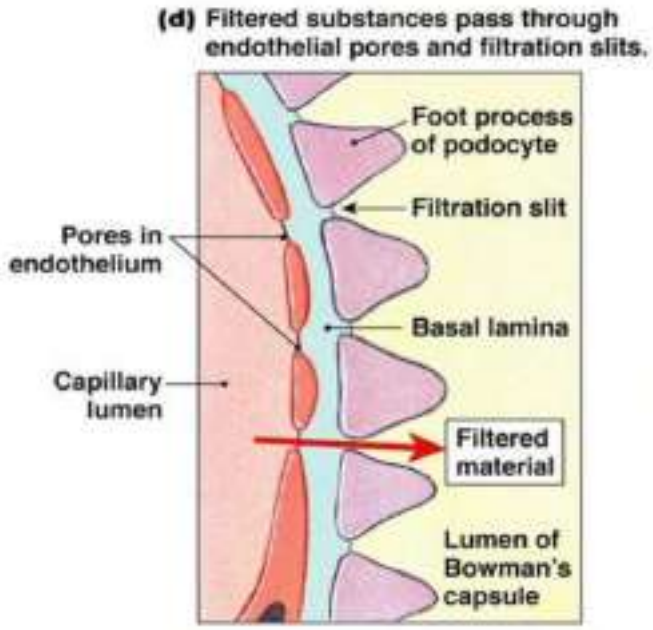
Filteration unit

Filtration occurs through the filtration unit

1) endothelial cells of glomerular capillaries, which are fenestrated. Fenestrae are large however prevent filtration of blood cells and most of plasma proteins. Permeability of glomerular capillaries is high (about 50 times that of capillaries of skeletal muscles).
2) Glomerular basement membrane, carry negative charges and allows the passage of substances that diameter is 4 -8 nm, and exclude substances more. Neutral substances less than 4 nm in diameter are always filtered, while substances more than 8 nm in diameter are never filtered.Negatively charged substances are filtered more difficult than the positively charged ones.
3) Epithelial cells of Bowman`s capsule (podocytes). (finger like projections forming slit pores).

- Albumin has a diameter of 6 nm but its glomerular concentration is 0.2% that of its plasma concentration that is because it carries negative charges.





Glomerular Filtration Rate (GFR)

Itis the volume of the glomerula r filtrate formed by the normal glomeruli of both kidneys per m inute.

Measuring GFR**:** by inulin clearance or creatinine clearance

(Forces drive the glomerular filtration (Starling's forces)):

A-Forces favoringfiltration:

1-Hydrostaticpressure of the capillary blood, it is about 55 mmHg.

2-Oncoticpressureof the fluid of the Bowman's capsule, it is nearly zero.

B- Forces opposing filtration **:**

Oncotic pressure of the plasma proteins in

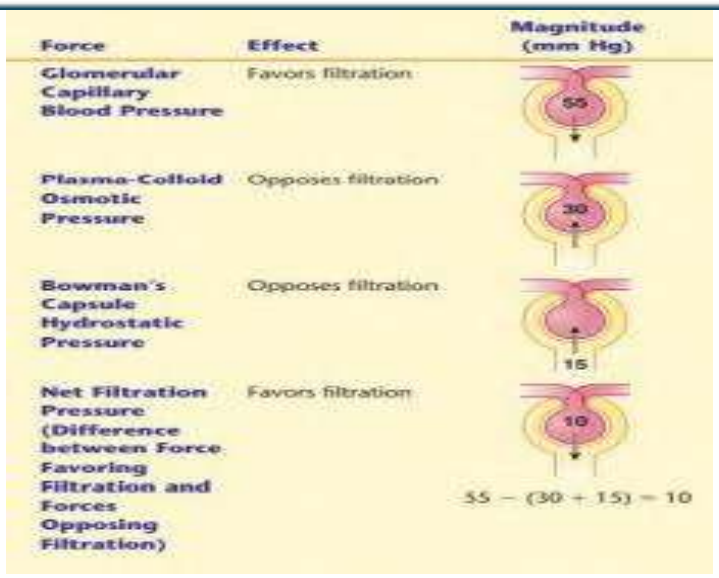
1- the glomerular capillary. It is about 30 mmHg.

2- Hydrostatic pressure of the Bowman's capsule, which also opposes filtration. It is about 15 mmHg.

The net pressure is as follows:

Hydrostatic pressure of glomerular capillaries - (Oncotic pressure of glomerularcapillaries + Hydrostatic pressure of the Bowman capsule):

55 - (35+10) = 55 – 45 = 10 mmHg.



Filtration co efficient (Kf)

= GFR / net filtration pressure =125/10=12.5ml/min/mmHg.

filtration coefficient (*Kf*) depends on:

a- surface area of the glomerular capil laries, which is affected by mesangialcells that are contractible cells. When they contract, they decreasethesurface area available for filtration.

b- permeability of the glomerular capillaries.

Regulation of glomerular filtration:

*1. RenalBlood Flow*It is themost important factor that is necessary for glomerular filtration. GFR is directly proportional to renal blood flow.

2. Glomerular Capillary Pressure: GFR is directly proportional to glomerular capillary pressure.

This Capillary pressure, in turn depends upon the RBF, ABP, diameter of afferent and efferent arterioles.

- Constriction of the **afferent arteriole** → ↓ RBF → ↓ the glomerular pressure → ↓GFR
- constriction of the **efferent arteriole** → ↑ the glomerular pressure → ↑ GFR.

If severe constriction occurs, GFR then ↓ due to ↓ RBF and ↑ colloidal osmotic pressure in the glomeruli due to high filtration.

3. Plasma Colloidal Osmotic Pressure: GFR is inversely proportional to colloidal osmotic pressure.

4. Hydrostatic Pressure in Bowman Capsule: GFR is inversely proportional to this.

.5. *Systemic ABP:*

RBF and GFR are not affected as long as the mean ABP is in between 60- and 180-mm Hg due to the autoregulatory mechanism. Variation in pressure above 180 mm Hg or below 60 mm Hg affects the renal blood flow and GFR.

6. *Sympathetic Stimulation:*

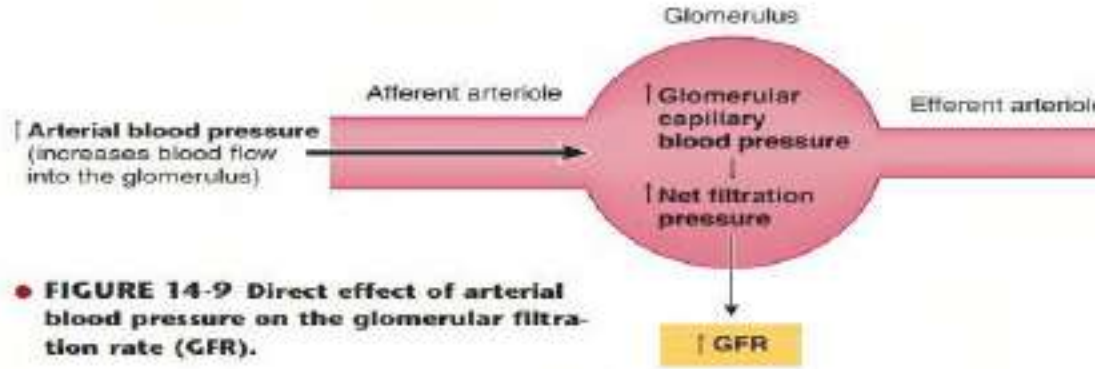
Sympathetic stimulation constricts both afferent and efferent arterioles but it affects the afferent arterioles more than the efferent arterioles → ↓ RBF & ↓ GFR.

7. Surface Area of Capillary Membrane: (controlled by mesangial cells)

GFR is directly proportional to the surface area of the capillary membrane.

8. Renal diseases:

- Albuminuria → ↑ colloidal osmotic pressure of Bowman capsule → ↑ GFR.
- Uncontrolled diabetes → ↓ GFR. → ↓ No. of functioning nephrons → ↓ glomerular surface area
- Renal failure or nephrectomy → ↑ GFR. → ↓ GFR.
- Nephritis increases the permeability of glomerular membrane → ↑ thickness of glomerular membrane



● **FIGURE 14-9 Direct effect of arterial blood pressure on the glomerular filtration rate (GFR).**



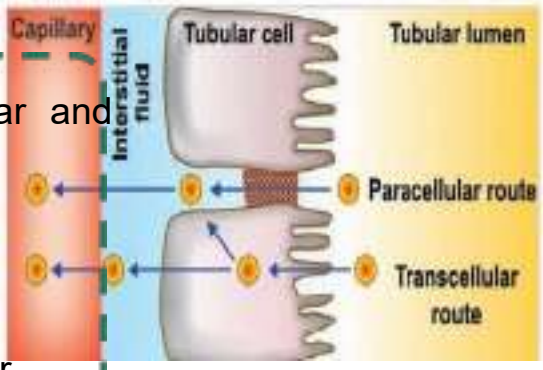
(a) Arteriolar vasoconstriction decreases the GFR



(b) Arteriolar vasodilation increases the GFR

,,,,

Tubular Reabsorbtion



- There are two routes of reabsorption: Paracellular and transcellular:

* Transcellular route: it depends on presence

of transporters (carriers and c hannels for example).

* Paracellular route: transport of substances fromtubular lumenintointerstitial fluid present in lateral intercellular space throughthetightjunction between the cells, then to peritubular capillaries.

Site of Reabsorption

Reabsorption of the substances occurs in almost all the segments of tubular portion of nephron

A. Substances Reabsorbed from Proximal Convoluted Tubule (PCT)

ThefluidthatentersthePCTisanisotonicultrafiltrate(300mOsm/L). Theconcentration of a freely filtered substance in this fluid equals its plasma concentration.

1- Sodium reabsorption:

- Approximately **two-thirds (65 %)** of the filtered sodium are reabsorbed in the PCT.

-1 The initial step is pumping of Na + out of the cell and K+ into the cell by Na + - K+ pump on the basolateral membrane. This will decrease the Na + concentration in the ICF (12 mEq compared to 140 mEq in the filtrate) and will also create a negative charge inside the cell. the Na + will pass from the lumen of the renal tubule into the cells

-2 down concentration and electrical gradient by many transporters as sodium - glucose symporter (SGLT1), sodium-amino acid symporter, sodium - phosphate symporter and by sodium- hydrogen antiporter and others. Sodium- hydrogen antiporter accounts for most of sodium reabsorption in the PCT.

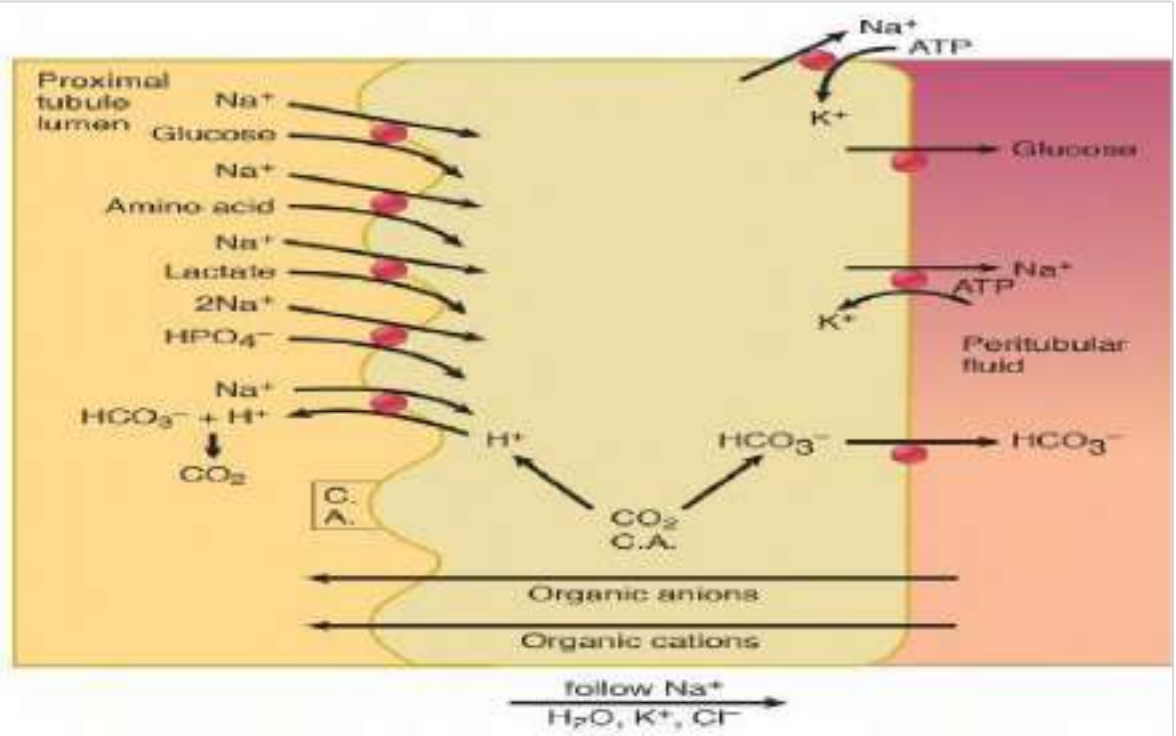
3- After crossing the luminal membrane, the Na + then passes to the interstitium by Na/K pump and by Na + /HCO3 - symporter.

4- Catecholamine and angiotensin II stimulate the basolateral ATPase and thus enhance the fraction of sodium reabsorbed in the PCT

2- Water (obligatory water reabsorption) and electrolytes

After reabsorption of sodium, an electrical gradient will be created, then 65% of filtered chloride is reabsorbed following the sodium. Chloride diffuses mainly inpassive paracellular route, and then 65% of water follows by osmosis through **"leaky" tight junctions** in the PCTs.

- **Two-thirds of** filtered K+ is reabsorbed in the late PCT.
- 75% of phosphates is reabsorbed by secondary active transport (only in PCT, inhibited by parathormone hormone).
- 60% of filtered Ca 2+ is reabsorbed (not dependent on parathormone)
- *The filtrate leaving the PCT is iso-osmolar* (300 mOsm/L).



3- Glucose and aminoacids Reabsorption

- Normally, all of the filtered **glucose** is reabsorbed in the PCT via secondary active transport linked to sodium.

- In addition, all **peptides**, **amino acids**, and **ketone bodies** are reabsorbed here via secondary active transport (linked to sodium reabsorption).

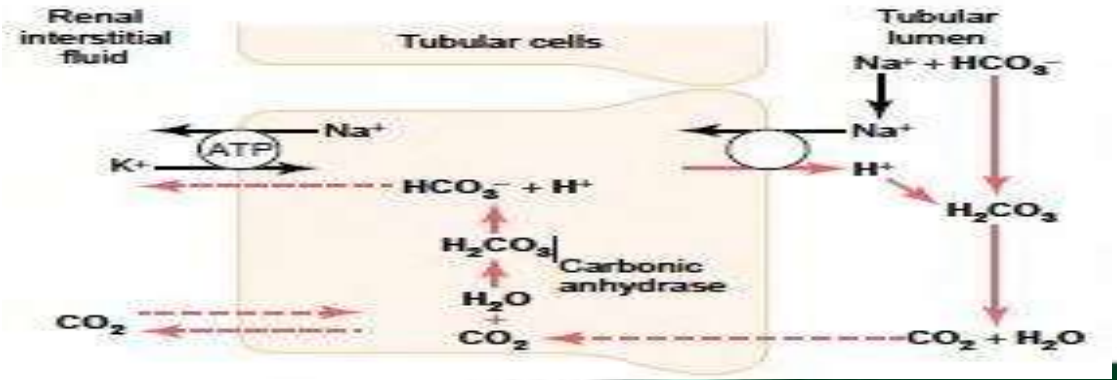
- Therefore, the concentration of the above metabolites should be zero in the tubular fluid leaving the proximal tubule (clearance is zero).

– Reabsorption of glucose starts by active transport of Na by the pumps on the basolateral membrane . This will create Na gradient which will cause Na + to cross the luminal membrane down its concentration gradient with glucose up its concentration gradient via SGLT2 (sodium-glucose transporter).
– The concentration of glucose will be increased in the cell and this will enable the glucose to pass down concentration gradient to the interstitium by glucose uniporter (GLUT2). Glucose will then pass to

the peritubular capillaries by simple bulk flow).
.

4.Bicarbonate reabsorption:

- About 80% of the filtered bicarbonate is reabsorbed in PCT. The mechanism for this reabsorption is:
- Bicarbonate combines with free H+ in lumen and is converted into CO2 and H2O, catalyzed by the luminal **carbonic anhydrase** enzyme (**CA**).
- H+ is pumped into the lumen in exchange with sodium (antiporter).
- CO2, being very soluble, crosses the luminal membrane where it combines with water to reform H+ and bicarbonate (note CA in the cell).
- The H+ is then pumped back into the lumen, while bicarbonate exits the basolateral membrane to complete its reabsorption.
- Because of this mechanism, bicarbonate reabsorption is dependent upon H+ secretion and the activity of CA.
- The most important factor for H+ secretion is the concentration of H+ in the cell. Thus, H+ secretion and bicarbonate reabsorption are increased during an acidosis and they decrease with an alkalosis.



5. Other substances

- 50% of urea is reabsorbed by solvent drag.
- 100% of filtered proteins and vitamins are reabsorbed.
- **Renal handling of proteins:** Plasma proteins are not filtered in Bowman's capsule but some proteins and peptides in blood may pass the filtration membrane and then reabsorbed. Some peptides are reabsorbed paracellularly, while others bind to the luminal membrane and then enter the cells by endocytosis, where they will degrade by peptidase enzymes to amino acids.

Secretion in PCT:

- **Organic acids and bases** are secreted into PCT. **PAH,** creatinine, many drugs such as **penicillin**

.

Summary

- 65% of Sodium, potassium, (calcium, bicarbonates, chlorides and water 2ry to reabsorption of Na)
- All glucose, amino acids (reabsorbed only in PCT),
- 75% phosphates (only in PCT by active process, inhibited by parathrmone hormone)
- 50% of urea (2ry to reabsorption of water "called back diffusion")
- **Na+** is reabsorbed via **cotransport** with amino acids, glucose, lactate, and phosphate, and via **countertransport** with H+ through the Na+–H+ exchanger.

Substances Reabsorbed from Loop of Henle

- Fluid entering the loop of Henle is isotonic (300 mOsm/kg), but the volume is only one-third the volume originally filtered into Bowman's space.

In the Thin de scending limb

-The wall of thin descendinglimb of LH is permeable to water but poorly permeable to solutes.
-15% of filtered water diffuses out by the **renal medullary hyperosmolarity** through leaky "tight" junctions on the epithelium.
-No solute reabsorption occurs in this segment.

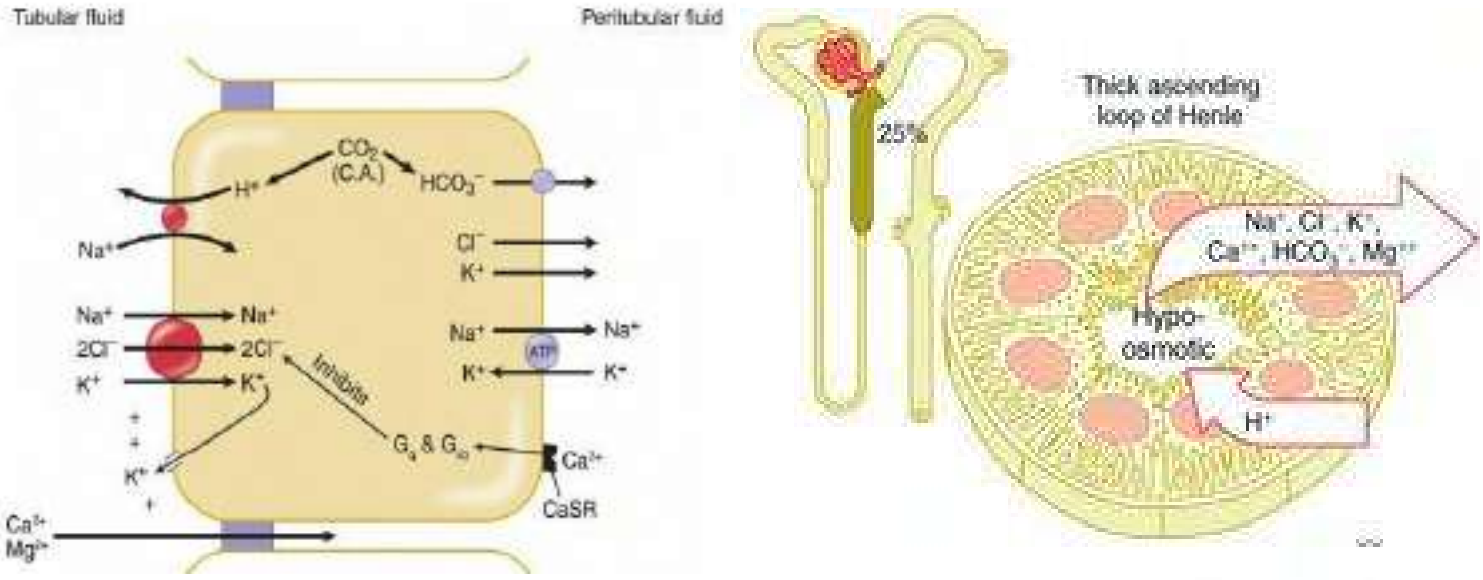
Theosmolarity of thefiltrate is increaseddue to reabsorption of water.

In the Thin As cending limb

- Impermeable to water but permeable to solutes.
- Passive reabsorption of sodium chloride occurs.

In the Thick A scending limb

is a **diluting segment** of the tubular system
- It is Impermeable to water and solutes.
- Actively reabsorbs Na + , K+ , and 2 Cl– via the **Na +–K+–2Cl– symporter.**
- The luminal membrane contains a K+ channel, allowing diffusion of this ion back into the tubular fluid. This back diffusion of K+ into the lumen creates a positive luminal potential, which in turn, promotes calcium and magnesium reabsorption via a paracellular pathway (primarily).
- Solute in this segment is transported without water resulting in a drop in the **filtrate osmolality** to low levels (**150 mOsm**) as compared to normal **plasma osmolality** of **300 mOsm**. So this segment is called the **diluting segment**, and the movement of solutes into the medullary interstitium creates the hypertonicity of tissue in this area.



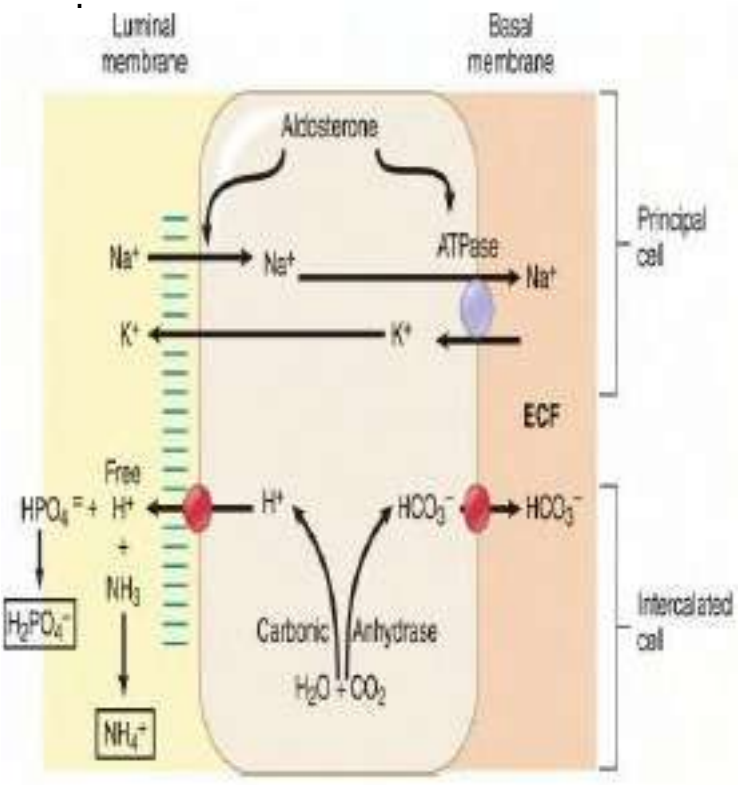
Reabsorption in thick ascending limb

Substances Reabsorbed from Distal Convoluted Tubule(DCT)

A. Early Distal Convoluted Tubule:

- This has the same characteristics as the thick ascending limbs of the LH.
- Responsible for reabsorption of 7% of sodium.
- NaCl crosses the luminal membrane via Na + - Cl -- symporter.
- The Na + is pumped across the basal membrane via the Na + /K- ATPase and Cl - diffuses down its electrochemical gradient through channels.
- This section is impermeable to water. Thus, osmolality decreases further. The ultrafiltrate in the early distal tubule has the lowest osmolality of the entire nephron

.



B. The late Distal convoluted tubule and collecting ducts:

(1)Principal cells:

Reabsorption:

a)Na reabsorption (3%): This occurs by ***primary active transport***. It is increased by the ***aldosterone*** *hormone* and is followed by *passive* reabsorption of both Cl - and water.

b) Ca ++ reabsorption: This also occurs by ***primary active transport***, and is increased by the ***parathyroid*** *hormone* (which thus increases Ca ++ blood level).

c) Water reabsorption under the effect of ADH. This hormone induces the insertion of **aquaporins** on the cell's apical surface to increase water permeability. In the absence of ADH, the late distal and collecting ducts are less permeable to water.

d) Urea reabsorption only occurs in the inner medullary collecting ducts. ADH increases urea permeability in this segment.

Secretion:

K+ : by Na **+** -K**+** pump and increased by aldosterone.

NH3: synthesis and secreted, help in acid–base balance.

2) Intercalated cells:

- *Secretion of H+* : by ***primary active transport.***

Regulation of tubular reabsorption:

Tubular reabsorption is regulated by:
1. Glomerulo-tubular balance
2. Hormonal factors
3. Nervous factors.

1. Glomerulo-tubular Balance

It is the balance between the filtration and reabsorption of solutes and water in the kidney. When GFR increases, the tubular load of solutes and water in the PCT is increased. It is followed by anincrease in the reabsorption of solutes and water.

Mechanism:

When GFRincreases, more amount of plasma proteins accumulate in the glomerulus. → the osmotic pressure increases in the peritubular capillaries increases → reabsorption of sodium and water from the tubule into the capillary blood.

.

2. Hormonal Factors.

Hormone	Action
Aldosterone	Increases sodium reabsorption in DCTs and collecting duct
Angiotensin II	Increases sodium reabsorption in PCT mainly.
ADH	Increases water reabsorption in DCT and collecting duct
ANP	Decreases sodium reabsorption
Parathormone	Increases reabsorption of calcium in DCT
Calcitonin	Decreases phosphate reabsorption in PCT
	Decreases calcium reabsorption

3.NervousFactor

Activation of the sympathetic nervous system leads to:
a) increases the tubular reabsorption (particularly of sodium) from renal tubules.
b) Stimulation of secretion of renin from juxtaglomerular cells.

Threshold substances:

Renal Threshold

is the plasmaconcentrationat which a substance appears inurine. Depending upon the degree of reabsorption, various substances are classified into:

1. High threshold substances
2. Low threshold substances
3. Non-threshold substances.

1. High-threshold Substances:

- substances that do not appear in urine under normal conditions e.g. glucose, amino acids, and vitamins. These substances are completely reabsorbed from renal tubules.
- These substances can appear in urine, only if their concentration in plasma is abnormally high or in renal diseases when reabsorption is affected.
- The renal threshold of glucose is 180 mg/dl. Glucose appears in urine when its plasma level exceeds 180 mg/dl.

2. Low-threshold Substances:

- substances thatappear inurineeven under normal conditions e.g. urea, uricacid, and phosphate.

3. Non-threshold Substances:

-substancesthat arenot reabsorbed at all and are excreted in urine irrespectiveoftheir plasma level e.g. The metabolic end products such as creatinine.

Transport maximum – Tm:

- Tubular transport maximum or Tm is the rate at which the maximum amount of a substance is transported by the renal tubule in certain time.
It is applied for every actively transported substance.

*For example, the transport maximum for glucose (TmG) is 375 mg/minute in adult males and about 300 mg/minute in adult females..

Load of substance

It is the amount of substance applied to renal tubules in one minute.

Load of substance (amount) **=** concentration of the substance in plasma X volume of filtered plasma in one minute (GFR)

Load of glucose at plasma concentration 180 mg/dl = $180 \times \frac{125}{100} = 225$ mg/min

but at this plasma concentration glucose begins to appear in urine although it is far fromTmG Because not all the 2 million nephrons have the same number of carriers and so they differ in their TmG.

Causes of glucosuria:

1. Diabetes mellitus.

2- Phloridzine glucosuria: Phloridzine blocks reabsorption of glucose as it blocksglucoseaccesstothecarrier protein → glucosuria without hyperglycaemia.

3- Alloxane glucosuria: Alloxane destroys ß cells of islets of Langerhans → hyperglycaemia → glucosuria.

4- Nephrogenic glucosuria: In some persons, the renal threshold of glucose is congenitally low, and glucose appears in urine without hyperglycaemia or symptoms of diabetes.

3 Renal Handling of important substances

1 Renal Hand ling of Sodium

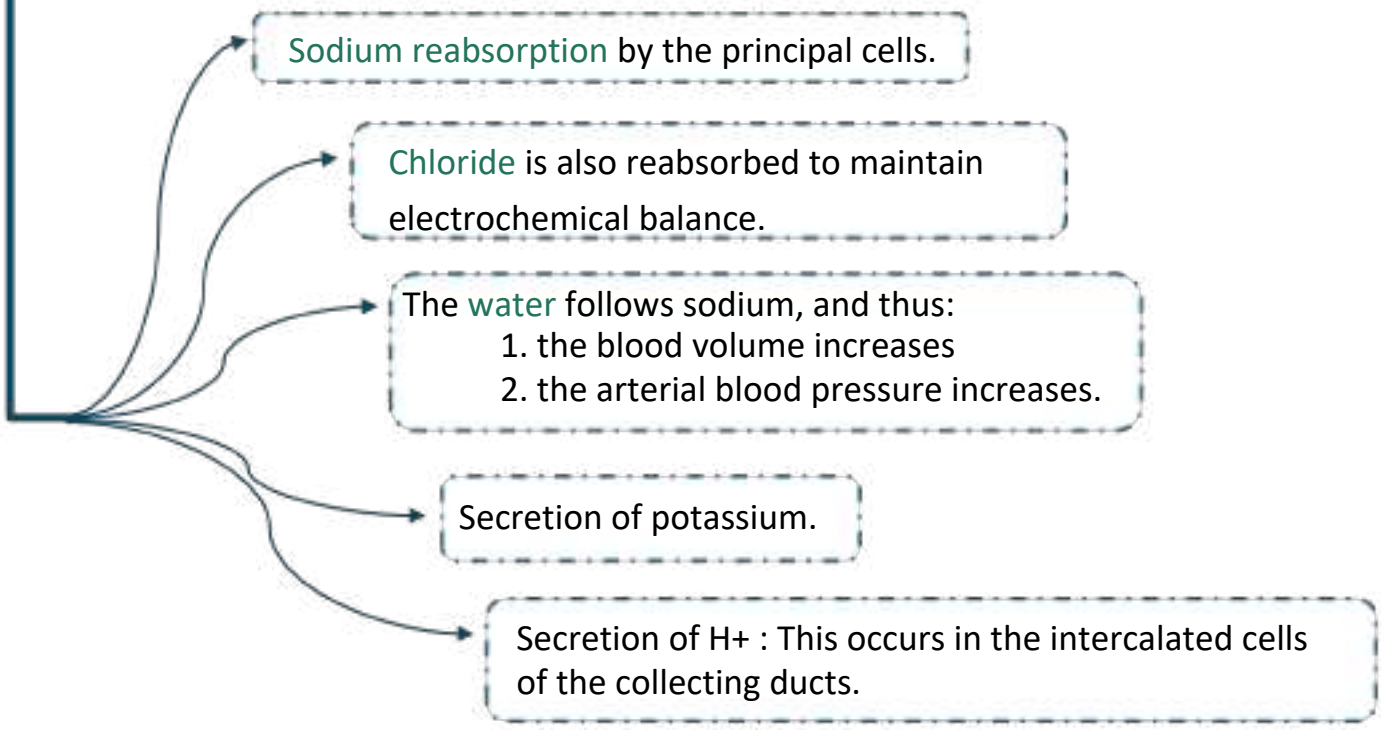
Reabsorption of sodium takes place in different portions of the nephron as follows:

- 65 % of sodium is reabsorbed in the PCTs.
- 25% are reabsorbed in the thick ascending limb of loop of Henle.
- 5% is reabsorbed in the distal tubule.
- 4 % in the collecting tubules and collecting ducts.
- 1% is excreted in urine

NOTE

97% of sodium reabsorption occurs independently from its plasma level (unregulated), This is true for sodium reabsorbed in PCT and LH, and early part of DCT, while the remaining 3% that is reabsorbed in distal ,collecting tubules and collecting ducts is regulated by Aldosterone.

Functions of aldosterone in the kidney



2 *Renal Hand ling of water*

(Urine concentration and dilution)

Daily intake of water is about 2500 ml :

1. 1500 ml as drunk water
2. 700 ml is within the food
3. About 300 ml of water is synthesized as an end-product of metabolic processes.

Water input has to equal water output. So,

1. 1500 ml is excretedinurine.
2. 900 ml of water is eliminated as insensible water loss via the respiratory system and skin.
3. 100 ml is excreted by the stool.

Every day, about 180 L of plas ma is filtered.

If this much of water is excreted in urine, body will face serious threats. So, the concentration of urine is very essential.

After being filtered in the glomerulus,

65% of water is reabsorbed in PCT.

15% in the thin descending limb of LH (obligatory water reabsorption).
The remaining in the distal nephron depending on the hydrated status of the body & controlled by ADH (facultative water reabsorption).

The kidney either excrete :

1. Concentrated urine (if the plasma is hypertonic like in dehydrated status).

OR

2. Diluted urine (if the plasma is hypotonic).

Formation of diluted urine:

When water content in the body increases, the kidney excretes diluted urine.

This is achieved by inhibition of ADH secretion.

So, ↓

Water reabsorption from DCT & CTs does not take place leading to the excretion of large amounts of water.
This makes the urine dilute.

Formation of concentrated urine

It is produced when circulating ADH levels are high (e.g., water deprivation, hemorrhage, SIADH).
Formation of concentrated urine is not as simple as that of dilute urine,

it involves two processes:

1. Development and maintenance of medullary gradient by countercurrent system
2. Secretion of ADH.

NOTE

Concentrated urine also called hyperosmotic urine,
in which urine osmolarity > blood osmolarity.

Medullary gradient (medullary hyperosmolarity)

- It plays an important role in the concentration of urine.
- Cortical interstitial fluid is isotonic to plasma with an osmolarity of 300 mOsm/L.
- Osmolarity of medullary interstitial fluid near the cortex is also 300 mOsm/L.
- However, while proceeding from the outer part towards the inner part of medulla:
    1. Osmolarity increases gradually
    2. reaches the maximum at the innermost part of the medulla near the renal sinus.

Here, the interstitial fluid is hypertonic with an osmolarity of 1,200 mOsm/L.

- This gradual increase in the osmolarity of the medullary interstitial fluid is called the **medullary gradient.**

steps of urine concentration

3. Bowman capsule:

- Glomerularfiltratecollected at the Bowman capsule is isotonic to plasma (300 mOsm/L).
- This is because it contains all the substances of plasma except proteins.

1. Proximal convoluted tubule:

- When the filtrate flows through PCT, there is active reabsorption of sodium and chloride followed by obligatory reabsorption of water.
- So, the osmolarity of fluid remains 300 mOsm/L.
- Thus -in PCT- the fluid is isotonic to plasma.

2. Descending segment of LH:

- When the fluid passes from PCT into the descending segment, water is reabsorbed from the tubule into the outer medullary interstitium by osmosis.
- The osmolarity of the fluid inside the upper part of this segment is between 450 and 600 Osm/L (slightly hypertonic to plasma).
- In the short loops of cortical nephrons, the osmolarity of fluid at the hairpin bend of LH becomes 600 mOsm/L
- but, in the long loops of juxtamedullary nephrons, at the hairpin bend, the osmolarity is 1,200 mOsm/L. Thus, in this segment, the fluid is (much hypertonic than plasma).

6. Thin ascending segment of LH:

- When the thin ascending segment of LH ascends upwards through the medullary region, osmolarity decreases gradually due to concentration gradient, sodium chloride diffuses out of tubular fluid and osmolarity decreases to 400 mOsm/L.
- The fluid in this segment is (slightly hypertonic to plasma).

5. Thick ascending segment of LH:

- This segment is impermeable to water.
- But there is active reabsorption of sodium and chloride that decreases the osmolarity of tubular fluid to a greater extent.
- The osmolarity is between 150 and 200 mOsm/L.
- The fluid inside becomes hypotonic to plasma.

4. Distal convoluted tubule and collecting duct:

- In the presence of ADH, the DCT & the P-cells of the CTs become permeable to water resulting in water reabsorption and final concentration of urine.
- This increases the osmolarity to 1,200 mOsm/L.
- The urine becomes hypertonic to plasma.

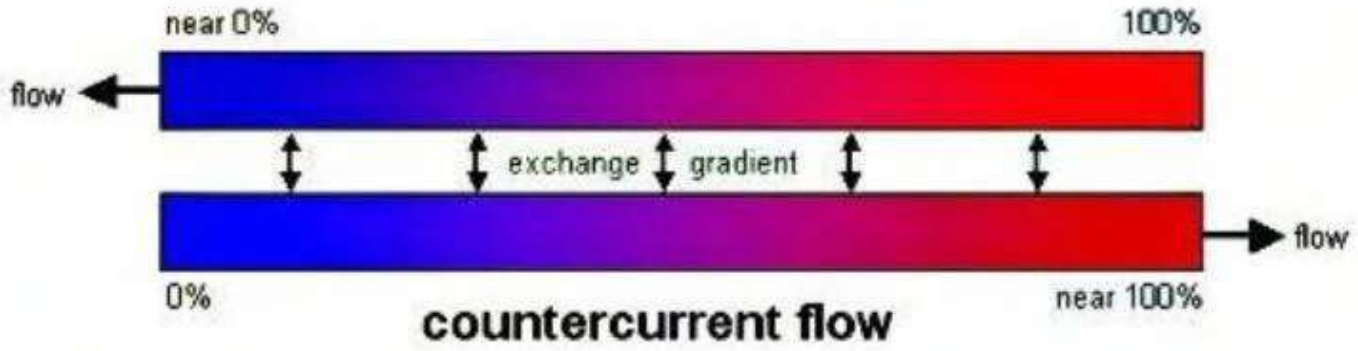
NOTE

ADHreceptors in the kidney are V2 receptors.
Binding to them ADH activates cAMP Which opens the aquaporin II channels → increase the water permeability.

3 Countercurrent mechanism (countercurrent flow)

- A system of 'U' shaped tubules (tubes) in which, the flow of fluid is in opposite direction, parallel and very close in two limbs of the 'U' shaped tubules.
- Countercurrent system has two divisions:

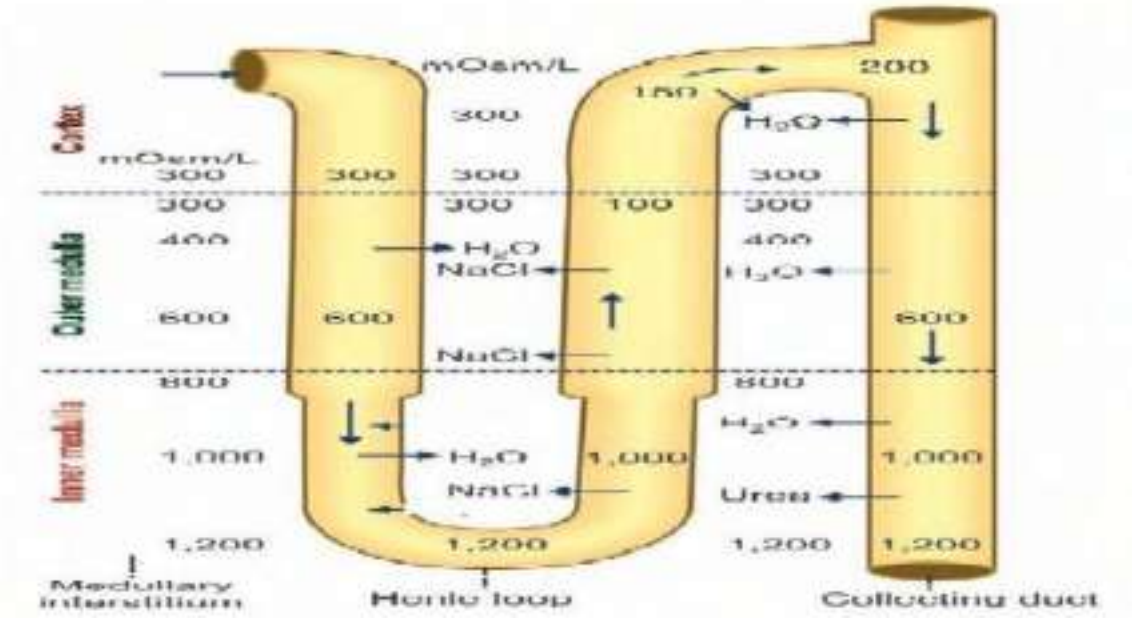
1. Countercurrent multiplier formed by loop of Henle of juxtamedullary nephrons.
2. Countercurrent exchanger formed by vasa recta.



Countercurrent multiplier

Descending limb of loop of Henle is permeable to water.
Ascending limb of LH is permeable to electrolytes, but impermeable to water. So, fluid will not follow electrolytes by osmosis. And thus, Ascending limb creates hypertonic interstitium that will attract water from descending limb.

- Sodium and chloride ions diffuse from medullary interstitium into the descending limb of LH and reach the ascending limb again via hairpin bend.
- Thus, the sodium and chloride ions are repeatedly recirculated between the descending limb and ascending limb of LH through medullary interstitial fluid leaving a small portion to be excreted in the urine.
- Apart from this, there is regular addition of more and more new sodium and chlorine ions into descending limb by constant filtration. Thus, the reabsorption of sodium chloride from ascending limb and addition of new sodium chloride ions into the filtrate increase or multiply the osmolarity of medullary interstitial fluid and medullary gradient. Hence, it is called countercurrent multiplier.

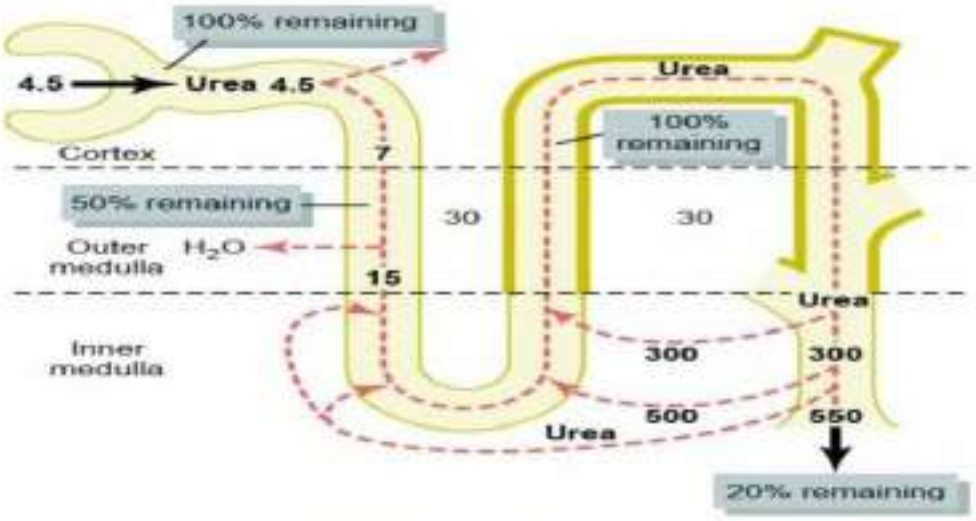


Other Factors Responsible for Hyperosmolarity of Medullary Interstitial Fluid

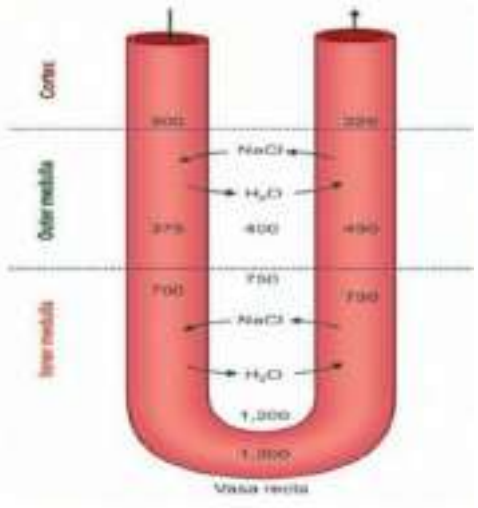
Reabsorption of sodium from medullary part of collecting duct intothemedullaryinterstitium,addstotheosmolarityofinnermedulla

Recirculation of urea :

- 50% of urea filtered in glomeruli is reabsorbed in PCT. Almost an equal amount of urea is secreted in the LH. So, the fluid in DCT has as much urea as amount filtered.
- Collecting duct is impermeable to urea. However, due to the water reabsorption from DCTs and collecting duct in the presence of ADH.
- Diffusion of urea from collecting duct into medullary interstitium is carried out by urea transporters, UTA1 and UTA3, which are activated by ADH.
- Due to continuous diffusion, the concentration of urea increases in the inner medulla resulting in hyperosmolarity of interstitium in inner medulla.
- Again, by concentration gradient, urea enters the ascending limb. From here, it passes through DCT and reaches the collecting duct. Urea enters the medullary interstitium from collecting duct. By this way urea recirculates repeatedly and helps to maintain the hyperosmolarity of inner medullary interstitium.
- Only a small amount of urea is excreted in urine.
- Urea recirculation accounts for 50% of hyperosmolarity in inner medulla.



Countercurrent exchanger (vasa recta)



- Vasa recta functions as countercurrent exchanger.
- It is responsible for the **maintenance of medullary gradient**, which is developed by countercurrent multiplier.
- Vasa recta acts like countercurrent exchanger because of its position.
- It is also 'U' shaped tubule with a descending limb, hairpin bend and an ascending limb.
- Vasa recta run parallel to LH. Its descending limb runs along the ascending limb of LH and its ascending limb runs along with descending LH.

- The sodium chloride reabsorbed from ascending limb of LH enters the medullary interstitium. From here it enters the descending limb of vasa recta. Simultaneously water diffuses from descending limb of vasa recta into medullary interstitium.
- The blood flows very slowly through vasa recta. So, a large quantity of sodium chloride accumulates in descending limb of vasa recta and flows slowly towards ascending limb. By the time the blood reaches the ascending limb of vasa recta, the concentration of sodium chloride increases very much. This causes diffusion of sodium chloride into the medullary interstitium. Simultaneously, water from medullary interstitium enters the ascending limb of vasa recta. And the cycle is repeated.
- If the vasa recta would be a straight vessel without hairpin arrangement, blood would leave the kidney quickly at renal papillary level. In that case, the blood would remove all the sodium chloride from medullary interstitium and thereby the hyperosmolarity will be decreased.
- Recycling of urea also occurs through vasa recta.

4 Renal Acid-Base Control Mechanisms

Normal pH: 7.4 with normal range (7.35-7.45)

Acidosis: A state resulting from abnormally low plasma pH (pH < 7.35)

Alkalosis: A state resulting from abnormally high plasma pH (pH > 7.45)

The kidneys are the third line of defense against wide changes in body fluid pH as their action is much slower in regulation of acid base balance. They take hours to days for correction. However, they are the most powerful and most efficient buffering mechanism.

The kidneys control acid-base balance by excreting either acidic or basic urine. Because of the daily production of metabolic acids, urinary excretion of acids is an essential process, and, as a result, the urine pH is normally slightly acidic (pH about 6.5).

The kidneys regulate blood pH through three fundamental mechanisms:

(1) Secretion of H+ .

(2) Reabsorption of filtered HCO3 .

(3) Production of new HCO3 -: which is accompanied by excretion of titratable acids (acid phosphate) or ammonia secretion.

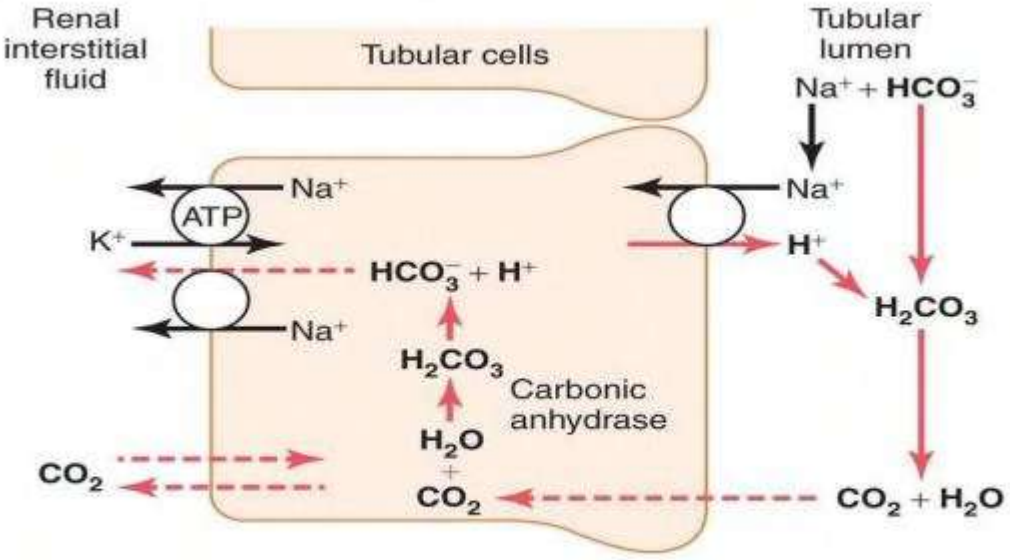
NOTE

The process of HCO3 - **reabsorption** and **excretion** are closely related to H+ secretion in the renal tubules.

1) In acidosis	2) In alkalosis
• the kidneys secrete more hydrogen and the urine becomes more acidic. It also reabsorbs all the filtered HCO3 and produce new HCO3 which is added back to the extracellular fluid. This reduces the extracellular fluid H+ concentration back toward normal.	• the kidneys don'treabsorb all the filtered HCO3 - thus increasing the excretion of HCO3

About 80 to 90 % of the HCO3 - reabsorption and H+ secretion occurs in the PCT.

Renal reabsorption of bicarbonate and secretion of H^+ in the PCT:

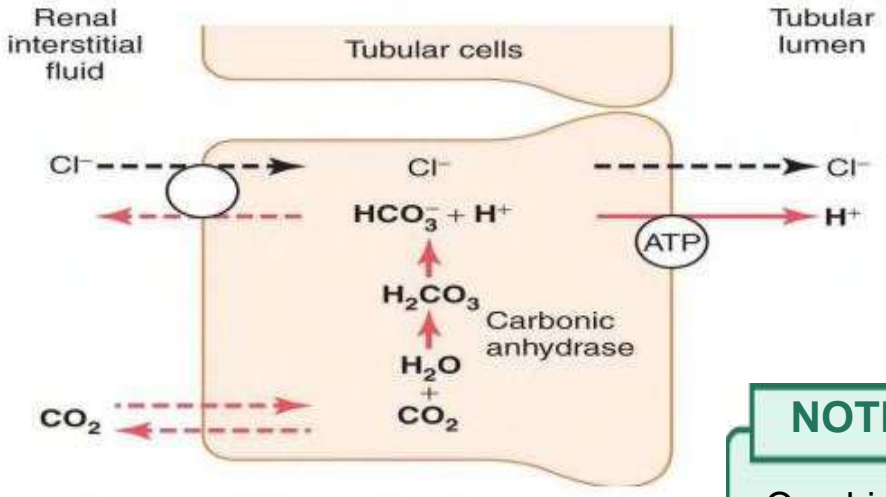


This pattern of hydrogen ion secretion occurs in the PCT, the thick ascending segment of the LH, and the early DCT.

N. B.

- The antiport utilized in PCTs can transport either Na + or K+ in exchange for H+ secretion.
- In DCTs and CDs, H+ and K+ compete for secretion.
- In PCTs, H+ can be secreted till the pH drops to about 6.9, while in the DCTs and CDs, it can be secreted till the pH drops to 4.5 .

Primary Active Secretion of H^+ in the Intercalated Cells of Late DCT and Collecting Tubules:



NOTE

One bicarbonate ion is reabsorbed for each hydrogen ion secreted, and a chloride ion is passively secreted along with the hydrogen ion

Factors that affect renal secretion of hydrogen :

1.Arterial CO2 tension: When this increases, the CO2 content in the renal tubular cellsalso increases leading to more formation of H2CO3 which, in turn, leads to moreH+ secretion.

2.K+concentration: When this increase, H+ secretion is decreased (and vice versa)since both compete for secretion in the DCTs & CCDs.

3.Carbonic anhydrase activity: Blocking of this enzyme by drugs e.g. acetazolamide (diamox) inhibits H+ secretion.

4.Aldosterone: This hormone increases both H+ and K+ secretion in exchange for Nareabsorption in the DCTs and CCDs.

5.Thetubular fluid pH: If this decreases below 4.5, H+ secretion stops.

Acid – base regulation during normal metabolism:

The products of normal metabolism are mainly acidic and include:

(a) CO2, which is mostly excreted at the lungs.
(b) Fixed acids: e.g. sulphuric and phosphoric acids. These are mostly derived from amino acid metabolism, and they are the only source of excess H+ during normal metabolism.

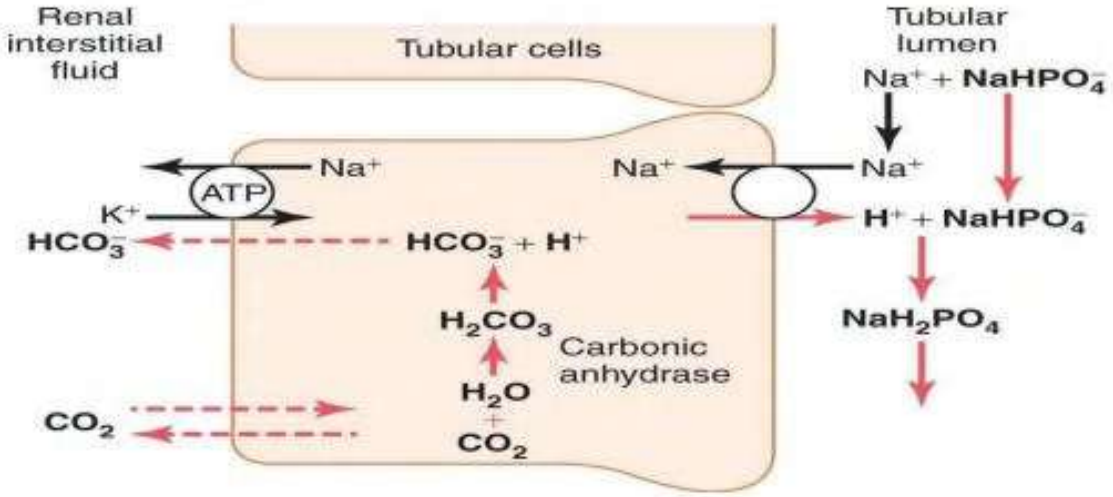
- They are immediately buffered mainly by the bicarbonate buf er in the blood.
- CO2 is mostly excreted at the lungs but it is also increased in the renal tubular cells leading to H+ secretion, which is accompanied by HCO3 - reabsorption.

Normally, the amount of H+ secreted in the renal tubules slightly exceeds that of the filtered HCO3 -, and about 90 % of H+ secretion occurs in the PCTs. This amount is buffered by the filtered HCO3 - , resulting in complete reabsorption of bicarbonate which conserves the alkali reserve of the body.

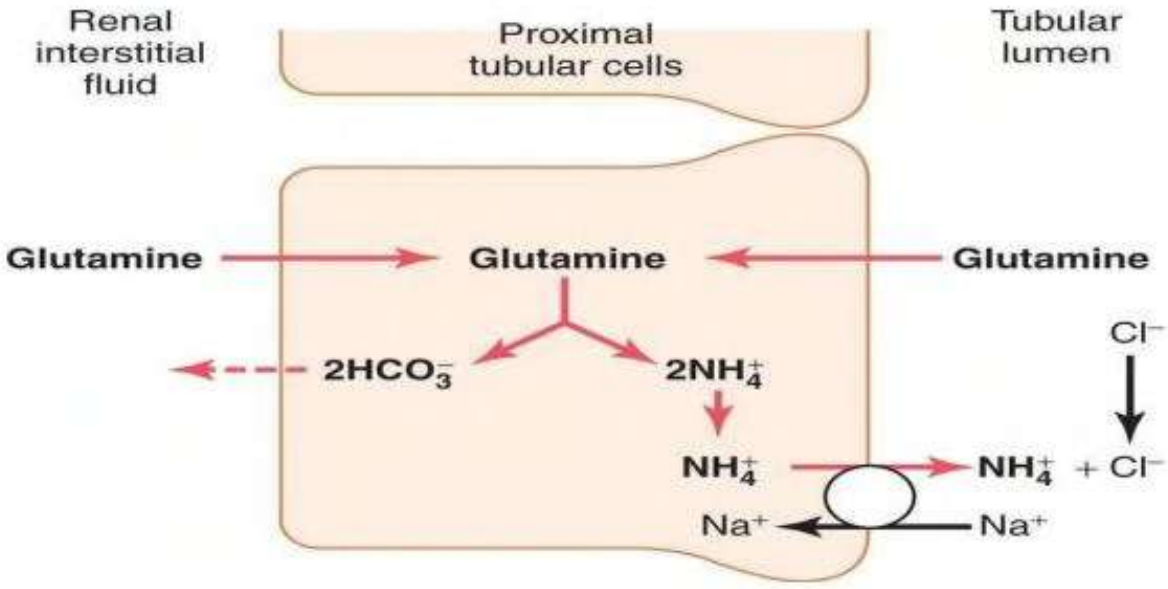
On the other hand, the remaining 10 % of H+ secretion occurs in the distal renal segments, and is buf ered by phosphate and ammonia resulting in 2 effects:

(1) Formation & reabsorption of a new amount of HCO3 - which compensates what has been used during the buffering process. Accordingly, the blood HCO3 - content increases, so the ratio (R) rises to its normal value (about 20) and the blood pH will thus be corrected and kept close to 7.4.
(2) Formation and excretion of acid phosphate & ammonium salts leading to urine acidification (so the normal urine pH is about 6).

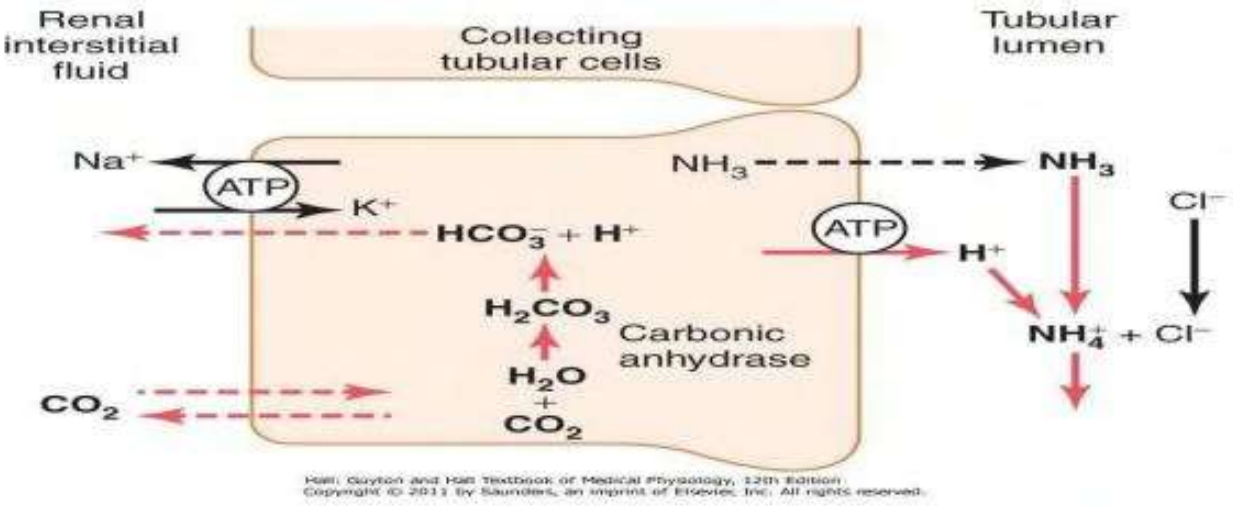
Buffering of Secreted Hydrogen Ions by Filtered Phosphate (excretion of titratable acids)



Ammonia secretion in PCT:



Buffering of hydrogen ion secretion by ammonia (NH_3) in the collecting tubules:



Renal Correction of Acidosis-Increased Excretion of H^+ and Addition of Bicarbonate to the ECF

Acidosis decreases the ratio of Bicarbonate/Hydrogen ion in Renal Tubular Fluid. As a result, there is excess H + in the renal tubules, causing more hydrogen secretion, complete reabsorption of bicarbonate and still leaving additional H+ available to combine with the urinary buffers (phosphate and ammonia) that increases the formation of new bicarbonate .

Renal Correction of Alkalosis

- In cases of alkalosis, the acid-base balance is regulated by excreting alkaline urine .
- more amounts of HCO3- than normal are filtered in the renal glomeruli while less H+ is secreted in the renal tubules. Therefore, the filtered HCO3- is not completely reabsorbed in the PCTs and the excess will be excreted in the urine.
- urine alkalinization begins in the PCTs (as urine acidification in acidosis) and the urine pH may rise up to 8.

Four basic types of imbalance :

1. Respiratory acidosis (CO_2 is increased) □
2. Metabolic acidosis: addition of H^+ (not of CO_2 origin) and/or loss of bicarbonate from the body. □
3. Respiratory alkalosis: CO_2 is decreased □
4. Metabolic alkalosis: loss of H^+ (not of CO_2 origin) and/or addition of base to the body .

1) *Respiratory Acidosis*

PH ↓, PCO2 ↑

Caused by respiratory hypoventilation as lung, bronchial or pleural diseases
Correction (compensation) of respiratory acidosis: This occurs by renal correction only. The kidneys compensate by increasing HCO_3^- and eliminating H^+ .

2) *Metabolic Acidosis*

pH ↓, HCO3 ↓

Thedrop in pH stimulates ventilation via peripheral chemoreceptors (respiratory compensation).

Causes:

1- Renal failure.
2- Diabetic ketoacidosis.
3- Intake of acids.
4- Loss of alkali as in diarrhea
5- Addison disease.
6- Dehydration.

3) Respiratory Alkalosis

pH ↑,PCO2 ↓

Caused by: respiratory hyperventilation either voluntary or in high altitude for example.

The kidneys compensate by eliminating HCO3 – and conserving H+ .

4) Metabolic Alkalosis

pH ↑, HCO3- ↑

casuses:

1- Repeated vomiting
2- Diuretics
3- Intake of alkali
4- Hyperaldosteronism

Table 4.3 Summary of 'simple' acid–base disturbances

Disorder	*pH*	*Primary change*	*Compensatory response*
Metabolic acidosis	Decreased	Decreased HCO_3	Decreased pCO_2
Metabolic alkalosis	Increased	Increased HCO_3	Increased pCO_2
Respiratory acidosis	Decreased	Increased pCO_2	Increased HCO_3
Respiratory alkalosis	Increased	Decreased pCO_2	Decreased HCO_3

5 Micturition and Micturition reflex

- Micturition (urination) is a process, by which the final urine is eliminated out of the body.
- After being drained into the ureters, urine is stored in the urinary bladder until being eliminated.
- The ureters enter the bladder wall obliquely and thus do not have a sphincter. Contraction of the bladder wall prevents urine reflux back to the ureters.
- There are two phases of bladder function that depend on characteristics of its **muscular wall** and **innervation**:

1) Bladder filling:

- ✓ Urine is poured into the bladder through the orifices of ureters.
- ✓ The ureter has regular one to five peristaltic contractions per minute. These contractions facilitate the moving of urine from the ureter to the bladder to prevent reflux of urine into the ureter.
- ✓ The capacity of the bladder is about 400 ml.
- ✓ When the bladder starts filling its wall extends and thus the pressure is not increased with the increased urine volume.

2) Bladder emptying:

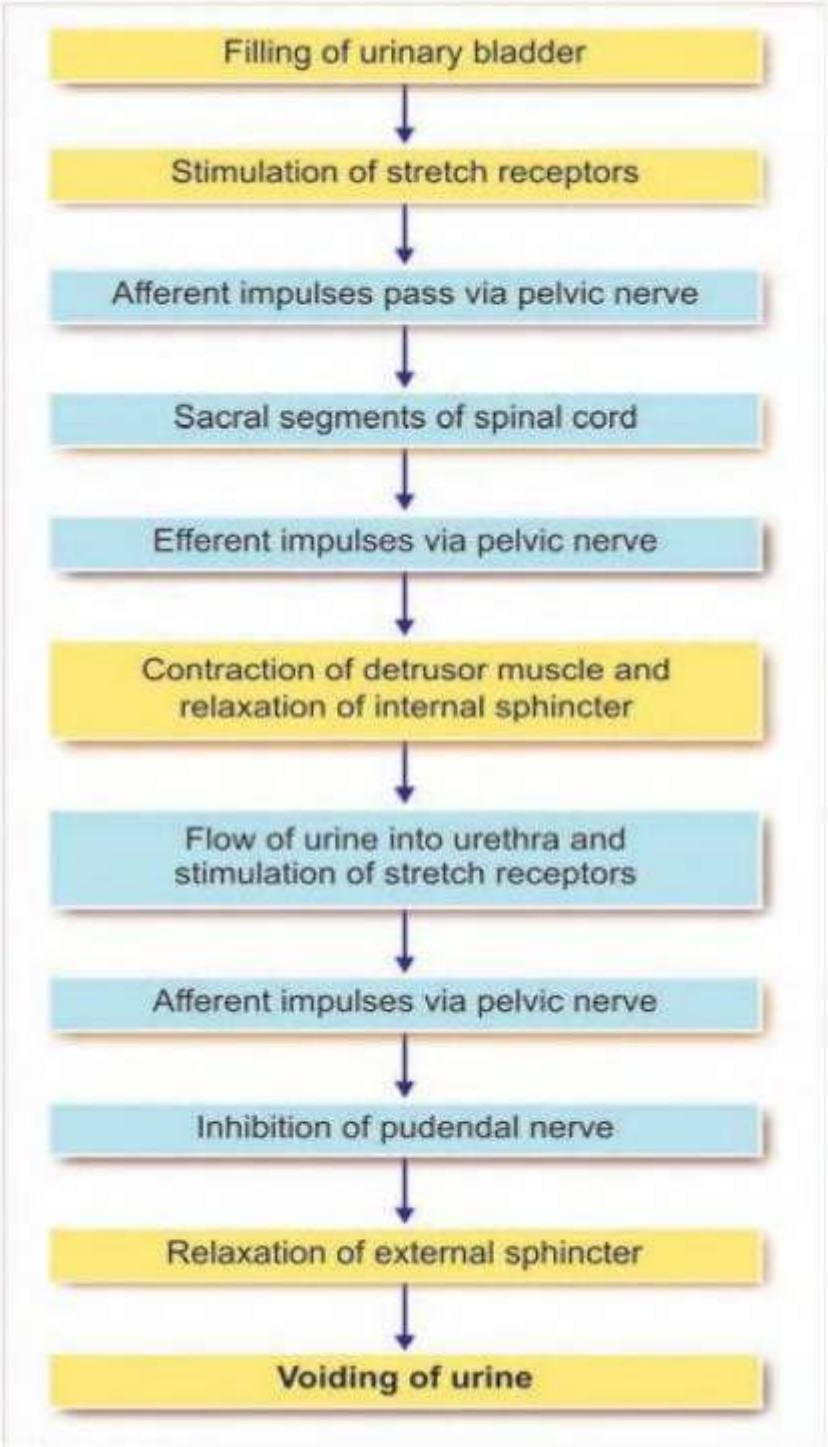
- ✓ When bladder is full stretch receptors in bladder wall are excited, and send signals via the sensory branches of pelvic nerves (parasympathetic) to the sacral plexus. (The first urge to void is felt at a bladder volume of about 150 ml.)
- ✓ In sacral portion of spinal cord, a motor signal is sent to the urinary bladder muscles and internal urethral sphincter through the efferent (motor) branches of pelvic nerve itself.

- ✓ Motor impulses cause contraction of detrusor muscle and relaxation of internal sphincter so that, urine enters the urethra from the bladder.
- ✓ Once urine enters urethra, the stretch receptors in the urethra are stimulated and send afferent impulses to spinal cord via pelvic nerve fibers.
- ✓ Now the impulses generated from spinal centers inhibit pudendal nerve. So, the external sphincter relaxes and micturition occurs.
- ✓ Once a micturition reflex begins, it is self-regenerative. The cycle continues repeatedly until the force of contraction of bladder reaches the maximum and the urine is voided out completely.
- ✓ During micturition, the flow of urine is facilitated by the increase in the abdominal pressure due to the voluntary contraction of abdominal muscles.

Supraspinal control of micturition :

- ✓ In adult people the neurons in sacral portion is influenced by nerve signals coming from brain (Micturition center in pons) that are also influenced by signals coming from cerebral cortex.

  So: The sensory signals, transmitted to the sacral region will also stimulate ascending pathway and the signals be also transmitted to the micturition center in the brain stem and then to the cerebrum to cause conscious desire for urination.
- ✓ If micturition is not convenient the brain sends signals to inhibit the parasympathetic motor neuron to the bladder via the sacral neurons. It also sends inhibitory signal via the somatomotor pudendal nerve to keep external urinary sphincter contracting.
- ✓ When micturition is convenient a brain signal via the sacral neurons stimulates the parasympathetic pelvic nerve to cause contraction of detrusor muscle via M-cholinergic receptors and causes relaxation of external urinary sphincter and the micturition occurs.
- ✓ Sympathetic hypogastric nerve does not contribute that much to the micturition reflex. It plays role in preventing reflux of semen into urinary bladder during ejaculation by contracting bladder muscles.


Filling of urinary bladder
Stimulation of stretch receptors
Afferent impulses pass via pelvic nerve
Sacral segments of spinal cord
Efferent impulses via pelvic nerve
Contraction of detrusor muscle and relaxation of internal sphincter
Flow of urine into urethra and stimulation of stretch receptors
Afferent impulses via pelvic nerve
Inhibition of pudendal nerve
Relaxation of external sphincter
Voiding of urine

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Renal System

Biochemistry

RNS-315

Page	Lecture
1	Urinary stones

1 Urinary Stones

What are urinary stones?

✓Renal calculi are formed in renal tubules, ureter or bladder .
✓Composed of metabolic products present in glomerular filtrate .
These products are in high concentration, near or above maximum solubility .

Conditions causing urinary stone formation:

➢ Highconcentration of metabolicproductsinglomerular filtrate .
➢ Changes in urine pH: The formation of various types of kidney stones is strongly influenced by urinary pH. An alkaline pH favors the crystallization of calcium- and phosphate-containing stones, whereas and acidic urine pH promotes uric acid or cystine stones.
➢ Urinary stagnation .
➢ Deficiency of stone-forming inhibitors in urine .

High conc. of metabolic products in glomerular filtrate is due to:

- Lowurinaryvolume (with normal renalfunction)duetorestrictedfluidintake .
- Increased fluid loss from the body .
- Increased excretion of metabolic products forming stones .

Changes in urine pH due to:

- Bacterialinfection
- Precipitation of salts at different pH

Urinary stagnation is due to:

- Obstructionofurinaryflow .

Deficiency of stone-forming inhibitors:

- Citrate, pyrophosphate, glycoproteinsinhibitgrowth of calcium phosphate and calcium oxalate crystals

Types of urinary stones :

- Calcium salts
- Uric acid
- Mg ammonium PO4
- Cystine
- Other (xanthine, etc.)

1 Calcium salt stones



80% of kidney stones contain calcium

The type of salt depends on :

– Urine pH (Calcium salt stones are formed in alkaline urine)

– Availability of oxalate

➢ General appearance:

–White, hard, radioopaque
–Calcium PO4: staghorn in renal pelvis (large)
–Calcium oxalate: present in ureter (small)



Calcium oxalate stones

➢Causes of calcium salt stones:

✓ **Hypercalciuria:**

–Increased urinarycalciumexcretion
–Men: > 7.5 mmols/day
–Women > 6.2 mmols/day
–May or may not beduetohypercalcemia .

✓ **Hyperoxaluria:**

–Causes the formationofcalcium oxalates without hypercalciuria
–Diet rich in oxalates
–Increased oxalateabsorptionin fat malabsorption .

✓ **Primary hyperoxaluria:**

–Due to inborn errors
–Urinary oxalate excretion:>400 mmols/day.

■ Treatment:

– Treatment of primary causes such as infection, hypercalcemia, hyperoxaluria
– Oxalate-restricted diet
– Increased fluid intake
– Acidification of urine (by dietary changes).

2 Uric acid stones



- About 8% of renal stones contain uric acid
- May be associated with hyperuricemia (with or without gout)
- Form in acidic urine

- **General appearance:**
  - ➢ – Small, friable, yellowish – May form
  - ➢ staghorn – Radiolucent (plain x-rayscannot
  - ➢ detect) – Visualized by ultrasoundor i.v.
  - ➢ pyelogram
- **Treatment:**



–Purine-restricted diet
–Alkalinization of urine (by dietary changes)
–Increased fluid intake

3 Mg ammonium PO4 stones

- About 10% of all renal stones contain Mg amm. PO4
- Also called struvite kidney stones
- Associated with chronic urinary tract infection

– Microorganisms (such as from Proteus genus) that metabolize urea into ammonia .
– Causing urine pH to become alkaline and stone formation .

- Commonly associated with staghorn calculi
- 75% of staghorn stones are of struvite type .

- **Treatment:**

–Treatment ofinfection
–Urine acidification
–Increased fluid intake



Mg ammonium phosphate (struvite) stone

4 Cystine stones



- A rare type of kidney stone
- Due to cystinuria Formed in
- acidic urine Soluble in
- alkaline urine Faint radio-
- opaque
- **Treatment:**
  - – Increased fluid intake
  - – Alkalinization of urine (by dietary changes)
  - – Penicillamine (binds to cysteine to form a compound more soluble than cystine).

Laboratory investigations of urinary stones:

If stone has formed and removed:

➢ Chemical analysis of stone helpsto:
- – Identify the cause .
- – Advise patient on preventionandfuture recurrence.

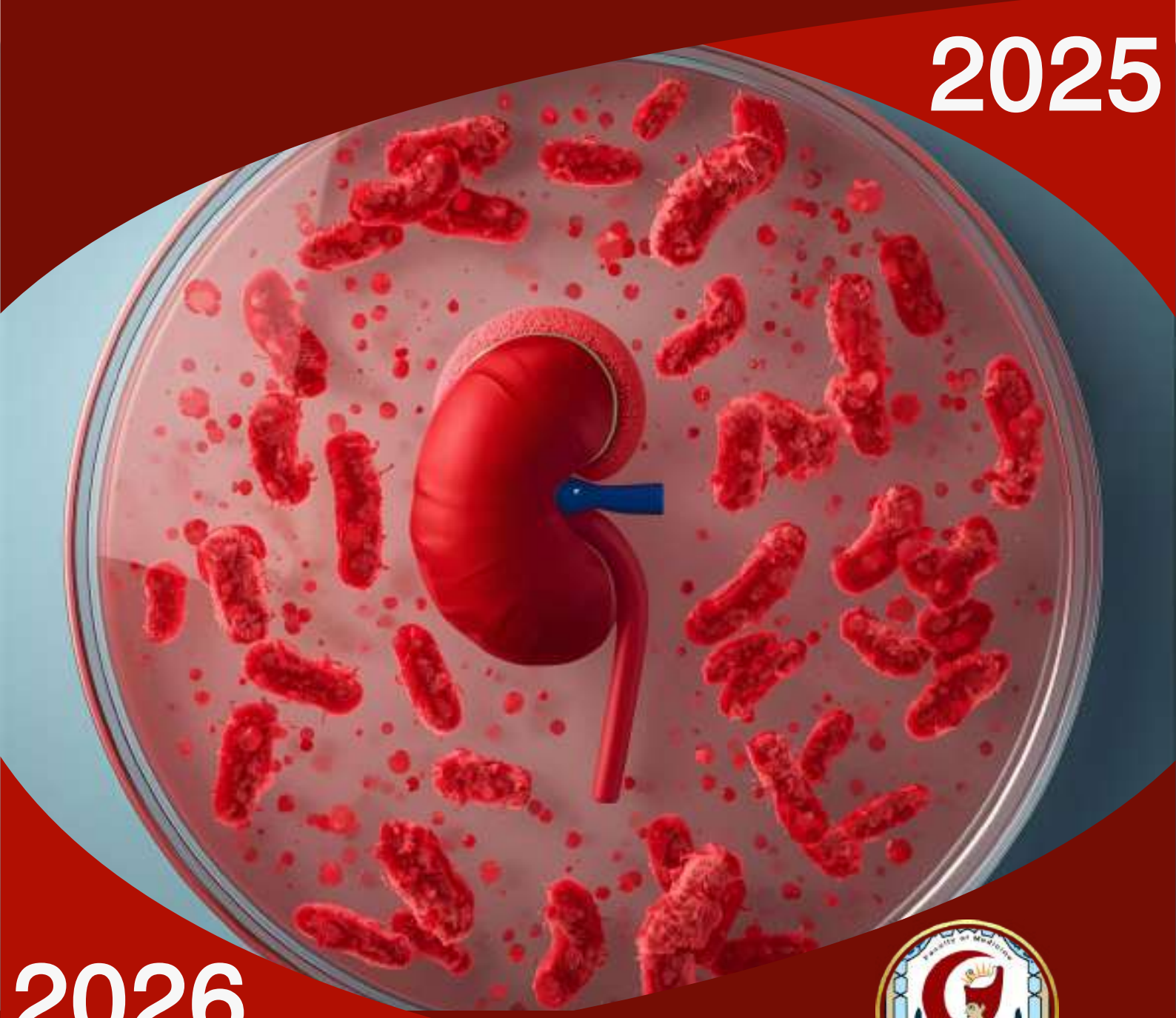
➢ If stone has not formed :

■ This type of investigation identifies causes that may contribute to stone formation .

– Serum calcium and uric acid analysis .
– Urinalysis: volume, calcium, oxalates and cystine levels .
– Urine pH > 8 suggests urinary tract infection (Mg amm. PO4) .

■ Urinary tract imaging:
– Ultrasound and i.v. pyelogram

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Renal System

Microbiology & immunity

RNS-315

Page	Lecture
1	Urinary tract infection (UTI)

1 Urinary tract infection (UTI)

Intended Learning Outcomes

1. Definitions
2. Risk factors for UTI
3. Classification of UTI
4. Pathogenesis and microbiology of UTI
5. Clinical manifestations and complications of UTI
6. Diagnosis and treatment of UTI

1 Definitions

- Urinary tract infection is an infection in any part of the urinary system; kidneys, ureters, bladder and urethra. Most infections involve the lower urinary tract; the bladder and the urethra
- Women are at greater risk of developing UTI than are men due to 3 features facilitate ascending infection to the bladder; short urethera, proximity of the urethera to the anus, and colonization of the vagina by members of faecal flora

Urethritis

- infection of anterior urethral tract

Cystitis

- infection to urinary bladder

Acute pyelonephritis

- infection of one/both kidneys

Chronic pyelonephritis

- particular type of pathology of kidney; may/may not be due to infection

Uncomplicated UTI

- UTI without underlying renal or neurologic disease. It is usually considered to be cystitis or pyelonephritis that occurs in premenopausal non-pregnant adult women with no structural or functional abnormality of the urinary tract and have no significant co- morbidity that could lead to more serious outcomes

Complicated UTI

- can involve either sex at any age

UTI is considered complicated if:

1. The patient is a child or pregnant
2. The patient has any of the following:

- A structural or functional urinary tract abnormality and obstruction of urine flow
- A co-morbidity that increases risk of acquiring infection or resistance to treatment, such as poorly controlled diabetes, chronic kidney disease, or immune-compromised
- Recent instrumentation or surgery of the urinary tract

Recurrent UTI

- >3 symptomatic UTIs within 12 months following clinical therapy

Reinfection

- recurrent UTI caused by a different pathogen at any time

Relapse

- recurrent UTI caused by same species causing original UTI within 2 weeks after therapy

Significant bacteriuria

- presence of ≥ 105 CFU/ml of urine

Asymptomatic bacteriuria

- bacteriuria with no symptoms

Pyuria

- is the presence of pus in urine (more than 10 cells/HPF). It usually accompanies bacteriuria

2 Risk factors for UTIs

- Anything that disrupts normal urine flow or complete emptying of the bladder or facilitates access of organisms to the bladder will predispose an individual to infection

1-The shorter female urethra is a less effective deterrent to infection than the male urethra

2-Sexual intercourse facilitates the movement of organisms up the urethra, particularly in females, so the incidence of UTI is higher among sexually active women than among celibate women

3-Preceding bacterial colonization of the peri-urethral area of the vagina is perhaps important

4-Inmale infants, UTIs are more common in the uncircumcised, and this is associated with colonization of the inside of the prepuce and urethra with fecal organisms

5-Pregnancy, prostatic hypertrophy, renal calculi, tumors and strictures are the main causes of obstruction to complete bladder emptying

6-Loss of neurological control of the bladder and sphincters (e.g. in spina bifida, paraplegia or multiple sclerosis), andthe resultant large residual volume of urine in the bladder, causes a functional obstruction to urine flow, and suchpatients are particularly prone to recurrent infections

7-Vesico-ureteric reflux (reflux of urine from the bladder cavity up the ureters, sometimes into the renal pelvis or parenchyma) is common in children with anatomical abnormalities of the urinary tract and may predispose to ascending infection and kidney damage. Reflux may also occur in association with infection in children without underlying abnormalities, but tends to disappear with age

8-People with diabetes mellitus may have more severe UTIs, and if diabetic neuropathy interferes with norma l bladder function then persistent UTIs are common

9-Catheterization is a major predisposing factor for UTI. During insertion of the catheter, bacteria may be carried directly into the bladder and, while in situ, the catheter facilitates bacterial access to the bladder either via the lumen of the catheter or by tracking up between the outside of the catheter and the urethral wall

3 Classification of UTIs

1-Upper UTI

- pyelonephritis, intra-renal abscess and perinephric abscess (usually late complications of pyelonephritis)

2-Lower UTI

- cystitis, urethritis and prostatitis
- Both upper & lower UTI are further divided into complicated and uncomplicated
- UTIs occur in two general settings: **community acquired** and **hospital (nosocomially) acquired**, the latter most often being associated with catheterization. Hospital-acquired UTIs, although less common than community acquired, contribute significantly about 40% to overall nosocomial infection rates

4 Pathogenesis of UTI

Routes of bacterial entry to the urinary tract

1)Ascending infection

- most common route. Organisms ascend through urethra into bladder

2)Blood borne spread

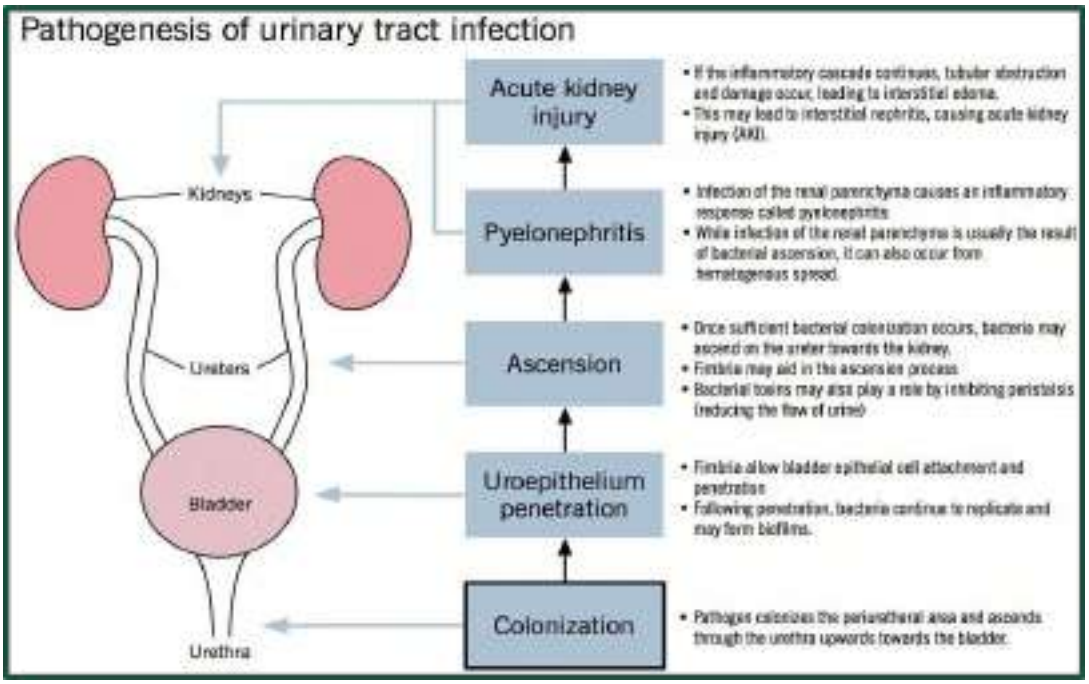
- Blood borne spread to kidneys. It occurs in bacteraemia mostly by S.aureus

3)Lymphatogenous spread

- Men- through rectal and colonic lymphatic vessels to prostate and bladder
- Women- through periuterine lymphatics to urinary tract

4)Direct extension from other organs

- Pelvic inflammatory diseases
- Genito-urinary tract fistulas



Acquisition and etiology

Infectious causes of urethritis

1-Gonococcal urethritis

- due to Neisseria gonorrhoea (**the most common cause**)

2-Non-gonococcal urethritis

- due to Chlamydia trachomatis (serotypes D-K)
- Ureaplasma urealyticum
- Mycoplasma hominis

3-Pyogenic bacteria

- as *E. coli*, Staph. and Enterococci
- They usually occur in the presence of uretheral stricture or post- traumatic following catheterization, instrumentation or foreign body insertion

Infectious causes of cystitis and pyelonephritis

1-Gram-negative aerobic bacteria (most often)

- Escherichia coli; 75 to 95% of cases, Klebsiella, Proteus mirabilis and Pseudomonas aeruginosa

2-Gram-positive bacteria (less often)

- Staphylococcus saprophyticus is isolated in 5 to 10% of bacterial UTIs, Enterococcus **faecalis and Staph. aureus**

3-Mycobacterium tuberculosis

4-Adenoviruses

- acute haemorrhagic cystitis in children

5-Candida albicans

Bacterial infection is usually acquired by the ascending route from the urethra to the bladder

Escherichia coli

- is the commonest cause of ascending UTI. In hospitalized patients, *E. coli* accounts for about 50% of cases. Source of infection is the patient own colonic flora that colonize the urogenital area. Certain O serotypes of *E. coli* cause UTI called uropathic strains characterized by:

1. Pili with adhesin proteins bind to specific receptors on urinary tract epithelium
2. Motility of E. coli aid to ascend through urinary tract

Proteus mirabilis

- is often associated with urinary stones (calculi), probably because this organism produces a potent urease enzyme, which acts on urea to produce ammonia, rendering the urine alkaline which favors the growth of organisms. Vigorous motility of the organism may contribute to their ability to invade the urinary tract

Klebsiella, Enterobacter, Proteus and Pseudomonas aeruginosa

- are more frequently found in hospital-acquired UTI because of their resistance to antibiotics

Staphylococcus saprophyticus (coagulase-negative staph) is an opportunistic pathogen, found primarily on the mucosa of the genital tract in young women and from that site can ascend into the urinary bladder to cause urinary tract infections. This organism is **second** to Escherichia coli as a cause of community-acquired UTI, particularly in **sexually active young women**. Infection usually occurs 24 hr after intercourse

5 Clinical manifestations of UTIs

- depend on the site of infection and age of the patients

1-Acute infections of the lower urinary tract

characterized by a rapid onset of:

- dysuria (burning pain on passing urine)
- urgency (the urgent need to pass urine)
- frequency of micturition
- However, UTIs in the elderly and those with indwelling catheters are usually asymptomatic
- Acute bacterial prostatitis causes systemic symptoms (fever) and local symptoms (perineal and low back pain, dysuria and frequency)

2-Pyelonephritis

causes fever and lower urinary tract symptoms. Asymptomatic infection can be detected in:

- Pregnant women and young children, where failure to treat may result in chronic renal damage
- People undergoing instrumentation of the urinary tract, in whom bacteriuria may proceed to bacteraemia
- The elderly and those with diabetes (both risk factors for asymptomatic bacteriuria)

Complications of UTI

Recurrent infections

- >3 symptomatic UTIs within 12 months following clinical therapy

Permanent kidney damage

- from an acute or chronic kidney infection (pyelonephritis) due to an untreated UTI

Increased risk in pregnant

- Increased risk in pregnant women of delivering low birth weight or premature infants

Urethral narrowing (stricture)

- inmen from recurrent urethritis, previously seen with gonococcal urethritis

Sepsis

- apotentially life-threatening complication of an infection, especially if the infection ascends from the lower urinary tract up to the kidneys

6 Laboratory diagnosis

Specimen

- A "**mid stream**" urine sample must be obtained under aseptic precautions, which include cleaning the external genitalia with tap water and drying, and then the middle sample of urine is collected in a sterile container
- In babies the urine is collected in sterile self-adhesive plastic bags. However, suprapubic needle aspiration of the full bladder may be necessary in infants and in pregnancy
- Catheter samples should be avoided as they carry the risk of causing ascending infection, unless the patient is already catheterized

Transport

- Samples should reach the laboratory within one hour after voiding or kept refrigerated at 4°C for 24 hours maximally, to avoid multiplication of bacteria in urine resulting in false high bacterial counts

Urine testing

1-Dipstick test

- which is a basic rapid diagnostic tool used to determine pathological changes in patient's urine
- Nitrate positive: is highly specific for UTI, but the test is of low sensitivity as it does not exclude infection
- The leukocyte esterase test: is very specific for the presence of > 10 WBCs/µL of urine and is fairly sensitive

2-Microscopic examination

- a wet film of urine deposit for the presence of pus cells, RBCs, bilharzial ova, casts or crystals
- **Pyuria**: most truly infected patients have >10 WBCs/µL. The presence ofbacteria in the absence of pyuria is due to contamination during sampling
- Microscopic hematuria (>4 RBCs/HPF) occurs in up to 50% of patients, but gross hematuria is uncommon

3-Viable bacterial count

is done within one hour by using a calibrated loop that carries a known volume of the uncentrifuged urine. It is spread on solid media and incubated at 37°C. The number of bacteria is estimated and interpreted as follows:

- ≥ 105 CFU/ml of urine indicates urinary tract infection
- Less than 10,000 bacteria per ml is regarded as contamination
- A count of 10,000 is considered significant if the organism is gram positive and only one type

4-Cultures

- are made on nutrient agar, blood agar and MacConkey's medium or CLED agar and incubated overnight at 37°C. The colonies are identified by morphology and biochemical reactions

E. coli is gram-negative bacilli. It is motile due to peritrichous flagella

- On MacConkey agar: pink colonies due to lactose fermentation
- On Eosin methylene blue (EMB): small, dry, dark colonies with green metallic sheen
- On CLED agar: opaque yellow colonies with a slightly deeper yellow center
- Biochemical testing: They ferment glucose, lactose, maltose, mannite, sucrose and salicin with production of acid and gas. They are indole positive, M.R. positive, V.P. negative and citrate negative (IMVC ++--)

➢ **Klebsiella** is gram negative, non-motile, capsulated bacilli
- On MacConkeyagar: pink colonies. Colonies are mucoid due to the production of abundant extracellular slime
- On EMB: large pale pink mucoid colonies due to abundant extracellular slime
- **Biochemical activities**: They ferment glucose, lactose, maltose, mannite, sucrose and salicin with production of acid and gas. They are indole negative, M.R. negative, V.P. positive and citrate positive (IMVC --++)

➢ **Proteus** is Gram negative bacilli, very pleomorphic and highly motile
- On nutrient agar; they give colonies which swarm in successive waves due to their high motility (fishy odour)
- On MacConkey, they produce pale non-lactose fermenting colonies
- **Biochemical activities**: They are phenylalanine deaminase positive and urease positive. They produce H2S which blackens the butt of triple sugar iron (TSI) agar

➢ **Pseudomonas aeruginosa** is Gram negative motile bacilli
- On nutrient agar leading to greenish colouration of the medium due to its diffusible exopigment which consists of pyocyanin (blue) and pyoverdin (yellow-green fluorescent). Cultures have a sweet grape-like odour
- **Biochemical activities**: P. aeruginosa is oxidase positive and does not ferment any sugar. Acid is produced from glucose by oxidation only

➢ **Enterococcus faecalis** is Gram positive cocci occur singly, in pairs and in short chains
- Grow in broth containing 6.5% NaCl
- On Bile-esculin agar: it can grow in presence of bile and hydrolyze esculin and turns the medium dark brown (bile- esculin-positive)
- **Biochemical activities:** Catalase negative

➢ **Staphylococcus saprophyticus** is Gram positive cocci arranged in grape-like clusters
- On blood agar, yield non-hemolytic white colonies
- **Biochemical activities**: Coagulase negative, catalase positive and novobiocin-resistant

-Antibiotic sensitivity test is done for the isolated organism

-When **TB** is suspected, five successive morning urine samples are examined by Z.N. stain and cultured on L.J. medium

TREATMENT

Urethritis

- Sexually active patients with symptoms are usually treated presumptively for STDs pending test results. A typical regimen is ceftriaxone 250 mg IM plus either azithromycin 1 gm once or doxycycline 100 mg bid for 7 days

Cystitis

- First-line treatment of uncomplicated cystitis is nitrofurantoin 100 mg bid for 3 days (it is contraindicated if creatinine clearance is < 60 mL/min), trimethoprim/sulfamethoxazole (TMP/SMX) 160/800 mg bid for 3 days

Acute pyelonephritis

➢ Antibiotics are required. Outpatient treatment with oral antibiotics is possible if all of the following criteria are satisfied:

- Patients are expected to be adherent
- Patients are immunocompetent
- Patients have no nausea or vomiting or evidence of volume depletion or septicemia
- Patients have no factors suggesting complicated UTI
- Ciprofloxacin 500 mg bid for 7 days
- A 2nd option is usually trimethoprim/sulfamethoxazole (TMP/SMX) 160/800 mg bid for 14 days

PREVENTION

- Recurrent infections in otherwise healthy women can be prevented by regularly emptying the bladder. This washes bacteria out of the urinary tract and is particularly important following intercourse
- The prophylactic use of antibiotics may also prevent recurrent infections, but in the presence of underlying abnormalities, there is a tendency to select antibiotic-resistant strains, which subsequently cause infections that are more difficult to treat
- Infection in catheterized patients is very common, but can be reduced by good catheter care procedures. Catheterization should be avoided if possible or kept to a minimum duration

Sterile pyuria

➢ It means the presence ofpus in urine, in the absence of bacterial growth on ordinary media. This can be due to:

1. Renal tuberculosis
2. Takingantibioticsthatsuppressgrowth oforganisms
3. Anaerobes, mycoplasma, ureaplasma, chlamydia or viruses

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Renal System

Parasitology

RNS-315

Page	Lecture
1	Parasitic infection of the renal system

1 Parasites Affecting Renal System

I Parasites affecting urinary tract

1. *Schistosomahaematobium.*
2. Urinary myiasis.

II Focal lesions in the kidney:

1. Hydatid disease.
2. Cysticercosis of the lungs.

III Parasites found in urine

1. *Trichomonas vaginalis* trophozoite.
2. *Strongyloides stercoralis* larva.
3. *Wuchereria bancrofti* microfilaria.
4. *Enterobius vermicularis* egg.

IV Others

1. *Plasmodium malariae* (benign quartan malaria
2. *Plasmodium falciparum* (malignant subtertian malaria)

Schistosoma haematobium (Urinary schistosomiasis)

Habitat: Vesical and pelvic venous plexus.
Mode of infection: penetration of skin by furcocercous cercaria.
Pathogenesis

The main pathological changes of *Schistosoma* occur mainly due to **egg granuloma** (the trapped eggs in tissues are surrounded by granulomatous reaction ending with fibrosis)

Clinical features

Stage of invasion	Skin penetration by the cercaria leads to dermatitis.
Stage of migration	-Due to the migrating Schistosomules through the lungs. - Loffler's pneumonia: Fever, cough, expectoration of blood-tinged sputum, eosinophilia.
Stage of egg deposition (acute stage)	• Terminal **hematuria** • dysuria, increased frequency •Granulomata surround trapped eggs in the wall of urinary bladder and surrounding tissues.
Stage of tissue proliferation (chronic stage)	•Urinary infection, hydroureter, hydronephrosis, pyoureter, pyonephrosis, kidney damage. • urinary bladder **carcinoma**.

Laboratory Diagnosis



- √ **Urine examination**: Eggs with terminal spines.
- √ **Cystoscopy and biopsy**: eggs.
- √ **Serological diagnosis**: ELISA.

Treatment: Praziquantel (single dose of 40mg/kg body weight).

Urinary myiasis

Definition:

Infestation of urinarytractby larvae of dipterous fly including ***Fannia*** and ***Musca domestica***. Itis**accidental** myiasis in which eggs are put around the orifices and larvaecrawl into the bladder).



Larva of *Musca domestica*

Malaria (*plasmodium*)

Definitive host (vector): female *Anopheles* mosquitoes.

Intermediate host: man.

Infective stage:

√ Bite of female *Anopheles* mosquitoes (sporozoite).

√ Blood transmission or congenitally (merozoite).

Effects on kidney:

Only two types of the malaria parasites, namely, *Plasmodium malariae* (quartan benign malaria) and *Plasmodium falciparum* (malignant

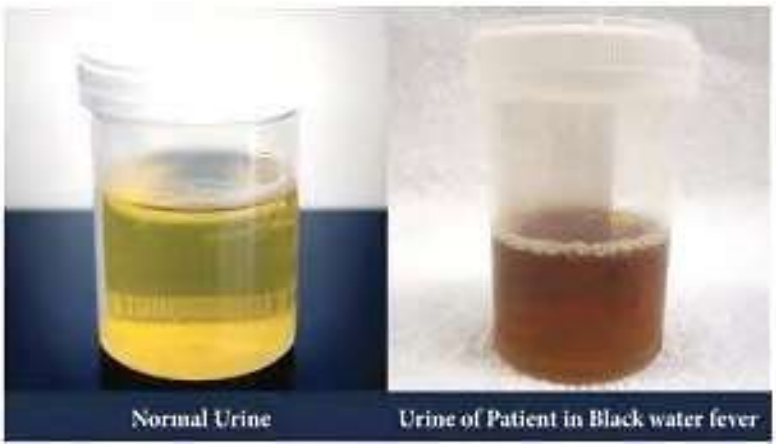
subtertian malaria), are clearly associated with renal disease, and this occurs only in a small percentage of patients.

1. *Plasmodium falciparum*

a. **Acute renal failure:** due to **cytoadherence** phenomenon impaired blood flow ⟶ tissue anoxia

b. **Black water fever:** It is **autoimmune** reaction (anti-erythrocyte antibodies leads to acute massive hemolysis with high levels of free hemoglobin and its breakdown products in the blood and urine). The urine is quite dark, hence the name of the condition. It is accompanied with acute renal failure.

(N.B: renal involvementinfalciparummalariaisusuallytransientanddisappears when the infection is treated)

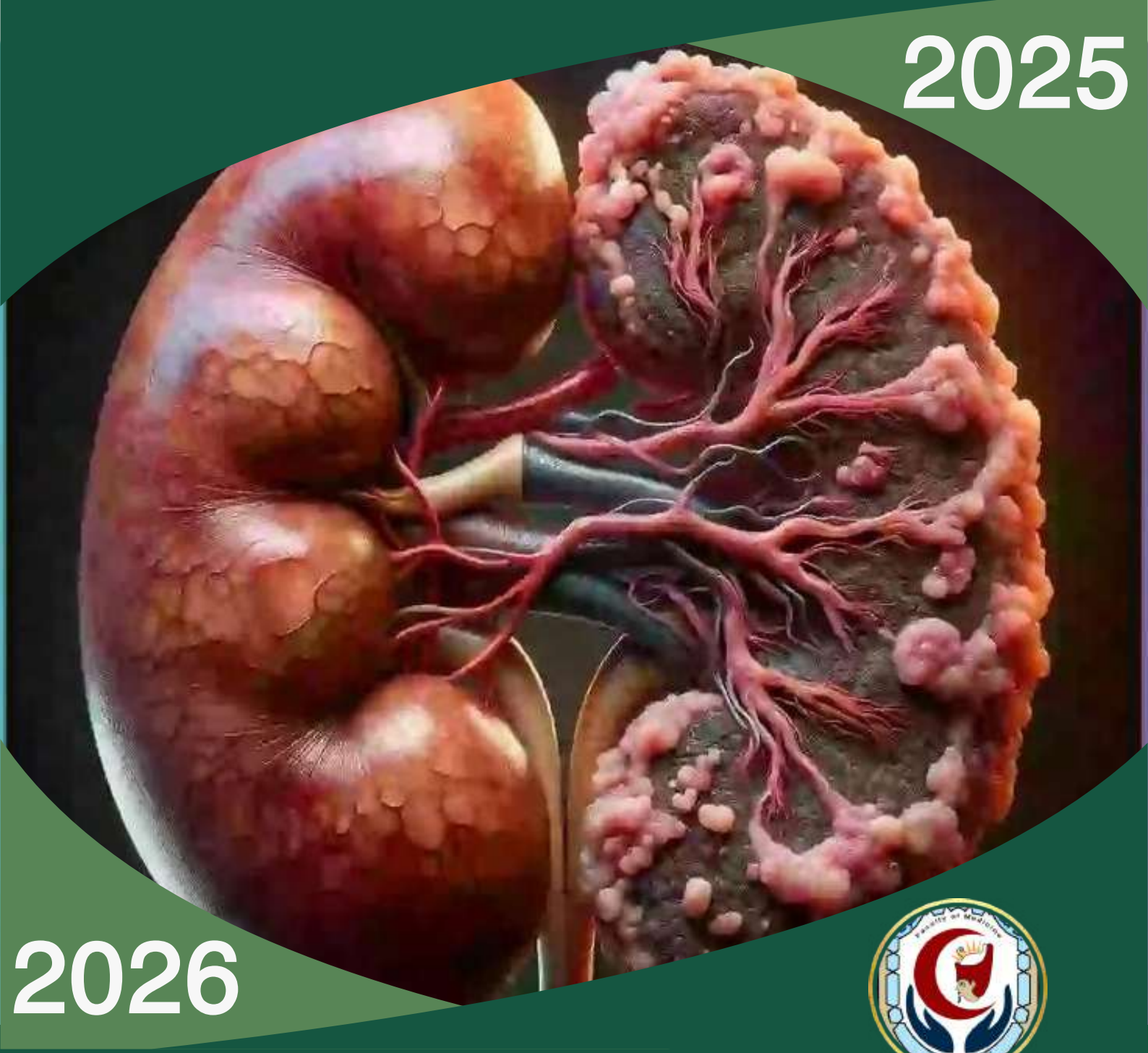


2. *Plasmodium malariae*

It may cause **nephrotic syndrome** in children due to deposition of immune complexes on glomerular capillary walls (type III hypersensitivity reaction).

(N.B: renal involvement in quartan malaria is irreversible)

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Renal System

Pathology

RNS-314

Page	Lecture
1	Congenital anomalies of the kidneys- Glomerular diseases (1)
5	Glomerular diseases (2)
18	Diseases of the tubules and interstitium
24	Urinary outflow obstruction - Diseases of the urinary bladder
31	Tumors of urinary tract

1 Congenital Anomalies of The Kidney & Glomerular Diseases I

Congenital Anomalies Of The Kidney

Agenesis
- ✓ Completeabsenceofonekidney.
- ✓ The other one shows compensatory hypertrophy.

Hypoplasia
- ✓ Failure of one kidney to develop to the normal size.
- ✓ Itremains small in size. The other one shows compensatory hypertrophy.

Polycystic Kidney Diseases
I. ***Autosomal Dominant Polycystic Kidney Disease.***
II. ***Autosomal Recessive Polycystic Kidney Disease.***

Ectopic Kidney
- ✓ The kidney fails to migrate up to the normal position and remains in the pelvis or at the pelvic brim.

Horse-Shoe Kidney
- ✓ Both kidneys are fused together usually along the lower poles by renal or fibrous tissue.

Congenital Double Ureters And Double Pelvis

1) *Autosomal Dominant Polycystic Kidney Disease (ADPKD)*

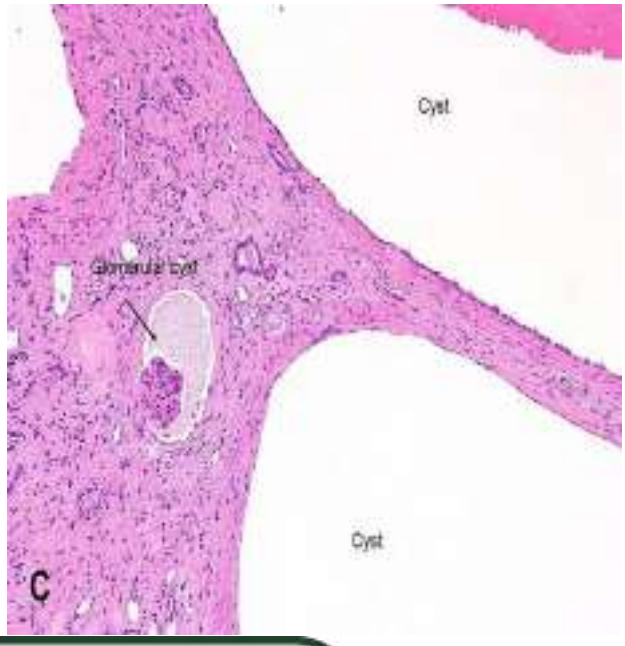
- ➢ It's a hereditary condition characterized by gradually expanding cysts that progressively destroy the renal **parenchyma (cortex and medulla)** of both kidneys final causing the renal failure.
- ➢ **Itis one of the most common hereditary human disorders.**
- ➢ Occurring in about **1:2 per 1000** live birth.
- ➢ Accounting for approximately **10% of cases** requiring **dialysis or renal transplantation**.
- ➢ Males and females are equally affected.
- ➢ There are no **racial predilection**.

Gross Picture



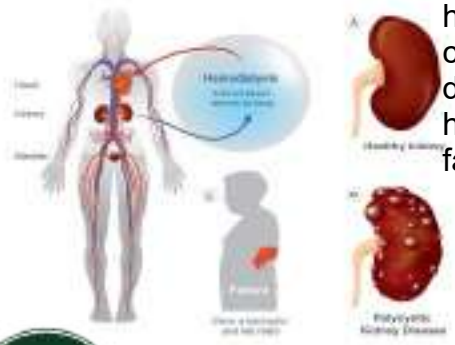
- **ADPKD is bilateral disease.**
- **Both kidneys are markedly enlarged by a large number of cysts which gradually enlarge throughout life.**
- **The cysts replace renal parenchyma which is in sever, advanced cases, can be detected only microscopically.**
- **Stones and haemorrhage inside the cysts may be seen.**

Microscopic Picture

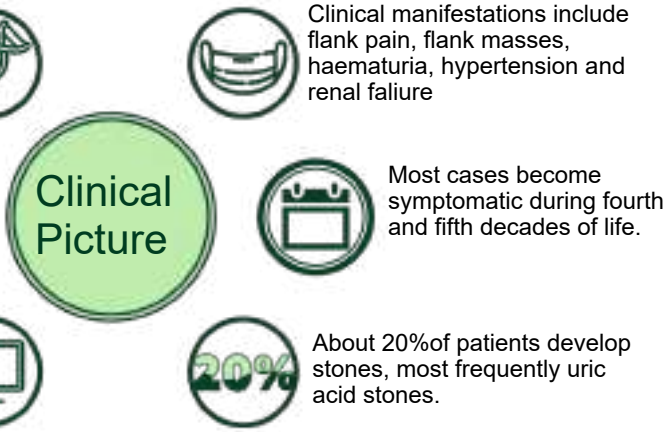


- **The cysts develop in all segments of the renal tubules and glomerular capsule.**
- **In early stages; the fluid of the cysts is derived from the glomerular filtrate.**
- **But when the cysts enlarge; they lose their attachment to the renal tubules and their fluid is derived exclusively from their lining epithelial cells.**
- **the cysts are lined by cuboidal, flattened epithelium, focal areas of hyperplasia may be seen.**

Clinical Picture



ADPKD caused by mutation in PKD2 gene has less sever disease course with later age at diagnosis. Late onset of hypertension and renal failure.



CompiainIctos

ADPKD is considered a **systemic disease** as more than **50%** of cases develop cysts in other organs as (**Liver – Pancrease – Spleen – Pineal Gland – Seminal Vesicles – Lungs)**.
Other anomalies including **(Cereberal – Coronary – Artery Aneurysms – Mitral Valve Prolapse – Skeletal Anomalies)**.

- **Hypertension**
- **Renal stones**
- **Chronic renal failure**
- **Renal adenoma in 20% of cases with ADPKD**



2) *Autosomal Recessive Polycystic Kidney Disease (AR-PKD)*

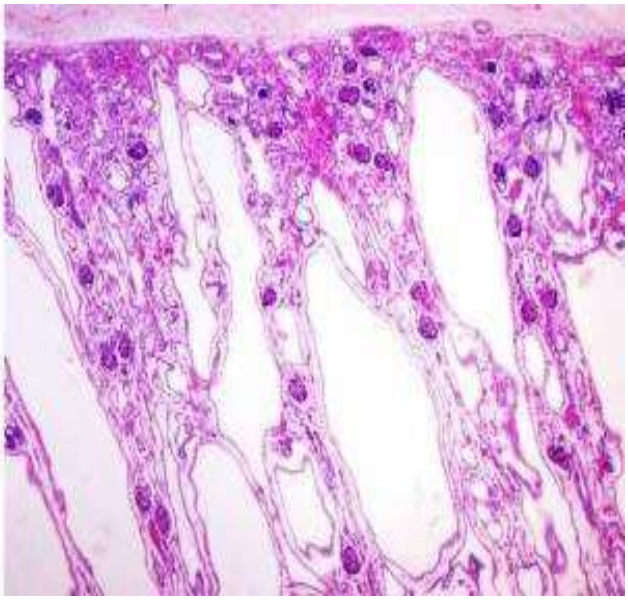
- Also called **Classic Infantile AR-PKD**.
- Itis a rare developmental anomaly, having autosomal recessive pattern of inheritance.
- Newborns with severe form of the disease **die shortly after birth** usually because of **respiratory insufficiency**.
- In other cases, death from renal failure may occur **in infancy or early childhood**.
- Occurring in about **1/ 6000-14000** live births. AR-PKD is a **rare** condition.
- All cases of classic AR-PKD have mutations in polycystic kidney and hepatic disease-1 **(PKHD1)** gene located on **chromosome 6**.

Gross Picture



- The kidneys are enlarged
- Both kidneys have a smooth external surface.
- Cut section revealed numerous small cysts in the cortex and medulla give the kidney a sponge appearance.

Microscopic Picture



- There is dilatation of all collecting tubules.
- The cysts have a uniform lining of cuboidal cells denoting their origin from collecting tubules.

NOTE

In almost all cases, there are multiple biliary cysts in enlarged portal tracts.
Some older children and adolescents with AR-PKD have asymptomatic, minor renal involvement (just few renal cysts) and progressive portal fibrosis, leading to portal hypertension.

2 Glomerular diseases

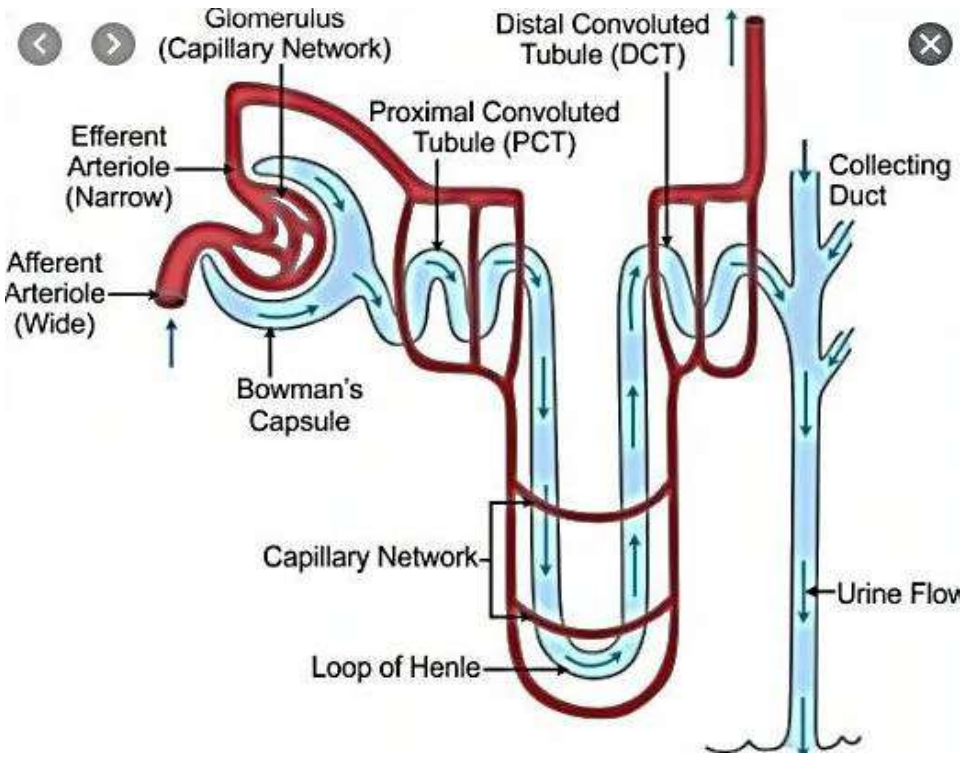
Important Items

- *I. The Glomerulus*
- *II. The Glomerular Capillary Wall*
- *III. Mesangeal Cells*

The Glomerulus

✓ is composed of an anastomosing network of capillaries incorporated in **Bowman's capsule**.

✓ It is lined by **two layers** of epithelial cells; the visceral epithelial cells that are incorporated into and become an intrinsic part of the capillary wall and separated from the endothelial cells by a **basement membrane**. **The parietal epithelial cells** line the Bowman's space.



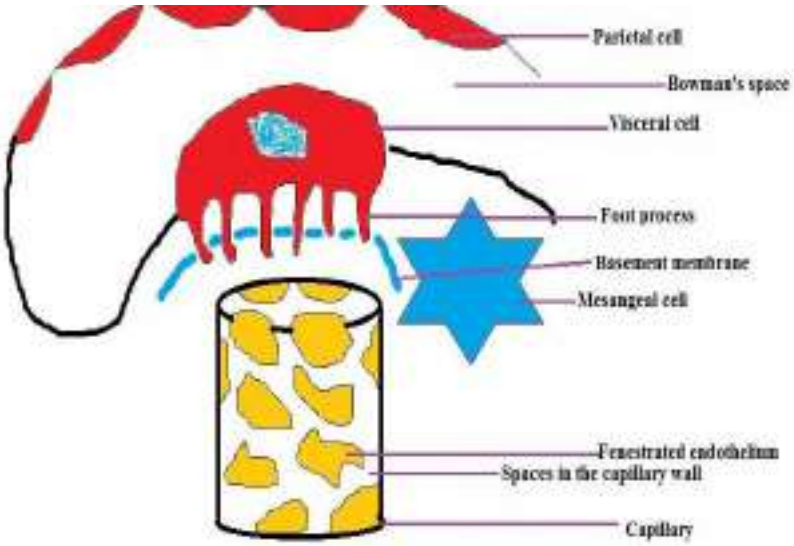
The Glomerular Capillary Wall

The glomerular capillary wall is the **filtering membrane** and consists of the following structures:

I. A thin layer of fenestrated endothelial.
II. A glomerular basement membrane.
III. Visceral epithelial cells **(Podocytes)**.

Mesangeal Cells

- ✓ The entire glomerular tuft is supported by **mesangeal cells**.
- ✓ They are lying between capillaries.
- ✓ Mesangeal cells produce **mesangeal matrix** that form a meshwork through which mesangeal cells are scattered.
- ✓ Mesangeal cells are **contractile, phagocytic and capable of proliferation** and lying down both matrix and collagen.



Glomerular Diseases

- Glomerular diseases constitutesomeof major problems in nephrology.
- Chronic glomerulonephritisisoneofthe most common causes of **chronic renal failure.**
- Glomeruli may be injuredbyavarietyof factors and in the course of number of systemic diseases.
- Here, the glomerular injuryiscalled **secondary glomerulonephritis**.
- In other conditions, the kidneyistheonly or the main organ involved.
- The latter constitutes varioustypesofprimary glomerulonephritis.

Glomerulopathy is a glomerulardisease that don't have a cellular inflammatory component

Glomerular diseases (G lomerulonephritis/ G.N.)

Primary glom erulonephritis:

- Acute diffuse proliferative G.N.
- Rapidly progressive G.N.
- Membranous G.N.
- Membrano-proliferative G.N.
- Minimal change G.N.
- Chronic G.N.

Secondary glo merulonephritis:

- Immunological as SLE.
- Vascular as hypertension.
- Metabolic as D.M

Clinical Manifestations of Glomerulonephritis

- Asymptomatic proteinuria, asymptomatic haematuria.
- Nephrotic syndrome.
- Neph ritic syndrome.
- Rapidly progressive glomerulonephritis causing acute renal failure.
- Chronic renal failure.

Pathogenesis of glomerular injury

- Most cases of G.N. are due to immune mechanisms

- In-situ immune complex deposition:

- Circulating antibodies react directly with antigens placed in the glomeruli either intrinsic or implanted as bacterial products or large protein particles. The antibody and the antigen form immune complex in the glomeruli that initiate inflammation and glomerular injury.

- Deposition of circulating immune-complexes within the glomeruli:

Inthis type , glomerular injury caused by trapping of circulating Ag/Ab complexes within the glomeruli.

• **Cell- mediated immunity.**

Histological Alteration of G.N.

• Various types of G.N. are characterized by one or more four basic tissue reactions:

Hypercellularity;

increase number of cells in the glomerular capillary tufts due to proliferation of Mesangeal cells, endothelial cells or parietal epithelial cells. in addition to leucocytic infiltration (neutrophils, monocytes and lymphocytes in some cases).

Thickening of the glomerular basement membrane.

Crescent formation:

Crescent formation: acrescent-shapedmassofcells may result from proliferation of parietal cells.

Hyalinization and sclerosis

• Other alterations:

* Intra-capillary thrombosis. * Fibrin deposition.

Accumulation of lipids.

Nephrotic syndrome:

• **Definition:** A group of renal glomerular diseases ; allarecharacterized by massive proteinuria, hypoalbuminaemia , generalized oedema and hyperlipidaemia.

➢ **Massive proteinuria:** daily loss of 3-5 gm/24 hours caused by increased permeability of the glomerular capillary tufts to protein.

➢ **Hypoalbuminaemia:** plasma albumin levels are less than 3 gm/ dl

➢ **Generalized oedema:** is caused by

1)hypoproteinaemia with decrease in plasma osmotic pressure.

2)hypoalbuminaemia causing decrease in plasma volume due to loss of osmotic effect.

This results in increased aldosterone release leading to sodium and water retention.

➢ **Hyperlipidaemia:** the exact cause is not compeletly understood , it may be due to protein loss.

Causes of Nephrotic Syndrome

Primary glomerular Diseases

- Membranous glomerulonephritis.
- Minimal change glomerulonephritis.
- Membrano-proliferative glomerulonephritis.
- Focal proliferative glomerulonephritis.
- Focal segmental glomerulosclerosis.

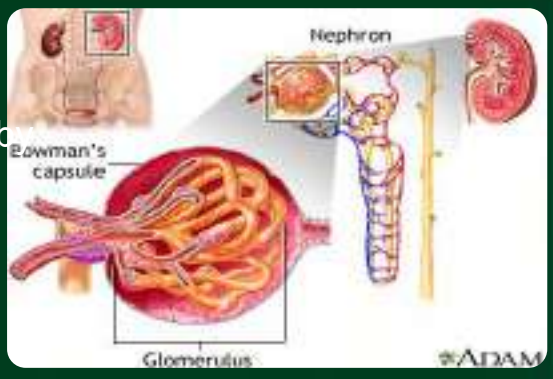
Systemic Diseases

- Diabetes mellitus.
- Amyloidosis.
- Systemic lupus erythematosis.
- Infections a s HBV, Malaria.
- Malignancy.

Acute Nephritic Syndrome

➢ it is a glomerular syndrome characterizedb acute onset of usually grossly visible haematuria (RBCs in urine). Mild to moderate proteinuria and hypertension.

- It is the classic presentation of acute post-streptococcal glomerulonephritis



Primary Glomerulonephritis

- Acute diffuse proliferative G.N.
- Membranous G.N.
- Minimal change G.N.

Acute diffuse proliferative glomerulonephritis/ post streptococcal glomerulonephritis.

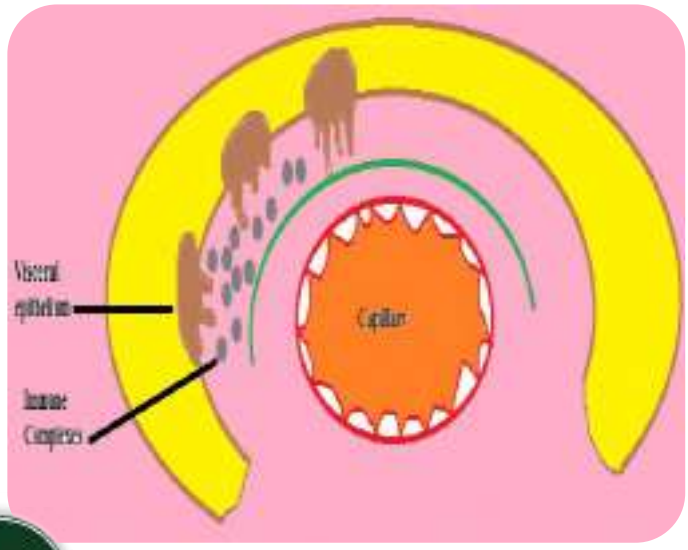
Aetiology:

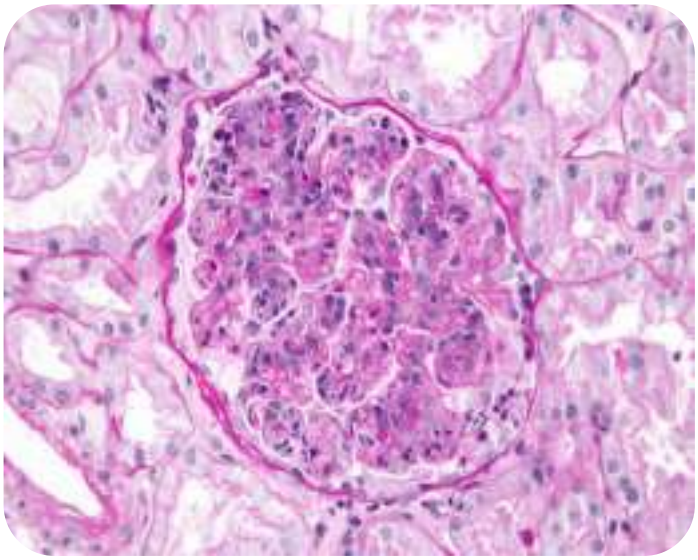
• A rather common auto-immune disease in children and young adults. The disease may follow acute tonsillitis, pharyngitis or rarely skin infection.

• It is caused by strains of group A beta haemolytic streptococci after a latent period of 1-4 weeks.

• **During the latent period;** antibodies mainly IgG are formed against streptococcal antigens.. an immune-complex reaction occurs between streptococcal antigens and the antibodies in patient's serum.

• The immune-complexes deposit between the visceral epithelial cells and glomerular basement membrane followed by complement activation and inflammatory injury to the glomerular capillaries



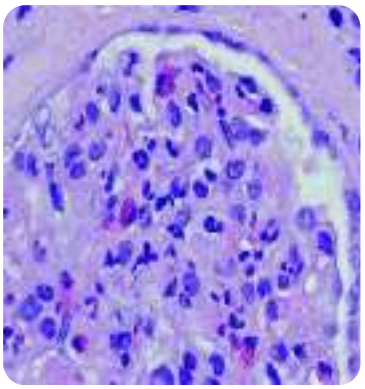


Gross Picture



- Both kidneys are slightly enlarged,
- the capsule strips easily. The
- kidney surface is smooth . Cut
- section is pale due to oedema.
- In severe cases, there is red spots due to due to capsular haemorrhage.

Microscopic picture



- All glomeruli show swollen, cellular, bloodless capillary tufts filling the capsular spaces.
- The increased cellularity is due to proliferation of capillary endothelium, glomerular epithelium, mesangial cells and infiltration by neutrophils and monocytes.
- The capsular space is narrowed and contains coagulated and albumin, fibrin threads, neutrophils, RBCs desquamated epithelial cells. The collecting tubules contain
- casts especially blood casts. The interstitial tissue is
- hyperaemic, oedematous and shows neutrophilic infiltrate.

Clinical picture

- Fever and malaise.
- Nephritic oedema: peri-orbital oedema which is marked in the morning. The oedema may involve the rest of the face and may become generalized.
- Hypertension: mild to moderate degree. Both hypertension and oedema are due to salt and water retention.
- Blood changes: elevation of blood urea and creatinine.

➢ **Urine changes:**

- ❑ Theurine is brownish and turbid (turbid or Coca-Cola like) due to presence of altered red cells.
- ❑ Oliguria, moderate proteinuria and increased specific gravity up to 1035.
- ❑ Microscopically; the sediment contains red cells, neutrophils, hyaline, granular, cellular and blood casts.

➢ Haematuria, proteinuria, hypertension, oedema, oliguria and uraemia constitute acute nephrtic syndrome.

course

- ➢ Complete recovery in over 95% of children and 65% of adults.
- ➢ The disease may progress to rapidly progressive or chronic glomerulonephritis.
- ➢ Death from acute uraemia or acute heart failure caused by sever hypertension

Rapidely progressive glomerulonephritis/ Crescentic glomerulonephritis, RPGN

> It is a primary glomerulonephritis characterized clinically by progressive deterioration in renal function that progress to renal failure within weeks to few months.

Aetiology:

I- It may complicate acute post-streptococcal proliferative glomerulonephritis, this complication is common in adults rather than children.

II- It may occur in association with systemic (immunologically-based) diseases as systemic lupus erythematosis, polyarteritis nodosa or scleroderma.

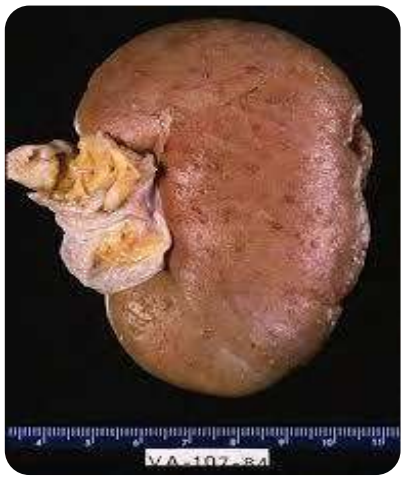
III- It may be idiopathic.

Pathological Features

GrossPicture

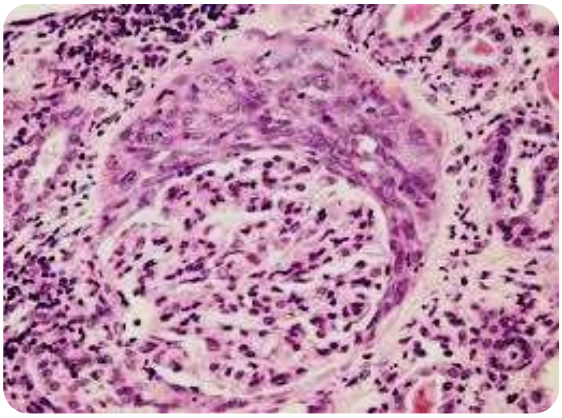
- Both kidneys are normal in size or slightly enlarged due to edema.

Outer surface is smooth . Cut section reveals pale cortex and interstitial hemorrhage.



Microscopic picture

- Glomeruli are enlarged and shoe prolifiration of endothelial celland mesangial cells
- Proliferation of the parietal cells of Bowman's capsule with formation of epithelial crescents in the capsular spaces.
- Leucocytic infiltration. Glomerular capillaries show thrombosis,
- necrosis and hemorrhage.
- Tubular epithelial cells may show necrosis, renal tubules contain



Clinical picture

- Hematuria, sever oliguria or anuria, hypertension.
- Both edema and proteinuria are variable.

Fate of RPGN

- The disease is rapidly progressive.
- Acute renal failure.
- Hypertention.

Minimal change glomerulonephritis/ lipoid disease/ light negative GN/ Foot process

> It is a disease of small children (2-6 years),

*but may occur in older children and adults.

*The exact aetiology is unknown but the disease may follow respiratory infection.

*This disease is the commonest cause of nephrotic syndrome in children.

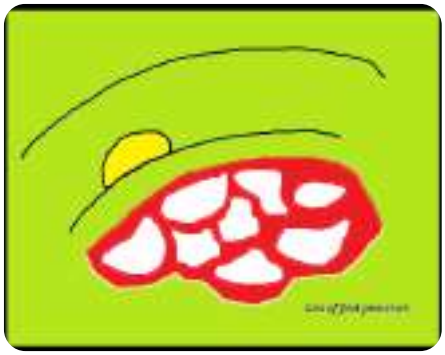
Pathologic al Features

➢ The kidneys don't show any gross abnormalities.

By Light Microscope

- the glomeruli show no pathological changes.
- However, the proximal convoluted tubules show lipid deposition in the lining cells.
- this lipid reflects glomerular leakage and reabsorption of lipoproteins

By electron Microscope



- diffuse loss of foot processes of the visceral epithelial cells.
- Most cases cure with corticosteroid therapy.

Membranous glomerulonephritis

Aetiology:

➢ Idiopathic in 85% of cases.

➢ May occur in association with:

❑ Systemic lupus erythematosis.

❑ Infections as hepatitis B, bilharziasi, malaria, syphilis.

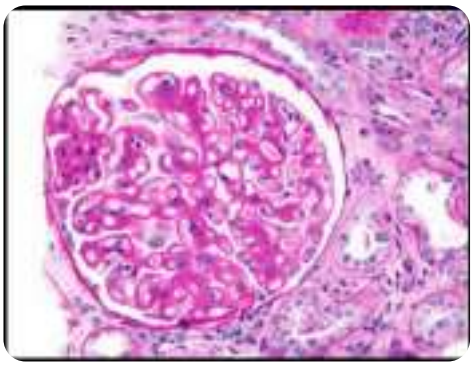
❑ Metabolic diseases as diabetes mellitus.

❑ Malignant epithelial tumors as carcinoma of lung, carcinoma of colon.

Pathologic al Features

➢ **The kidneys are enlarged and pale**.

By Light Microscope



• diffuse thickening of the glomerular basement membrane

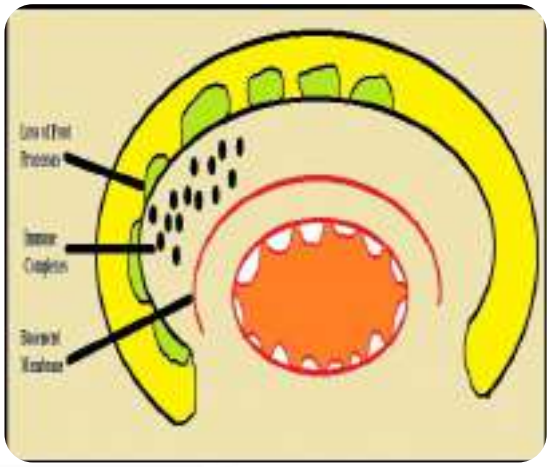
easily demonstrated by PAS stain. No cellular proliferation of the glomeruli.

• In advanced cases, glomerular sclerosis and hyalinosis

occur, atrophy of tubules and interstitial fibrosis.

Patients suffer from nephrotic syndrome. In advanced cases, renal insufficiency and hypertension develop.

By electron Microscope



• the epithelial cells are swollen andloss their foot

processes, dense IgG deposited between basement membrane and overlying epithelial cells

Secondary Glomerulosnephritis

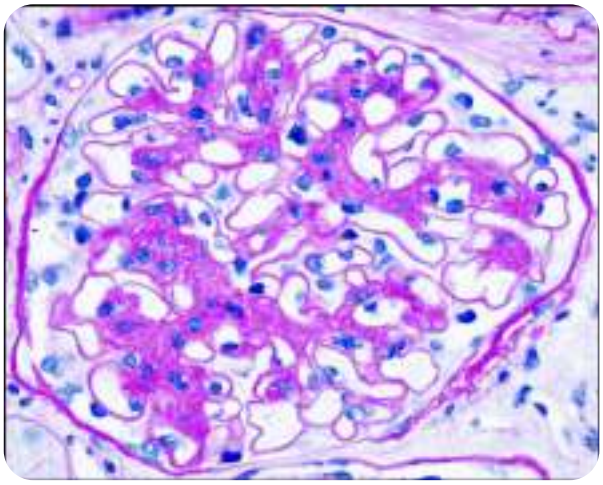
Diabetic Glom erulosclerosis

The characteristic renal changes in DM are best seen in the glomeruli and arterioles.

By Light Microscope

•1) Diffuse diabetic glomerulosclerosis:

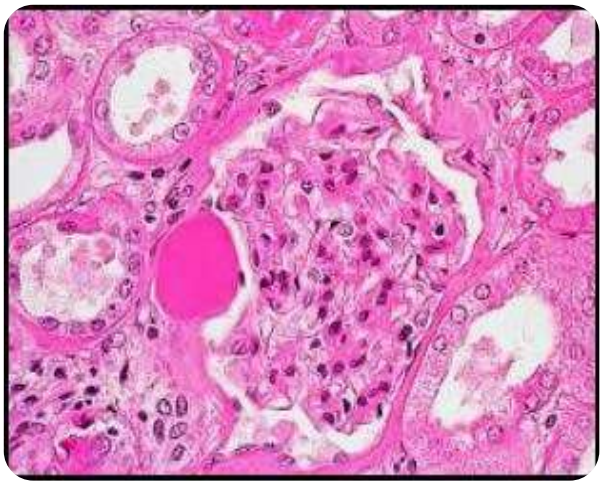
This is due to increase amount of mesangeal matrix in all msangeal regionsin all glomeruli



•2) Nodular diabetic glomerulosclerosis (Kimmelstiel-Wilson nodule):

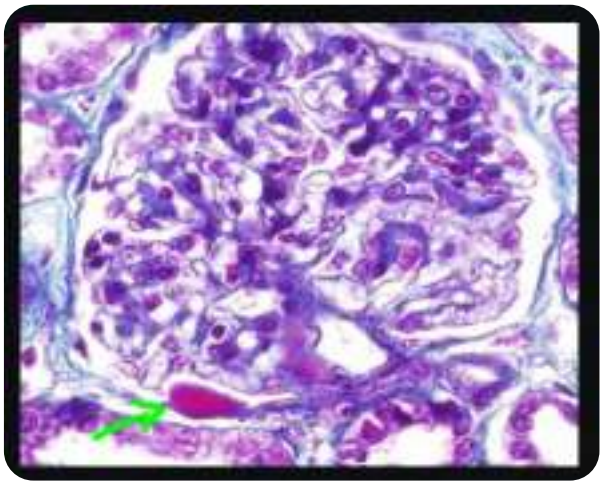
- This is a localized sclerotic acellular nodule located at one side of the glomerular capillary tuft.
- It is almost always superimposed on a pre-existing diffuse diabetic glomerulosclerosis.

Itappears as amorphous, eosinophilic material. It has the same staining properities as mesangeal matrix

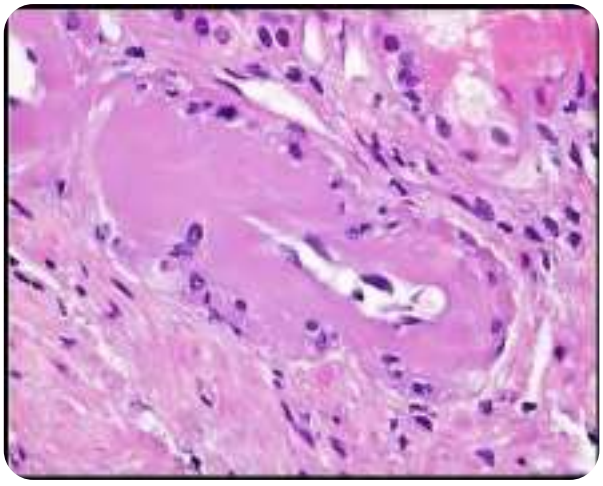


3) Insudative glomerular lesions (fibrin caps and capsular drops):

- These insudativelesions resulting from accumulation of plasma proteins in differentregions in the glomeruli.
- If plasma proteinsare accumulated between glomerular capillary endothelium andglomerular basement membrane; the resulting lesion is called **fibrin cap.** If plasma proteinsare accumulated
- between parietal and visceral epithelial cells (inBowman's space); the resulting lesion is called **capsular drops.**



4) Prominent arteriolar hyalinosis.



3 Disease of renal tubules & renal interstitium

1 DISEASES OF RENAL TUBULES (ACUTE TUBULAR NECROSIS)

1.Toxic Tubular Necrosis

Causes

Tubulardamage by toxic substances as

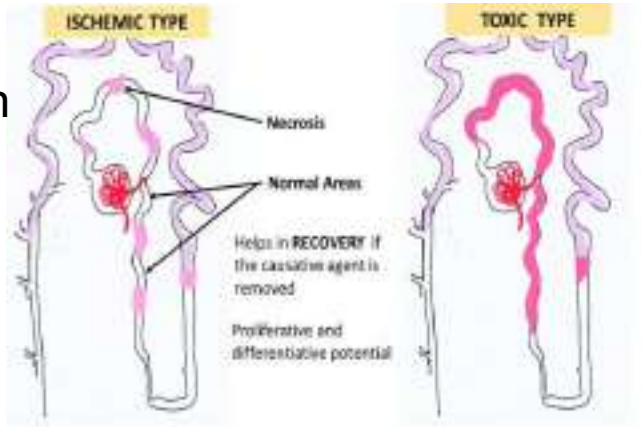
- carbon tetrachloride,
- phenol,
- mercurials,
- some antibiotics
- anesthetics.

Pathology

- The tubular epithelium especially in the proximal tubules shows extensive necrosis. - The necrotic tissue blocks the tubular lumen.

-The interstitial tissue shows edema.

- Both kidneys are enlarged and soft and the cut surface shows a pale cortex.



Effects

➢ **Anuria due to:**

(a) Blockage of the tubules by the necrotic cells.

(b) Complete reabsorption of the glomerular filtrate by the peri-tubular plexus or vessels exposed by the necrotic epithelium.

➢ Acute uraemia

➢. Acute cardiac failure caused by hyperkalaemia.

➢ Tubular regeneration and recovery rarely occurs.

2. Ischaemic Tubular Necrosis

Causes

• Tubular damage due to renal ischemia. The latter may be due to shock,

severe hemorrhage, crush injuries, burns and incompatible blood transfusion..

Pathology

• Focal necrosis at multiple points along the nephron.

- The tubular lumens are blocked by the necrotic cells and casts of hematin

Effects

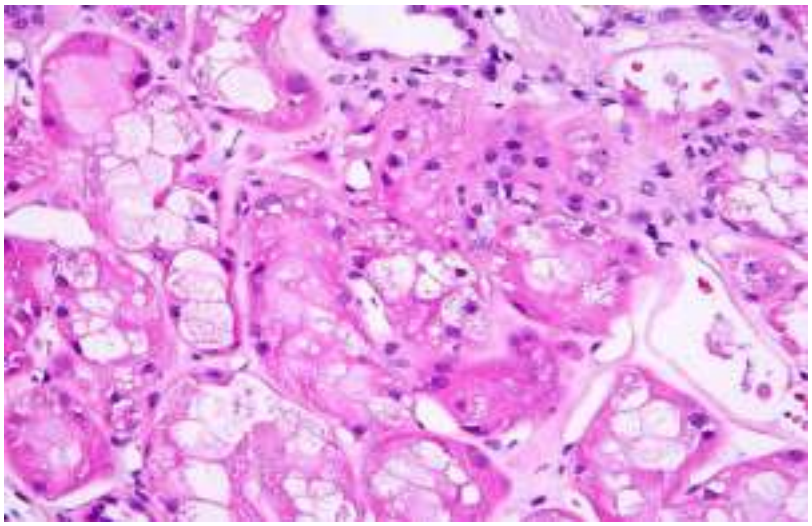
➢ Anuria due to:

(a)Decrease in the glomerular filtrate secondary to renal ischaemia.

(b) Blockage of the tubules by the necrotic cells and casts.

(c) Complete reabsorption of the glomerular filtrate by the peri-tubular plexus of vessels exposed by the necrotic epithelium.

➢ The rest of the effects are similar to those of toxic necrosis.



Pyelonephritis

Definition

• It is bacterial inflammation of the interstitial tissues of the renal pelvis, medulla and cortex.

- Pyelonephritis is a very common kidney disease

Aetiology:

Causative organisms:

E.Coli is the commonest causative organism. Others as staph.aureus, strpt. Foecalis, bacillus proteus and typhoid bacilli, mixed infection is common.

.F

Predisposing factors:

- Urinary tract obstruction: causes stasis of urine, facilitating infection. Obstruction is caused by urinary bilharziasis, stricture, stones, tumors, enlarged prostate.
- Diabetes Mellitus
- Instrumentation of the urinary tract.
- Pyelonephritis is more common in females due to short wibe female urethra and pressure of the enlarged pregnant uterus on the ureters at the pelvic prim.

Routes of infection:

➢ Haematogenous infection: occur in most cases. Bacteria enter the blood from septic focus as tonsillitis, sinusitis, otitis media, bronchiectasis,…. .

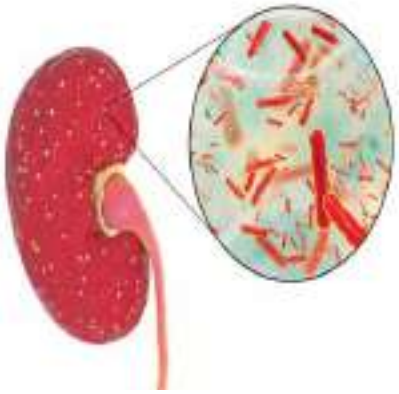
➢ Ascending infection: from the bladder by the following routes

- The lumen of the ureter.
- Through periureteric lymphatics.

2 Types of Pyelonephritis

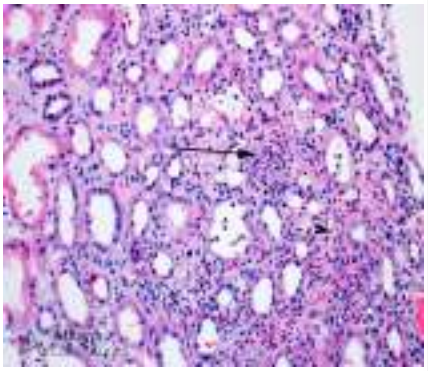
Acute Pye lonephritis

Grossly



- commonly the lesion is bilateral;
- the kidney is enlarged and congested.
- The capsule strips easily.
- The outer surface shows multiple, tiny abscesses.
- Cut section shows multiple abscesses in cortex and medulla.
- The pelvi-calyceal system is congested and contain purulent exudate.

Microscopic Picture



- The interstitial tissue is infiltrated by polymorphonuclear leucocytes and pus cells.
- the tubules show tubular degeneration, necrosis and leucocytic casts.
- The calyces and pelvis are hyperaemic, oedematous and infiltrated by acute inflammatory cells.

Clinically

- fever, renal tenderness, dysuria and pyuria.

- Course of acute pyelonephritis:

Resolution and recovery in mild cases.

Change to chronic pyelonephritis.

In sever, bilateral affection; death from acute renal failure.

Chronic Py elonephritis

May follow acute pyelonephritis but commonly starts as a chronic inflammation.



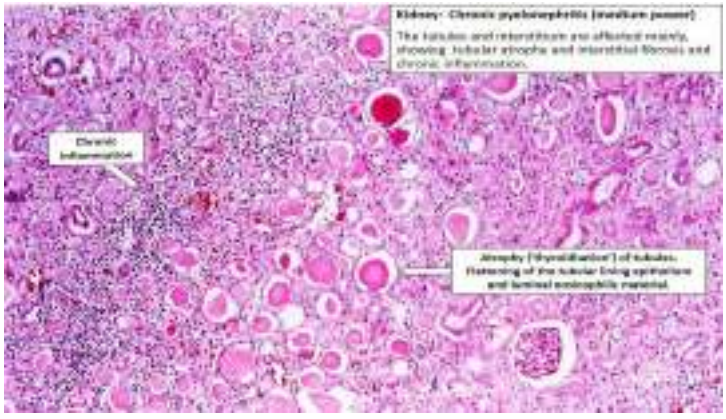
Grossly

- The kidney is reduced in size,
- the capsule is thick and adherent.
- The surface shows irregular depressions due to cortical scarring.
- The pelvi-calyceal system is thickened, distorted and contain purulent material.

Microscopic Picture

- The interstitial tissue is infiltrated by lymphocytes, plasma cells, macrophages, neutrophils and shows broad bands of fibrosis.
- some tubules are atrophic while others are dilated and contain large hyaline casts (thyroidization).
- The blood vessels are thickened

.



-Effects of chronic pyelonephritis

✓ Chronic renal failure.

✓ Secondary hypertension.

4 Urinary Outflow Obstruction & Diseases of The Urinary Bladder

Aims of this lecture

Definition of hydronephrosis and causes of unilateral and bilateral hydronephrosis and their complications.
Etiology and pathological consequences of urinary calculi.
Definition of cystitis, etiology and complications of cystitis

UrinaryTractOutflowObstruction

Urinary outflow obstruction (obstructive uropathy)

- Recognition of urinary obstruction is important because obstruction increase the liability to infection and stone formation.

- Unrelieved obstruction almost always leads to permanent renal atrophy (a condition named hydronephrosis or obstructive uropathy).

- Obstruction may be sudden or insidious, partial or complete, unilateral or bilateral.

- Obstruction can occur at any level from **urethra** to **renalpelvis** .

- It can be caused by lesions **intrinsic** to the urinary tract or **extrinsic** as compression on the ureter from outside.

The common causes of urinary tract obstruction

- Congenital anomalies as posterior urethral valve, urethral stricture, bladder neck obstruction, narrowing of the uretro-pelvic junction.
- Urinary calculi.
- Benign prostatic hyperplasia (BPH).
- Tumors: prostatic carcinoma, bladder cancer, carcinoma
- Inflammation: prostatitis, uretritis, urethritis.
- Pregnancy: causes compression on the ureter against pelvic prim.
- Neurogenic causes as spinal cord damage.

Hydron ephrosis

Hydronephrosis ⇨ is a term used to describe dilatation of the renal pelvis and calyces, associated with chronic atrophy of the renal parenchyma due to obstruction of the outflow of the urine.

- Dilatation of the ureter due to distal obstruction is called **hydroureter** .
- Because of this continuing filtration; the affectedcalyces and pelvis become progressively dilated.

High pressure in the renal pelvis is transmitted back onto the renal parenchyma, causing renal atrophy.

Obstruction also triggers interstitial inflammatory reaction, which in turn will result in interstitial fibrosis and destruction of the renal parenchyma .

NOTE

Even with complete obstruction, glomerular filtration persist for some time because the filtrate diffuse back to the renal interstitium and peri-renal spaces and then return back to the lymphatics and venous system

Causes of unilateral hydronephrosis

Uretral causes : uretritis, uretral stricture, uretral stone, sever vesico-uretral reflux, pressure by gravid uterus or malignant obstruction.

Renal causes : stone in the renal pelvis, tumor of the renal pelvis, pressure of aberrant renal artery on the renal pelvis .

Causes of bilateral hydronephrosis

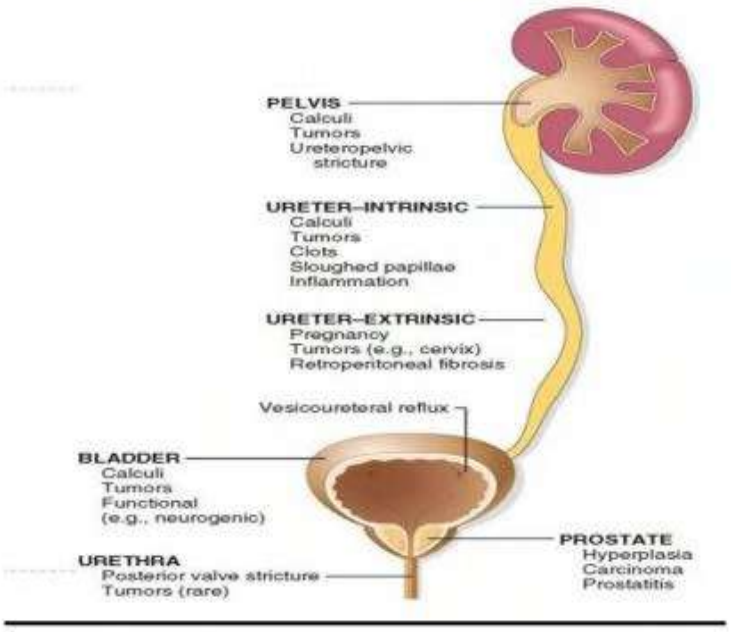
Urethral lesions : posterior urethralvalve, posttraumatic orpost inflammatory urethral stricture .

Prostaticlesions : BPH, prostatitis, prostatic carcinoma.

Bladderlesions : bladder calculi or tumors obstructing urethral or both uretral orifices

Complications of Hydronephrosis

- Renal failure due to renal atrophy.
- Superimposed infection (it will change hydronephrosis to pyonephrosis).
- Secondary hypertension



Renal Cal culi (stones)

Aetiology

several factors are responsible for precipitation of urinary crystalloids and stone formation.

1.Stasis: facilitates precipitation of urinary salts and and predispose to bacterial infection.

Stasis may results from:

- **Obstruction** to urinary outflow as in BPH, Prostatic Carcinoma
- Prolonged **recumbence** after bone fracture

2. Increased urinary concentration: as in cases of;

- Limited intake of water.
- Hot weather and excessive sweating

3. Infection: infection leads to stone formation by the following;

- changing PH of the urine causes precipitation of different urinary salts.
- Pyogenic bacteria decompose urea into ammonia causing alkalinity and phosphate deposition.
- The products of inflammation as shedded epithelial cells, fibrin, blood clots and pus cells act as nuclei for stone formation.

4. Metabolic disorders:

- Hypercalcuria in hyperparathyroidism.
- Increases oxalate secretion in oxaluria as in excess intake of tomato, spinach.

Types of renal stones

I- Primary or metabolic stones

1- oxalate stones

•Composed of calcium oxalate and develop primary in the renal pelvis in acidic urine.
•Oxalate stones are multiple, small, hard with rough, spiny surface. They cause mucosal injury and haemorrhage. The deposited blood pigments on the surface of the stone gives it a dark-brown or black color.

2- Uric acid and urate stones

•Composed of uric acid, sodium urate and ammonium urate. They develop in the pelvis and calyces in acidic urine. The stone is usually single, hard, round to oval, yellowish- brown with smooth outer surface.
The stone may enlarge and takes the shape of the renal pelvi-calyceal system producind a ***staghorn or coralline*** stone.

3- Cystine stones

•Rare, develop in cases of cystinuria. The stone is soft and yellowish-green

II- Secondary or infection stones

(Phosphate stone)

•Composed of calcium phosphate and dev

elop in bladder in infected, alkaline urine. The stone is rounded, white with smooth surface. It is friable, chalky and the internal structure is amorphous.
A small urate or oxalate stone reaching the bladder may form a nucleus for a bigger phosphate stone when infection occurs.

Effects and complications of renal stones

1.Pain:

- Dull aching pain (big stone).
- Renal colic resulting from contraction of ureteric muscle to move a small stone down the ureteric lumen.

2-Mechanical irritation cause:

- Haematuria.
- Mucosal ulceration with subsequent stenosis.
- Squamous metaplasia and leukoplakia.
- Bladder carcinoma.

3-Obstructive effects:

- Hypertrophy and dilatation of the bladder with diverticulae formation.
- Hydroureter and hydronephrosis.

4-Infection:

- Obstruction by stones facilitates infection with development ofcystitis, urethritis, pyelonephritis, pyoureter and pyonephrosis.

Cystitis

1) Acute Cystitis

Definition ⇨ Acute inflammation of the urinary bladder.

Aetiology ⇨ The normal mucosa of the urinary bladder is resistant to infection

Acute cystitis is predisposed to by

Stasis of urine due to:

- Urinary bilharziasis.

- Obstruction to the urinary flow by stones , nodular hyper plasia of the prostate urethral stricture and tumours of the bladder, prostate and urethra.

- Loss of nervous control of the bladder as in spinal cord injuries leads to incomplete emptying of the bladder and stasis of urine

Inflammation in adjacent organs (kidney, ureter, prostate, vagina, cervix and intestine).

Traumatic injury to the bladder mucosa as by a stone or a catheter.

Debilitating diseases *as diabetes mellitus*

- ***Organisms:*** E. coli, staphylococcus, streptococcus, gonococcus and bacillus proteus.

➢ ***Routes*** *of* ***infection:***

Direct spread from the ureter, kidney, prostate ... etc	By the way of the urethra through catheterization.	Lymphatic spread from the pelvic organs	Hematogenous infection from distant organs as the lung and tonsils.

➢ **Pathology of acute cystitis:**

- The mucosa is congested, oedematous and may be ulcerated.

The submucosa shows congestion, oedema and infiltration by polymorpho- nuclear leucocytes.

- If the bacte1ia are urea splitting the urine becomes alkaline.

➢ **Clinically:** Frequency (repeated micturation), dysuria (burning micturition) and pyuria (pus in urine)

2) Chronic Cystitis

➢ **Aetiology** ⇨ Followsacute cystitis or starts as chronic inflammation.

➢ **Pathology of chronic cystitis:**

- The surface urothelium shows hyperplasia, Cystitis cystica and cystitis glandularis and areas of ulceration.
- The lamina propria shows infiltration by lymphocytes, plasma cells and macrophages.
- The muscle layer shows hypertrophy and thickening.

In long standing cases the bladder wall shows fibrosis and the bladder volume is reduced (contracted bladder).

➢ **Complications:**

- Ascending ureteritis and pyelonephlitis.
- Phosphate stone formation.
- Leukoplakia.

➢ **Clinically:** Frequency, dysuria and pyuria.

Specific cystitis

•Bilharzial cystitis.

•Tuberculous cystitis

5 Tumors of the Urinary System

(WHO/ISUP) classification of tumors of the Kidney (2016)

Renal cell

1- Clear cell renal cell carcinoma. 2- Renal cell carcinoma, unclassified

3- Multilocular cystic renal neoplasm of low malignant potential.

4- Papillary renal cell carcinoma. 5- Clear cell papillary renal cell carcinoma.

6 -Hereditary leiomyomatosis and renal cell carcinoma- associated renal cell carcinoma.

7- Chromophobe renal cell carcinoma.

8- Collecting duct carcinoma.

9- Renal medullary carcinoma 10- MiT family translocation renal cell carcinomas.

11- Succinate dehydrogenase-deficient renal cell carcinoma.

12- Mucinous tubular and spindle cell carcinoma.

13- Tubulocystic renal cell carcinoma.

14- Acquired cystic disease-associated renal cell carcinoma.

15- Papillary adenoma. 16- Oncocytoma.

Nephroblastic and cystic tumors occurring

mainly in

1- Nephrogenic rests. 2- Nephroblastoma.

3- Cystic partially differentiatednephroblastoma. 4- Paediatric cystic nephroma.

Mesenchymal

1) Occurring mainly in children:

- Clearcell sarcoma.
- Rhabdoid tumor.
- Congenital mesoblastic nephroma.
- Ossifying renal tumor of infancy.

2) occurring mainly in adult:

- Leiomyoma/Leiomyosarcoma
- Haemangioma/ Lymphangioma
- Haemangioblastoma
- Angiosarcoma
- Rhabdomyosarcoma
- Osteosarcoma
- Synovial sarcoma
- Ewing sarcoma
- Angiomyolipoma
- Epithelioid angiomyolipoma
- Juxtaglomerular cell tumor
- Renomedullary interstitial cell tumor
- Schwannoma
- Solitary fibrous tumor

Renal cell carcinoma (Hypernephroma)

Itis a common tumor which arises from the epitheliumof the renal tubules.
The tumor is more common in males between the age of 40-70 years old.



Clinical picture:

1) Painlesshematuria.
2) Loin pain.
3) Palpableloinmass.
4) *Obstructionoflefttesticular vein if tumor located in left kidney produce varicocele.*

Pathological features:

- The tumor presents at one side of the kidney. It forms a variable sized, rounded sharply demarcated mass.
- Cut surface of the tumor has a characteristic yellow color due to high lipid content of the tumor cells.
- It usually shows areas of haemorrhage, necrosis and cystic change

Microscopically: RCCs may show different morphological pictures.

Types:

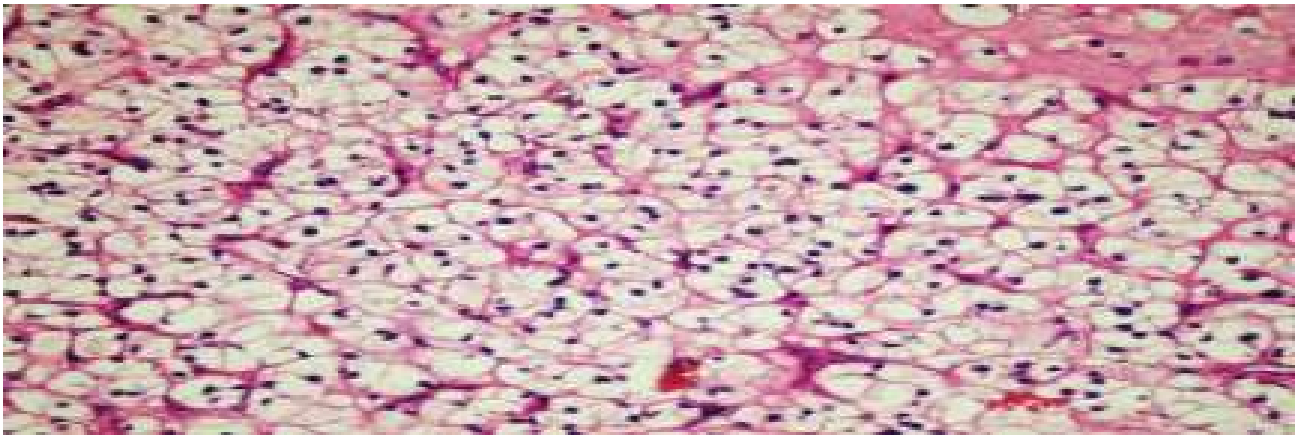
1) Clearcell RCC
2) Papillary RCC
3) Chromophobe RCC

Clear cell RCC

➢ It is the most commontypeof RCCs accounting for about 65-70% of all renal cancers.
➢ In 90% of cases, thesetumors exhibit alterations in the von Hippel–Lindau (VHL) tumor suppressor geneonchromosome 3.
➢ Multifocal tumors are rareand a recent study has shown that apparent multifocality may be duetoretrograde venous invasion from a single tumor.

Grossly:

➢ Clear cell RCCs characteristically is:
A solid, golden- yellow tumor with variable amounts of cystic change, haemorrhage, calcification and necrosis.
➢ The tumor is sharply separated from the surrounding tissue by a fibrous pseudocapsule.
➢ On occasion, the cystic degeneration is so advanced that a mural nodule remains as the only evidence of the real nature of the lesion.



Microscopically:

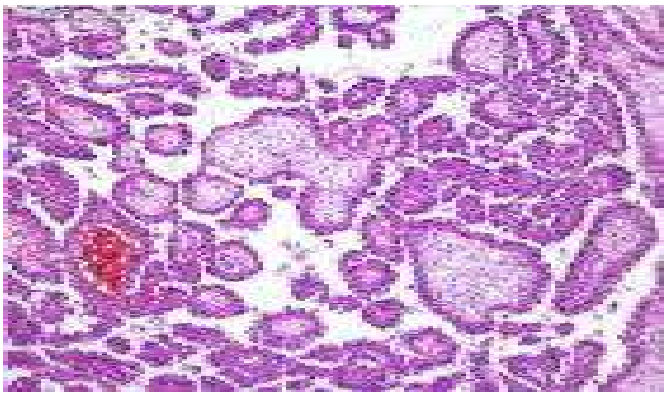
- The cells arelarge, rounded or polyhedral and arranged in acini or trabeculae separated byscanty stroma containing thin walled blood vessels.
- The cytoplasmis clear or vacuolated due to high content of lipid and glycogen.
- The nucleusissmall and darkly stained.

Papillary RCC

- It is the second commonest subtype accounting for 15-18%of all RCCs.
- It is more often multifocal and bilateral than the other common tumor types, as seen in approximately 10%of cases.

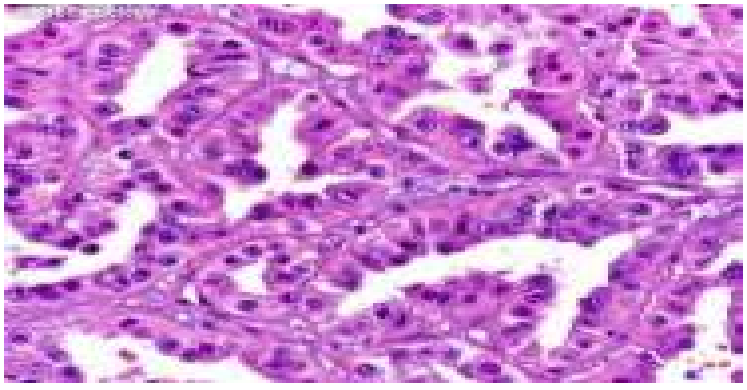
Type 1:

- the papillae are lined by a single layer of cells with scanty pale cytoplasm
- Collections of foamy macrophages are often present within the papillary cores
- Calcifications (psammoma bodies) and haemosiderin laden macrophages are common.



Type 2:

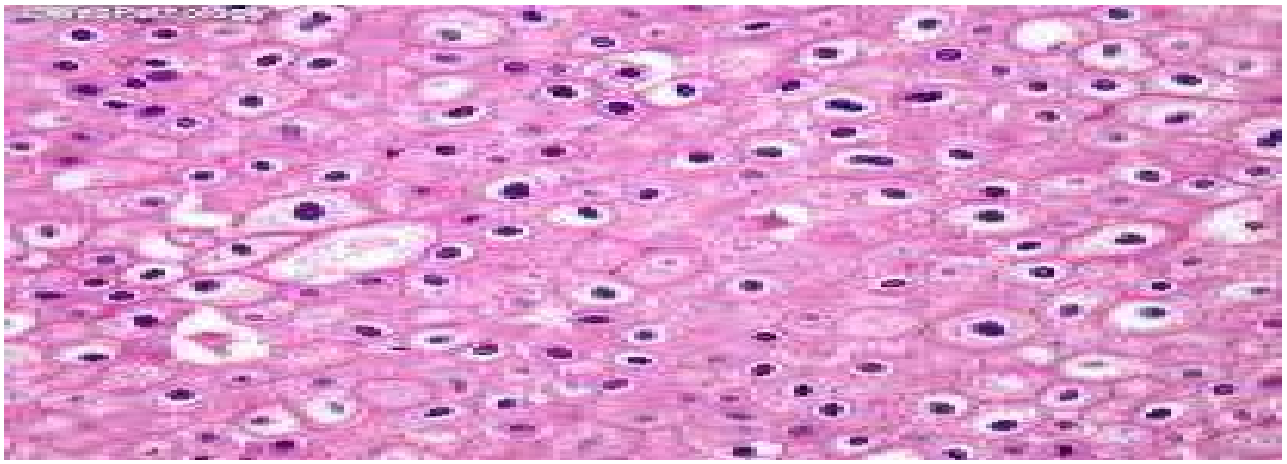
- RCC is characterized by its papillae which are lined a pseudostratified epithelium composed of cells with abundant esinophilic cytoplasm.



Chromophobe RCC

- It consists about 5% of all cases of RCCs.
- is usually sporadic and generally has a good prognosis
- Most of these tumors are confined to the kidney at diagnosis, though they may be largeat the time of presentation.
- They are characteristically tan in color.

- The tumor cells are small, cuboidal with eosinophilic granular cytoplasm and tend to form solid sheets.
- The stroma showamyloid-like material.



Spread of renal cell carcinoma:

1) **Direct spread:** to renal capsule, renal pelvis, renal vein or even inferior vena cava.
2) **Blood spread:** to lung, bone and liver.
3) **Lymphatic spread:** to para-aortic lymph nodes.

Wilm's tumor (Nephroblastoma or Embryoma)

It is seen primarily in infants, 50% of cases occur before the age of 3 years and 90% before the age of 6 years.
Equally occurs in both sexes.

Location:

- The classic location of Wilm's tumor is the kidney.
- Both kidneys are equally affected
- the incidence of bilateral involvement being 5-10%.
- Cases of Wilm's tumor have rarely been recorded in external sites as retroperitoneum, sacrococcygeal region, testis, uterus and cervix (as a cervical polyp), inguinal canal and mediastinum.

Clinical picture:

The classic presentation is ***abdominal mass.***

Hematuria and pain are rare.

Hypertension in some cases due to renin secretion by the tumor.

Proteinuria due to tumor-associated glomerular disease in the non-neoplastic kidney.

Traumatic rupture.

Pathological features

1) Grossly:

▪Solitary.However,multicentricity is present in 7%of cases. The tumor is well circumscribed, rounded, soft in consistency.
▪Cut section is predominantly solid, pale grey to tan and often exhibits areas of cystic degeneration, hemorrhage and necrosis.
▪Sometimes, the tumor is lobulated because of fibrous septation.

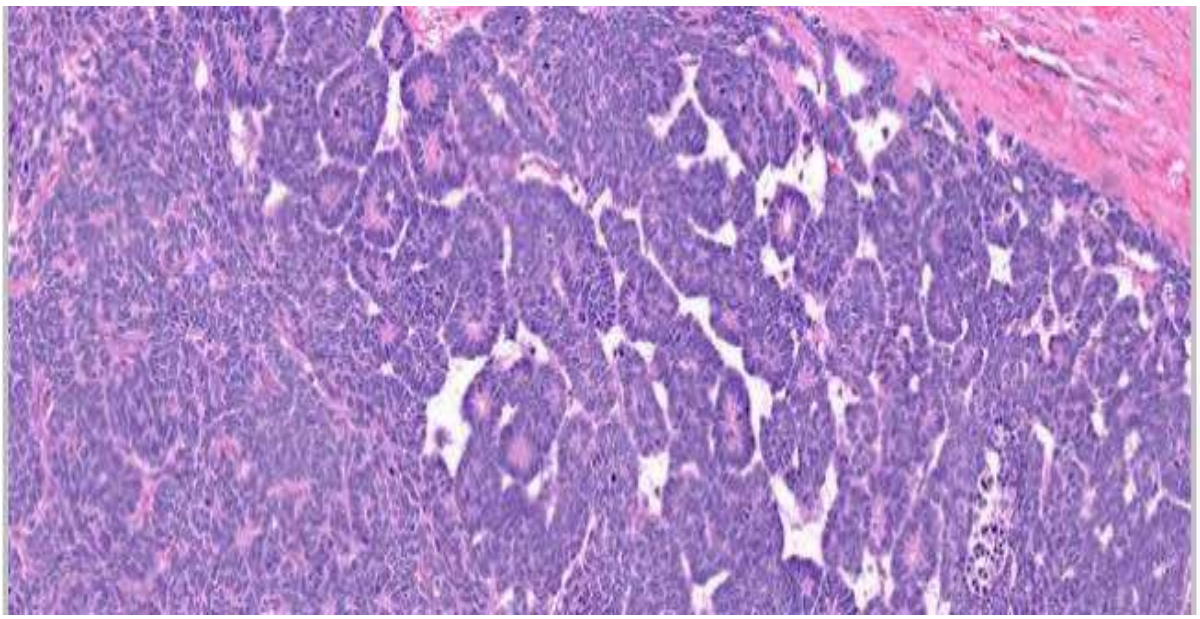
2) Microscopically:

Three majorcomponents are identified (triphasic tumor):

1) undifferentiated blastemal (primitive) elements
2) mesenchymal (stromal) elements
3) epithelial elements (it arises from pleuripotent stem cell).

Most cases show representation of all three components. However, the proportions vary widely fromcase to case and even from region to region within the same tumor. Some tumors are biphasic and others are monophasic (monomorphus).

The blastemal component: is extremely cellular and composed of small round to oval primitive cells, the cytoplasm is usually very scanty, the nuclei are commonly crowded.



Themesenchymalcomponents:
usually have **spindle cells** with **fibroblastic** or **smooth muscle** features.
they also may exhibit differentiation toward various cell types, especially skeletal muscles.

The epithelialcomponents :
Is characterized by formation of **embryonic tubular**, **glomerular structures** that recapitulate the appearance of normal developing tubules and glomeruli at the light microscopy.

Spread

- Local spread: itinfiltrates renal capsule and surrounding structures.
- Blood spread: tolung, liver, bone and brain.
- Lymphatic spread: to para-aortic lymph nodes.

2- *BladderCarcinoma*

- Urothelial Carcinoma
- Squamous Cell Carcinoma

A- UrothelialCarcinoma

- Historical name was **transitional cell carcinoma**, TCC.
- Urothelial carcinoma represents about 90%of all primary tumors of urinary bladder.

Riskfactors of urothelial carcinoma

Schistosoma haematopium induce urinary bladder cancer (both urothelial and squamous cell carcinomas).

Patients treated with radiotherapy for treatment of prostatic carcinoma.

Cigarette smoking.

Drugs as Cyclophosphamides.

Clinical picture of urothelial carcinoma

- Most patients are above 50 years old, but it can occur at a younger age group.
- **Men > women**.
- Patients usually presentedwithhematuriaandsymptomsofurinarytractinfection as dysuria, pyuria or necroturia.

Sites of urothelial carcinoma within the urinary bladder

Urothelial carcinomacan occuranywherein thebladder.However,thelocations were listed as follows;

- Lateral walls followed by posterior wall, trigone, neck, uretheric orifices, dome and lastly

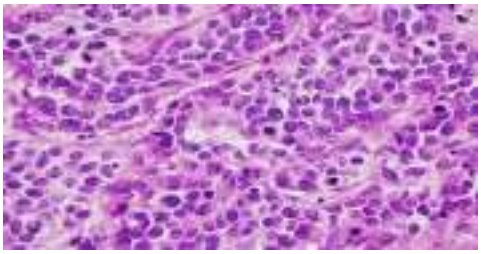
anterior wall.

Gross picture:

A polypoid mass arise from the bladder wall projecting inthebladder lumen and infiltrate the wall into variable distance. Thetumor shows areas of haemorrhage and necrosis.

Microscopical features

- Non-invasive Urothelial Neoplasm.
- Invasive Urothelial Carcinoma

A) Non-invasive Urothelial Neoplasm	B) Invasive Urothelial Carcinoma
▪ Herein, the neoplastic urothelial cells are restricted to the surface without invasion to the lamina propria or muscle layer. ▪ Non-invasive urothelial neoplasm is either flat or papillary. Non invasive papillary urothelial neoplasm •The surface urothelium is thrown into papillae, covered by several layers of neoplastic epithelial cells of urothelial origin, the cells show criteria of malignancy (pleomorphic and hyperchromatic nuclei with increases in nucleo- cytoplasmic ratio). •No invasion into the underlying lamina propria or muscles.	Herein, the neoplastic urothelial cells invade the underlying tissues. Microscopic picture: Invasive Urothelial carcinoma: Irregularly distributed neoplastic epithelial cells of urothelial morphology in the form of nests and sheets either with roundedor jagged contours. The cells showing criteria of malignancy (pleomorphism, hyperchromatism, increased N/C ratio and frequent mitotic figures). Stromal response vary widely but includes fibrosis, inflammatory reaction. The tumor cells invade the different layers of the bladder wall. Divergent differentiation is common, usually in the form of squamous or less commonly glandular differentiation. 

B- Squamous Cell Carcinoma

- It represents about 5% of all malignant bladder tumors.

Risk factors:

Drugs as cyclophosphamides.

Urinary bladder stones.

Bladder extrophy.

Chronic cystitis with extensive squamous metaplasia.

Schistosomal infection.

Gross picture

Friable tumor mass, with ulcerative and necrotic surface, infiltrates the bladder wall.

Microscopicpicture

Malignant featuring epithelial cells of squamous morphologyarranged in sheets and nests with **central keratin pearls formation**. The neoplastic cells **invade** the bladder tissue. The surrounding stroma shows chronic inflammatory cell infiltrate.

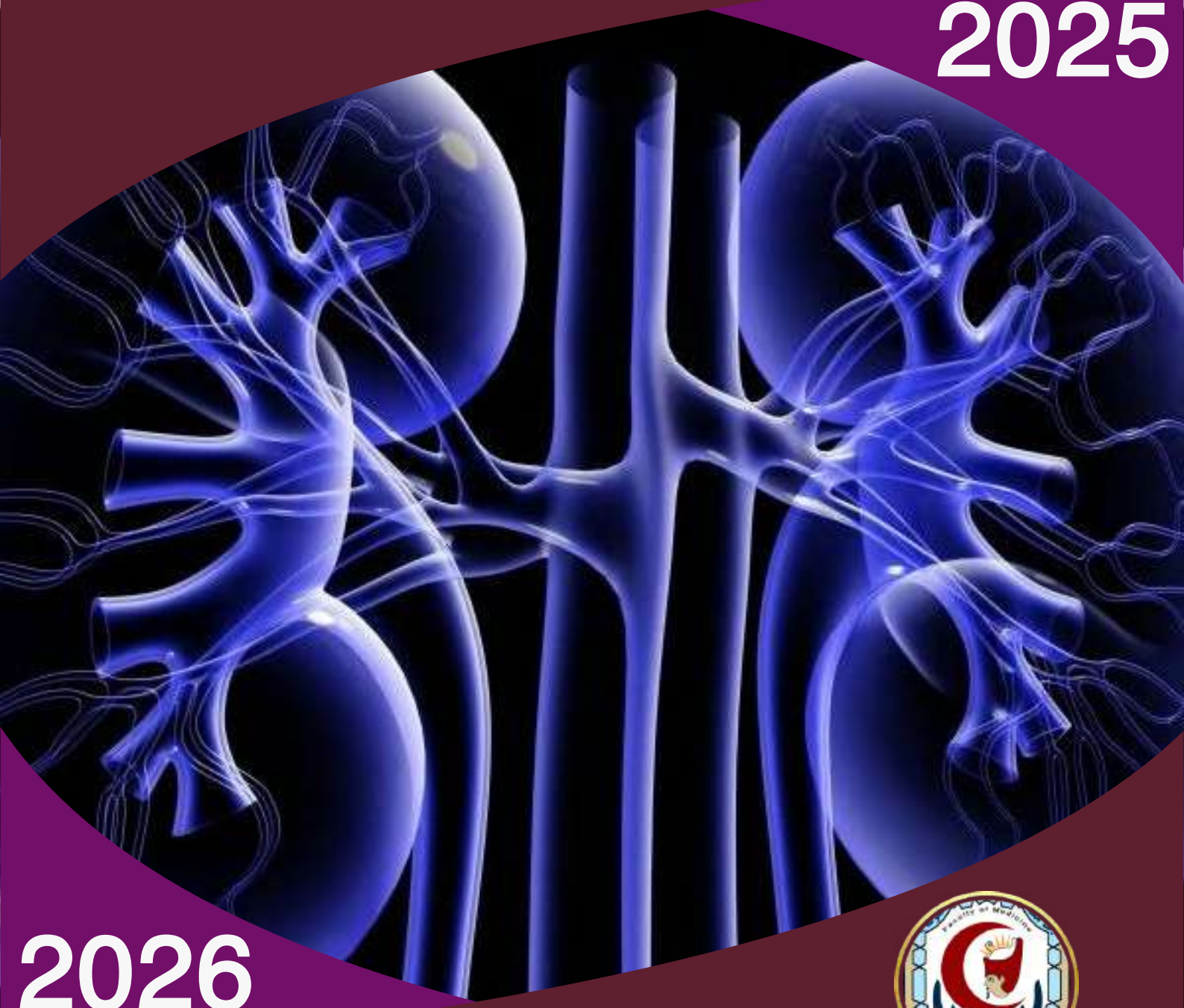
Spread ofbladdercarcinoma

Local spread: to the ureters, prostate, seminal vesicles uterus, vagina, and rectum. Local spread may result in a malignant fistula between the bladder on one side and vagina, colon or rectum on other side. Lymphatic spread: to the hypogastric and iliac lymph nodes. Blood spread: to the lung, liver and bone.

Complications:

Marked haematuria.
Hydroureter.
Hydronephrosis.
Pyelonephritis.
Malignant fistulas.

2025



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MEDICINE

Renal System

Pharmacology

RNS_315

Page	Lecture
1	Diuretics (1) &(2)
10	Nephrotoxic Drugs

1 Diuretics

"Diuresis"

- **Signifies** an **increase in urine volume.**

"Natriuresis"

- Denotes an**increase** in renal **sodium** excretion.
- Because natriuretic drugs almost **always** also **increase** water excretion, they are usually called **diuretics**.

Types of diuretics:

Accordingtotheir**natriuretic capacity**

I- High efficacy:

They allow **15-25%** of filtered **Na** tobeexcreted as:

- **Loop diuretics**

II-Moderate efficacy:

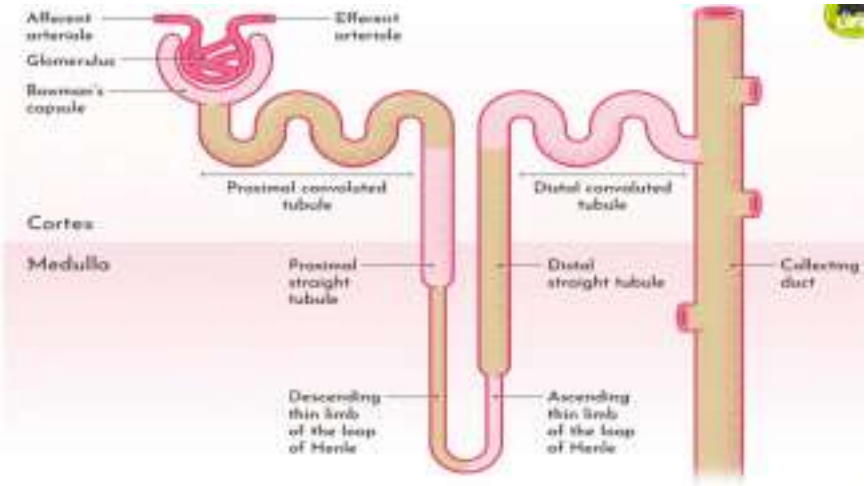
They allow **5-10 %** of filtered **Na** tobeexcreted as:

- **Thiazides.**

III-Low efficacy:

They allow **less than 5 %** of filtered **Na** to be excreted as:

- **K sparing diuretics**
- **Osmotic diuretics**
- **Carbonic anhydrase inhibitors**



Loop diuretics

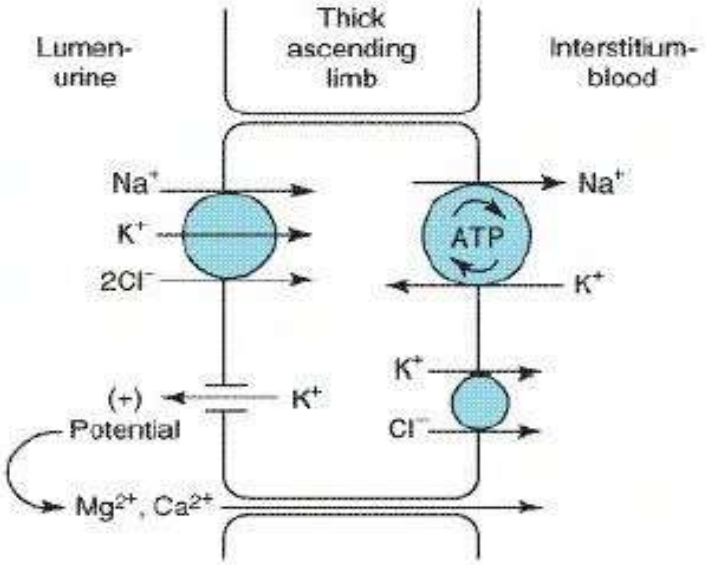
- Furosemide.
- Torsemide
- Bumetanide.
- Ethacrynic acid.

Mechanism of actions:

These drugs **inhibit**the**luminal Na+/K+/2Cl- transporter** in the **thick ascending limb** of the **loop of Henle**. So, loop diuretics lead to:

- **Reduce** the **reabsorption** of NaCl.
- **Increase** in K+ excretion.
- **Increase** in Mg2+ and Ca2+ excretion.

- **Induce (COX-2),** which **participates** in the **synthesis of PGE 2,** from arachidonic acid.



Ion transport pathways across the luminal and basolateral membranes of the thick ascending limb cell. The lumen positive electrical potential created by K^+ back diffusion drives divalent cation reabsorption via the paracellular pathway.

Hemodynamic action:

- Loop diuretics **stimulate** renal prostaglandin synthesis.
- These **PGs** participate in the **renal actions** of these drugs.
- **NSAIDs** (e.g., **indomethacin**) can **interfere** with the **actions of the loop diuretics** by **reducing PGs synthesis** in the kidney.
- Loop diuretics **increase** renal blood flow.
- **Increase** systemic venous capacitance.
- **Reduce** pulmonary congestion.
- **Decrease** left ventricular filling pressures in heart failure **before** a **measurable increase** in urinary output occurs.

Therapeuticuses:

1-**Acute pulmonary edema.**
2-Other **edematous** states as in
Cardiac, renal, or hepatic diseases,
In some cases, salt and water retention may become **severe** so a **combination** with thiazides is necessary.
3-**Essential hypertension:** • Useful in **hypertensive emergency** • In patients who have **hypertension** with **chronic renal failure**
4-**Hyperkalemia**: It can **significantly enhance** urinary excretion of K+.
5-**Symptomatic hypercalcemia:** It **increases** the urinary excretion of Ca++ but **replacement** of urinary losses of Na+ and Cl- is required.

Toxicity:

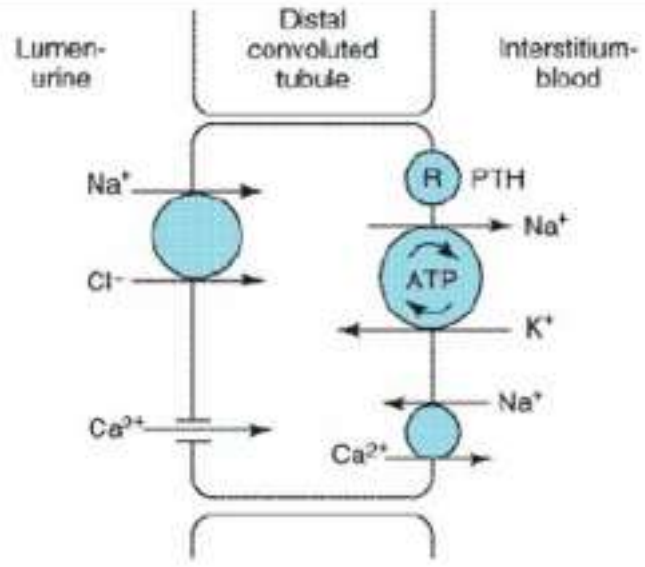
1-**Hypokalemia**:Can be **reversed** by K+ replacement.
2-**Ototoxicity**: Loop diuretics can **cause** dose-related hearing loss that is usually **reversible**.
3-**Hyperuricemia**: Loop diuretics can **precipitate** attacks of gout .
4-**Hypomagnesemia**: Can be **reversed** by the administration of oral magnesium preparations.
5-**Allergic Reactions:** Skin rash, eosinophilia, and **less often,** interstitial nephritis These usually **disappear** after **drug withdrawal.**
6-**Hyperglycemia**
7-**Nephrotoxicity**:Loop diuretics can **potentiate** the **nephrotoxicity** of **cephalosporins**.

Thiazide diuretics

- Chlorothiazide.
- Hydrochlorothiazide.
- Bendroflumethiazide
- Chlorthalidone.
- Indapamide.
- Metolazone

Mechanism of actions:

- Thiazides acton **distalconvolutedtubule.**
- They **block** the **Na+/Cl-transport** system.
- So, **inhibit** NaCl reabsorption from the **luminal side** of epithelial cells.



Ion transport pathways across the luminal and basolateral membranes of the distal convoluted tubule cell. As in all tubular cells, Na^+/K^+ ATPase is present in the basolateral membrane. (R, PTH receptor.)

Thiazides Ca2+ reabsorption in the DCT:

Thisenhancementisdue to:

- thiazides **increaseNa+/Ca2+ exchange** so,**increasing reabsorption of Ca2+.**
- **AffectCa2+absorption**by **parathyroid hormone** on **DCT**.

Therapeutic uses:

1-Hypertension

• Thiazides are **useful** in treating **allpatients** with essential hypertension.
Firstly, they **decreaseB.P** by **decreasing plasma volume andCOP.**
During **chronictherapy,** they **decrease B.P** by **vasodilator** effect.

2- Heart failure:

In **mild and moderate CHF.**
But **notuseful** in patients with **impairedrenal function.**

3- **Edema** due to chronic hepatic and renal diseases.

4- **Nephrolithiasis** due to **idiopathichypercalciuria**
This disease can be **treated** with **thiazidediuretics** , which enhance calcium reabsorption in the **distal convolutedtubule** and thus **reduce** the **urinary calciumconcentration.**

5- Osteoporosis

6- Nephrogenic Diabetes Insipidus:

• Thiazide diuretics can **reduce polyuria** in these patients.
• The effect is **paradoxical**.
• By **reduction** in **plasma volume GFR**
• **Proximal reabsorption** NaCl and water

Toxicity:

1- **Hypokalemic**: Can be reversed by K+ replacement.
2- **Hyperuricemia:** Thiazide's diuretics can **precipitate** attacks of **gout**
3- **Hyperlipidemia**. They increase in **serum cholesterol** and **increased (LDL).**
4- **Hyperglycemia**.
5- **Hyponatremia**.
6- **Allergic Reactions** As photosensitivity, dermatitis, hemolytic anemia, thrombocytopenia.
7-Hypercalcemia.

Potassium-Sparing Diuretics

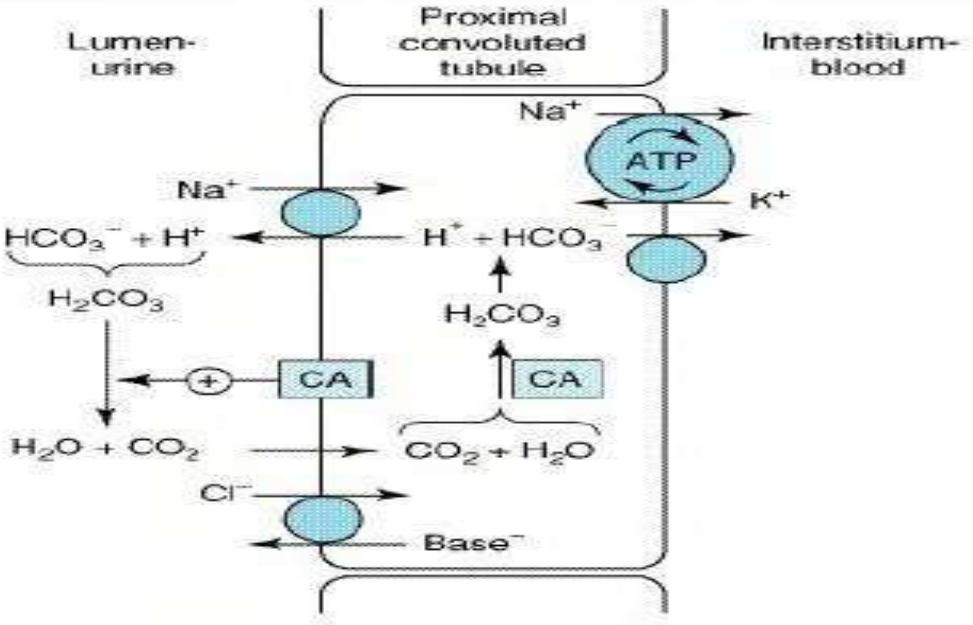
- Aldosterone receptors antagonist:
  - **Spironolactone.**
  - **Eplerenone.**
- Inhibitors of Na+ influx through ion channels in renal epithelial cells:
  - **Amiloride.**
  - **Triamterene.**

Aldosterone receptors

- **Spironolactone.**
- **Eplerenone.**

Mechanism of action:

- It antagonizes the effects of **aldosterone** in the collecting tubule.
- It reduces **Na+** absorption.
- It reduces **K+ and H+** secretion



Therapeutic uses:

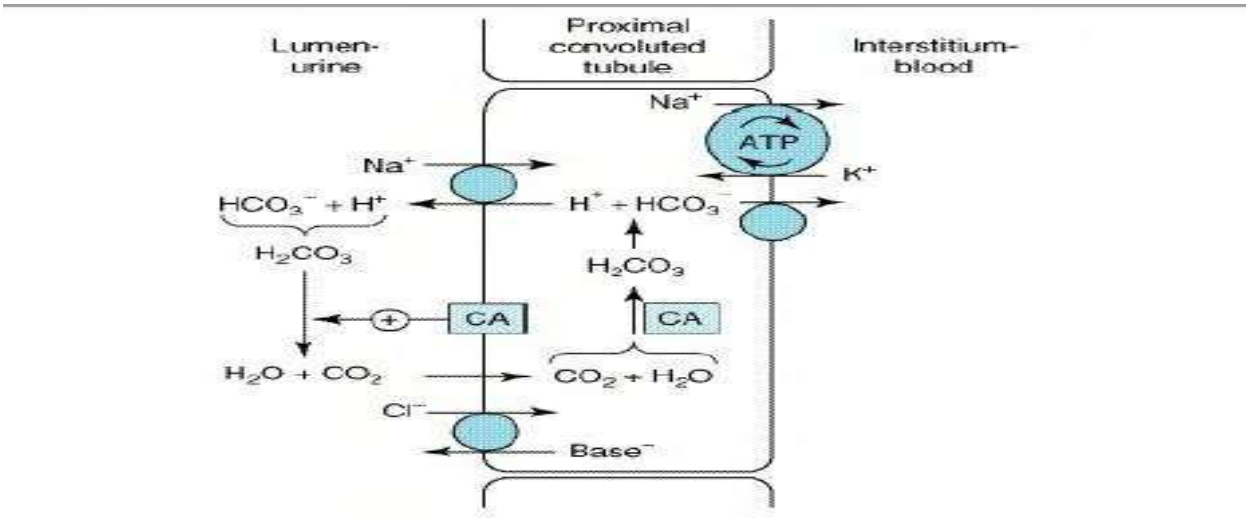
- In treatment of **hypertension**, usually associated with other diuretics. In
- states of **primary** hyperaldosteronism as Conn's syndrome, ectopic ACTH production.
- In cases of **secondary** aldosteronism in
  - o Hepatic cirrhosis.
  - o Nephrotic syndrome.
  - o Heart failure.
- It slows the progression of albuminuria in **diabetic patients** and
  - o Reduces myocardial perfusion defects after **myocardial infarction.**
  - o Low doses of eplerenone interfere with the **fibrotic and inflammatory** effects of aldosterone.

Adverse effects:

- ○ Hyperkalemia
- ○ Kidney stone with traimterene
- ○ Nausea, vomiting diarrhea.

❑ CarbonicAnhydrase (CA) Inhibitors

- ○ CA inhibitors block CA enzyme, so prevent the dehydration of H2CO3 to CO2 at the luminal membrane and cause diuresis.
- ○ CA inhibition causes significant HCO3 losses and metabolic acidosis



Clinical uses:

- **Glaucoma**: The reduction of aqueous humor formation by HCO_3^- losses lead to reduce in the I.O.P.
- **Urinary Alkalinization**: CA inhibitors can increase urinary pH, so decrease uric acid stones.
- They are used as adjuvants in the treatment of **epilepsy**.

Adverse Effects:

- Most adverse effects are secondary to urinary alkalinization or metabolic acidosis, including:
  - Drowsiness.
  - Increase the **absorption of ammonia** from urine into the blood leads to worsening hepatic encephalopathy, so the drugs are contraindicated in patients with hepatic cirrhosis.
  - Ureteral colic due to **precipitation of calcium** salts in alkaline urine.

Osmotic Diuretics

Mannitol

Site of action:

- **Proximal tubule.**
- **Descending limb of Henle's loop.**

Pharmacokinetics

❑ **Osmotic diuretics are characterized by:**

- They are poorly absorbed.
- They are not metabolized and are excreted primarily by glomerular filtration.
- They are not reabsorbed by renal tubules.
  They are pharmacologically inert.

Mechanism of acti on:

- They **prevent the water reabsorption** in the **Proximal tubule** and **Descending** limb of the loop of Henle.
- They also **antagonize** the action of **ADH** in the collecting tubule.
- The presence of mannitol prevents the normal absorption of water by increasing the osmotic force.
- So, urine volume **increases**.
- The **increase in urine flow** rate **decreases Na* reabsorption**.

Therapeutic uses:

- Prophylaxis and early treatment of acute renal failure in cases of:
  - Cardiovascular operations.
  - Severe traumatic injury.
  - Hemolytic transfusion reactions.

- Used to reduce intracranial pressure in neurologic conditions.
- To reduce intraocular pressure before ophthalmologic procedures
  - o Used in the treatment of
  - o **dialysis disequilibrium syndrome:**
  - o In the patient with chronic renal failure

Toxicity

- **Extracellular Volume Expansion**

- Mannitol is rapidly distributed in the extracellular compartment and extracts water from cells.

This effect can:

- o Complicate **heart failure.**
- o Produce **pulmonary edema**
- o Hypernatremia.
- o Dehydration.
- o Headache, nausea, and vomiting.

2 Nephrotic drugs

ILOS

- List of nephrotoxic drugs
- Common drugs that induce nephrotoxicity
- Their nephrotoxic action

Definition

- Nephrotoxic drugs are medications or chemical substances that cause structural or functional damage to the kidneys, leading to impaired renal function. Characterized

❑ **characterized by**

- Increase serum creatinine value
- Increase Blood urea nitrogen

Risk factors

❑ Patientrelatedfactors:

- Age
- sex
- race
- Pre-existent renal disease
- Specific disease (DM)
- Dehydration

- Drug-relatedfactors:
  - Inherent
  - Dose of the drug
  - Duration of drug administration
  - frequency of drug administration
- Drug interactions

Clinical presentation

General

Most common manifestation is a decline in glomerular filtration leading to a risenin serum creatinine and blood urea nitrogen.

Symptom

Malaise, anorexia, vomiting, shortness of breath or edema.

Signs

Decreased urine output

Mechanism of renal nephrotoxicity

1-Tubular cell toxicity

2-Rhabdomyolysis

3-Altered intraglomerular hemodynamics

4-Inflammations:

- √ Acute interstitial nephritis
- √ Chronic interstitial nephritis

5-Crystal nephropathy

6-Thrombotic microangiopathy

1)Tubular cell toxicity

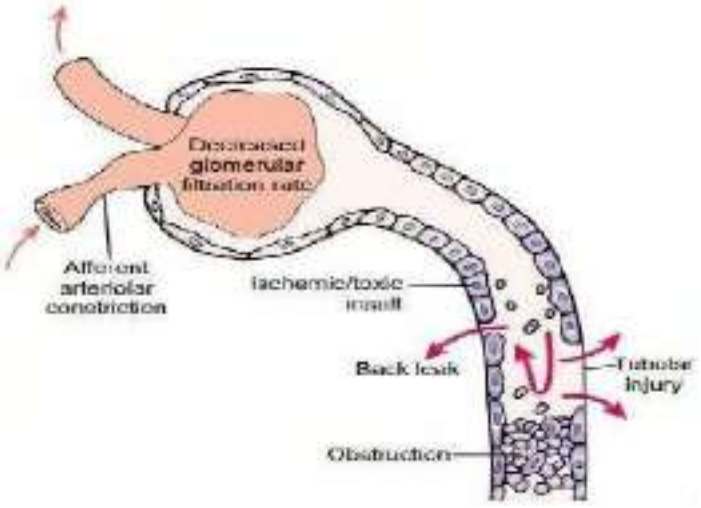
Renal tubular cells, in particular proximal tubules, are vulnerable to the toxic effect of drugs because of their roles in concentrating and reabsorbing glomerular filtrate exposes them to high levels of circulating toxins.

Drugs that cause tubular cell toxicity do so by:

- **Impairing mitochondrial function**
- **Interfering with tubular transport**
- **Increasing oxidative stress or forming free radicals.**

Drugs associated with tubular cell toxicity:

- Aminoglycosides
- Amphotericin B
- Antiretrovirals



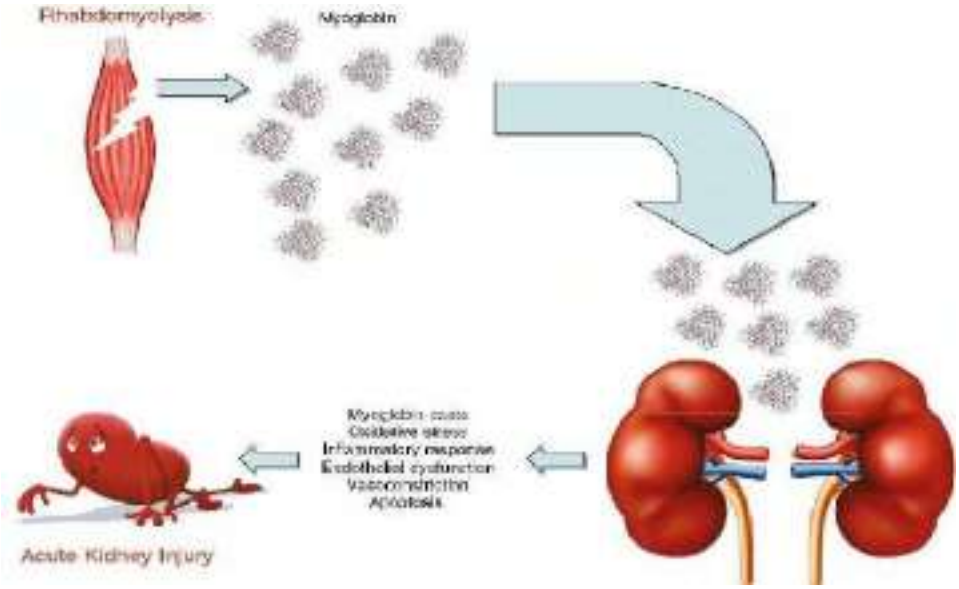
(Tubular cell toxicity)

2)Rhabdomyolysis

Rhabdomyolysis: A syndrome in which skeletal muscle injury leads to lysis of the myocyte, releasing intracellular contents including myoglobin and creatinine kinase into plasma. Myoglobin induces renal injury secondary to direct toxicity, tubular obstruction and alternation in GFR.

Examples:

Statins, fibrates, antipsychotics (haloperidol), cocaine & heroin



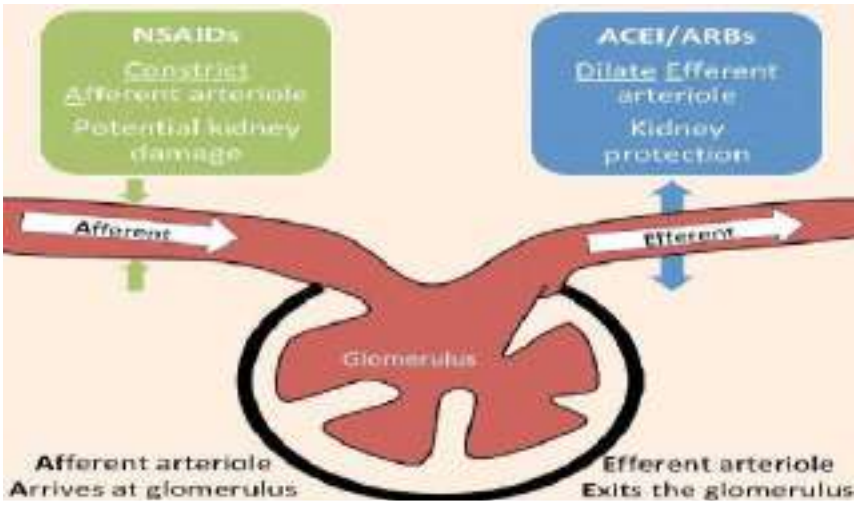
Rhabdomyolysis

3)Altered intraglomerular hemodynamics

Kidney is known to autoregulate the intraglomerular pressure by changing the afferent and efferent arterial tone to preserve GFR and urine output.

In case of volume depletion, following changes occur: Circulating prostaglandins vasodilate the afferent arterioles allowing more blood to flow through the glomerulus, while angiotensin I I causes vasoconstriction of the efferent arterioles. Some drug classes are known to interfere with kidneys ability to autoregulate glomerular pressures and thereby decrease GFR.

- **NSAIDs with antiprostaglandin activity**
- **Drugs with anti-angiotensin I I activity (ACEIs & ARBs)**



4-Inflammations

Drugs can cause inflammatory changes in the glomerulus, renal tubular cells, and the surrounding interstitium, leading to fibrosis and renal scaring.

❑ Drugs that cause Acute interstitial nephritis

- Sulfonamides
- Methicillin
- Ampicillin
- Rifampicin
- Thiazides
- NSAIDs
- Allopurinol
- Cimetidine

SMART Nephrons Are Crying



❑ Drugs that cause chronic interstitial nephritis

1. Acetaminophen
2. Aspirin

5-Crystal nephropathy

Renal impairment may result from using drugs that produce crystals which are insoluble in urine. The crystals precipitate, usually within the distal tubular lumen, obstructing urine flow and eliciting an interstitial reaction

Drugs associated with production of crystals include:

1. Antibiotics (ampicillin, sulfonamides)
2. Antiviral (acyclovir)

6-Thrombotic microangiopathy

Thrombotic microangiopathy, organ damage is caused by platelet thrombi in microcirculation, as thrombotic thrombocytopenic purpura.
Renal injury occurs due to immune mediated reactions or direct endothelial
toxicity

Examples:

Antiplatelet agents (clopidogrel)

Measures to prevent drugs induced nephrotoxicity

1. Adjust medication dosages
2. Avoid nephrotoxic combination.
3. Correct risk factors for nephrotoxicity before initiation of drug therapy.
4. Ensure adequate hydration before and during therapy.
5. Use equally effective non-nephrotoxic drugs whenever possible.
6. Measures to prevent drugs induced nephrotoxicity

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